

Rand Afrikaans University

SEMESTER TEST: 20 AUGUST 2002

COURSE: Computer Science
MEMORANDUM: Computer Science 6032

TIME: 1 hour
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.

Question 1

- a) What follows the line containing `Host` in an HTTP request?
- A:** Content-type
 - B:** GET / HTTP/1.0
 - C:** An empty line
 - D:** More than one of the above
- b) The HEAD request of HTTP
- A:** Supplies information about the file concerned, such as the file size.
 - B:** Contains elements such as the title of the page.
 - C:** Determines which socket should be used for the request.
 - D:** More than one of the above
- c) XML fits into layer ... of the ISO/OSI model.
- A:** 4
 - B:** 5
 - C:** 6
 - D:** 7

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d) The first line of an HTML 4 file should be

A: <html>

B: <HTML>

☒ **C:** <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"
"http://www.w3.org/TR/html4/strict.dtd">

D: More than one of the above

e) Optical fibre is regularly used to communicate at the following communication speed.

A: $3 \times 10^8 \text{ms}^{-1}$

☒ **B:** 1 Gbps

C: 36 000 km

D: 10 Mbps

E: None of the above

f) Given that the speed of light is $3 \times 10^8 \text{ms}^{-1}$, how long does it take a signal to travel from an earth station to a geostationary satellite and back to another earth station?

A: 0,12s

☒ **B:** 0,24s

C: 0,36s

D: 0,48s

g) Advantages of a LEO satellite for data communications include the following:

☒ **A:** Since it is placed closer to earth than geosynchronous satellites, a smaller transmitter can be used from earth.

B: It orbits around the earth.

C: It is not affected as much as geosynchronous satellites by mountains and other natural obstacles.

D: More than one of the above

E: All of the above

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h) Which of the following infrastructure is inherently able to communicate in full-duplex?

A: Ethernet

B: Token ring

☒ **C:** RS-232

D: CSMA/CD

E: More than one of the above

i) A modem

☒ **A:** Modulates a digital signal so that it can be transmitted via an analog medium over a long distance.

B: Demodulates a digital signal so that it can be transmitted via an analog medium over a long distance.

C: Modulates an analog signal so that it can be transmitted via a digital medium over a long distance.

D: Demodulates an analog signal so that it can be transmitted via a digital medium over a long distance.

j) PPP (*Point-to-Point Protocol*) uses ... to achieve transparency.

A: Bit stuffing

☒ **B:** Byte stuffing

C: Known field lengths

D: Special transmission techniques

E: It does not guarantee transparency.

k) Which of the following is not a function of the data link control layer?

A: Data delineation

B: Error control

☒ **C:** Encoding

D: Media access control

E: None of the above

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- l) If two stations using Ethernet transmit at the same time, a collision occurs. Why don't the two start transmitting later at exactly the same time, just to be involved in another collision?

A: The different lengths of the packets make a second collision unlikely.

B: The two stations won't notice the collision at precisely the same moment; even if they wait an equal time, they will not start transmitting at exactly the same time again.

C: It does not matter if they collide again; eventually the different clocks on the computers will cause the one to start transmitting slightly before the other.

D: Only one of the stations involved in the collision will stop transmitting; the other station will continue and complete data transmission; therefore later collisions are not a problem.

☒ **E:** None of the above

- m) Which IEEE standard specifies the operation of wireless local area networks?

A: IEEE 802.3

B: IEEE 802.5

C: IEEE 802.10

☒ **D:** IEEE 802.11

- n) FDDI differs from IEEE 802.5 because

A: FDDI has two rings.

B: FDDI can continue operating if one of the nodes is switched off.

C: FDDI is used over larger distances.

☒ **D:** More than one of the above

E: All of the above

- o) An Ethernet frame may contain ... octets of data.

A: 0 – 1500

B: 46 – ∞

☒ **C:** 46 – 1500

D: 2 – 1500

E: 150 – 4800

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p) A UTP cable used for Ethernet, uses the following connector:

A: DB25

B: Telkom

☒ **C:** RJ-45

D: X-45

q) A switch (used in the place of a wiring hub)

A: Connects different networks and can convert protocols.

☒ **B:** Makes a connection only between computers that are busy communicating.

C: Routes packets between nodes.

D: Operates on ISO OSI layer 3.

E: More than one of the above

r) ADSL

A: Is used between the exchange and the user of the telephone service.

B: Is faster than BRI ISDN.

C: Carries traffic at equal speeds in both directions.

☒ **D:** More than one of the above

E: All of the above

s) IP address 172.16.5.79 is a ... address.

A: Class A

☒ **B:** Class B

C: Class C

D: Class D

E: Class E

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t) How many class B IP addresses can (potentially) exist?

A: $\pm 16\,000$

B: $\pm 64\,000$

C: $\pm 1\,000\,000$

D: $\pm 4\,000$

E: $\pm 8\,000$

[20]

Question 2

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

a) 011011001010 **[8]**

b) 110011101110 **[11]**

c) 100001010101 **[12]**

d) 101001011110 **[1]**

e) 100101011100 **[0]**

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

[10]

7./...

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Question 3

- a) What are the following fields in an Ethernet packet used for?
- (i) Preamble [Synchronisation]
 - (ii) Source address [Identifies sender]
 - (iii) CRC [Error control]
 - (iv) Frame type [Identifies layer 3 protocol] (4)
- b) What are *multicast* and *broadcast*? Ensure that the difference is clear. [Broadcast: message to everyone on network; multicast: message to group of nodes on network.] (2)
- c) (i) What is Telnet? [Virtual terminal (protocol)]
- (ii) Which port is normally used by Telnet? [23] (2)
- d) What is the purpose of the following two HTML tags?
- (i) <HR> [Horizontal line]
 - (ii) <TR> [Row in table] (2)

[10]

Question 4

Assume that the costs between nodes of a network are as follows:

A — B: 8 C — F: 7
B — D: 1 D — E: 1
B — E: 4 D — F: 7
B — F: 10 E — F: 4
C — E: 3

Use Dijkstra's algorithm to determine the routes from A.

[10]

TOTAL

[50]

END OF PAPER

[																					
Cost										Prior											
Completed					Candidates					X	B C D E F					B C D E F					
+-----					+-----					+	+-----					+-----					
1		A				B			B		8	?	?	?	?		A	?	?	?	?
2		AB				DEF			D		8	?	9	12	18		A	?	B	B	B
3		ABD				EF			E		8	?	9	10	16		A	?	B	D	D
4		ABDE				FC			C		8	13	9	10	14		A	E	B	D	E
5		ABDEC				F			F		8	13	9	10	14		A	E	B	D	E
]																					

Rand Afrikaans University

SEMESTER TEST: 8 OCTOBER 2002

COURSE: Computer Science
MEMORANDUM: Computer Science 6032

TIME: 1 hour
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) ADSL in South Africa supports a download speed of

- A: 6 Mbps
- B: 1 Mbps
- ☐ C: 512 kbps
- D: 128 kbps
- E: 56 kbps

b) The MTU (*Maximum Transmission Unit*) is

- ☐ A: The size of the largest packet that can be transmitted via the network.
- B: The highest speed at which packets can be transmitted through the network.
- C: The size of the largest TCP segment that can be transmitted via the network.
- D: The cost of the most expensive communications circuit that can be used.
- E: The highest speed at which bits can be transmitted through the network.

2./...

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- c) Where are fragmented IP datagrams reassembled?
- A:** At the first router that notices that it is fragmented.
 - B:** At the first router that can handle the total IP datagram.
 - C:** At the final destination.
 - D:** Whenever the TTL becomes 0.
 - E:** Fragmented IP datagrams are discarded, rather than reassembled.
- d) An IPv6 address consists of
- A:** 128 bits.
 - B:** 16 octets.
 - C:** 8 hexadecimal numbers, separated by colons.
 - D:** More than one of the above
 - E:** All of the above
- e) ping works by sending the following ICMP packet:
- A:** *Ping request*
 - B:** *Echo request*
 - C:** *Information request*
 - D:** *Source quench*
 - E:** All of the above
- f) traceroute (or tracert) can work using
- A:** IP's TTL and ICMP's *Time exceeded*
 - B:** IP's TTL and ICMP's *Destination unreachable*
 - C:** ICMP's Router advertisement
 - D:** IP's *Source route*
 - E:** All of the above

3./...

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g) The path MTU can be determined by using

A: ICMP's *Echo request* and *echo reply*

B: IP's *Don't fragment* and ICMP's *Destination unreachable: Fragmentation required but DF bit is set*

C: ICMP's *Get path MTU*

D: TCP's three-way handshake

E: HTTP's GET /MTU

h) The three messages sent by TCP when a connection is established, are

A: SYN, SYN+ACK, ACK

B: HELO, HELO, HELO

C: HELO, HELO, ACK

D: HELO, HELO+OK, OK

i) If a TCP segment is lost, TCP reacts by

A: Retransmitting the segment.

B: Decreasing the rate at which segments are sent.

C: Performing another three-way handshake.

D: More than one of the above

E: All of the above

j) DNS normally uses the following port:

A: 22

B: 25

C: 53

D: 80

E: 110

4./...

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- k) FTP's **passive** command has the effect that, in contrast to its usual operation,
- A:** The data connection is also used to transfer files.
 - ☒ **B:** The control connection is also used to transfer files.
 - C:** The data connection is also used to transfer commands.
 - D:** The control connection is also used to transfer commands.
 - E:** None of the above
- l) To use anonymous FTP, one uses the following user name:
- A:** *Anonymous*
 - B:** *FTP*
 - C:** Leave the name field empty
 - ☒ **D:** More than one of the above
 - E:** All of the above
- m) What is the name of the server that is commonly used on Unix systems to share files with Windows clients?
- A:** Tango
 - B:** Foxtrot
 - ☒ **C:** Samba
 - D:** Waltz
 - E:** Jive
- n) In which of the following programming languages can CGI programs be written?
- A:** Perl
 - B:** C++
 - C:** COBOL
 - D:** More than one of the above
 - ☒ **E:** All of the above

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- o) Which of the following statements regarding DHCP is false?
- A:** DHCP was designed as an extension of BOOTP.
 - B:** DHCP can supply the address of the default router.
 - C:** DHCP uses IP to carry data.
 - D:** DHCP will not assign an address assigned to computer A to a computer B before B has been switched of.
 - E:** DHCP uses UDP to carry data.
- p) Which of the following am I allowed to put on my Web page without permission?
- A:** Khachaturian's *Sabre Dance*, performed by the LSO, from the CD that I bought yesterday.
 - B:** The text of Shakespeare's *Hamlet*.
 - C:** A story from last week's *You*, that I find useful for my academic studies.
 - D:** More than one of the above
 - E:** None of the above
- q) If one uses a program running on a server from a workstation and X Windows is used to display the results graphically, the X server will execute on the
- A:** Workstation
 - B:** Server
 - C:** Other Unix computer
 - D:** MIT computers
 - E:** More than one of the above
- r) To terminate an FTP session, the following command(s) may be used:
- A:** quit
 - B:** exit
 - C:** bye
 - D:** More than one of the above
 - E:** All of the above

6./...

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s) To have text display larger in WML, one can use the following:

- A:** <big>
- B:** <bigger>
- C:** <font++>
- D:** <sizeup>
- E:** More than one of the above

t) PGP may be used for

- A:** Public key cryptography
- B:** Digital signatures
- C:** Symmetric encryption
- D:** All of the above
- E:** None of the above

[20]

Question 2

a) Briefly explain what each of the following DNS entry types is used for. (Do not just say what the abbreviation stands for.)

- (i) A [Gives IP address of specified name]
- (ii) CNAME [Associates another name with specified name]
- (iii) MX [Gives details about machine(s) handling e-mail on behalf of specified machine]

(3)

b) What is the difference between iterative and recursive name resolution? [iterative: resolving machine queries once server after another up the DNS tree; recursive: DNS server passes query on to next server up the tree on behalf of resolving machine]

(2)

[5]

7./...

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Question 3

Assume that a *command* in a protocol is a message sent from a client to a server. (As an example, GET, HEAD and POST may be seen as commands of HTTP.) Name any five (5) commands of POP3 and briefly say what each is used for. **[5]**

Question 4

Assume that the following six letters occur with the given frequencies:

- A 16%
- B 17%
- C 7%
- D 8%
- E 19%
- F 33%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- a) Write your answer in the space provided on your answer sheet. **[A : 101 B : 00 C : 1000 D : 1001 E : 01 F : 11]** (6)
- b) What message is encoded by the following bit string?
1001111011001101101 [DFADAA] (2)
- c) How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? **[18%]** (2)

[10]

8./...

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Question 5

Explain what each of the following Unix commands is used for. Also explain what the purpose is of every option and/or parameter that is given. (Do not just say what the command, parameter or option is an abbreviation for.)

- a) `ls -l abc` [Displays list of files, -l: displays details about files, abc: name of file for which details are to be displayed] (3)
- b) `passwd def` [Changes password, def: name of user whose password is to be changed] (2)
- c) `vi ghi` [text editor, ghi: file to be edited] (2)
- d) `mount /dev/fd0 /mnt/jkl` [Adds disc to file system, /dev/fd0: first floppy, /mnt/jkl: mount point] (3)

[10]

TOTAL

[50]

END OF PAPER

University of Pretoria

SEMESTER TEST: 3 MAY 2004

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 1 hour
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.

Question 1

a) The physical path used by a data communications message is known as the

☒ **A:** Medium

B: Protocol

C: UTP cable

D: Transport layer

E: All of the above

b) Which ISO OSI layer provides process-to-process delivery?

A: 2

B: 3

☒ **C:** 4

D: 5

E: 6

F: 7

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MEMORANDUM: COS332

- c) When a message moves through the protocol stack towards the physical layer, headers are
- A:** Appended
 - B:** Removed
 - ☒ **C:** Added
 - D:** Modified
 - E:** Rearranged
- d) A header added to a message at ISO OSI layer n will be removed at the destination at layer ...
- A:** $n - 1$
 - ☒ **B:** n
 - C:** $n + 1$
 - D:** 1
 - E:** 7
- e) Manchester encoding has a transition ... each bit interval.
- A:** between
 - ☒ **B:** in the middle of
 - C:** All of the above
 - D:** None of the above
- f) Assume ASK is used for data transmission at 600 bps. The baud rate is
- A:** 150
 - B:** 300
 - ☒ **C:** 600
 - D:** 1200
 - E:** 2400

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MEMORANDUM: COS332

- g) An E1 line communicates at
- A:** the same speed as a T1 line
 - B:** 1.5 Mbps
 - ☒ **C:** 2.048 Mbps
 - D:** 10.000 Mbps
 - E:** All of the above
- h) A CRC is used to ensure that data is transmitted correctly. We know that no transmission errors have occurred if the remainder (after division) is
- A:** Even
 - B:** Odd
 - ☒ **C:** 0
 - D:** 1
 - E:** The same as the remainder of the original data
- i) IEEE 802.11g (at full speed) communicates at
- ☒ **A:** 54 Mbps
 - B:** 11 Mbps
 - C:** 10 Mbps
 - D:** 1 Mbps
 - E:** 128 kbps
- j) The REJ frame of HDLC will cause the ... frame(s) to be resent.
- A:** Previous
 - B:** Indicated
 - C:** Indicated and all prior
 - ☒ **D:** Indicated and all subsequent

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MEMORANDUM: COS332

k) HDLC's DISC (*Disconnect*) is an example of a(n) ... frame.

A: Handshake

B: Unnumbered

C: Supervisory

D: Information

E: TCP

l) Which PPP authentication method is more secure?

A: CHAP

B: PAP

C: SSH

D: Telnet

E: LCP

m) When no monitor's presence is detected on an 802.5 network

A: The hub appoints a new monitor.

B: The previous monitor resumes its duties.

C: The station with the highest priority starts acting as monitor.

D: A new monitor is elected by the remaining nodes.

E: The networks simply continues operating because it does not really need a monitor.

n) Three stations are waiting to transmit data on a medium where .5-persistent CSMA/CD is used. What is the probability of a successful transmission the moment when the medium becomes available?

A: $\frac{1}{8}$

B: $\frac{2}{8}$

C: $\frac{3}{8}$

D: $\frac{4}{8}$

E: $\frac{5}{8}$

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MEMORANDUM: COS332

- o) The Ethernet address 7B-10-3F-90-8C-B3 could be
- A:** A unicast destination address.
 - B:** A multicast destination address.
 - C:** A source address.
 - D:** More than one of the above
 - E:** All of the above
- p) In CSMA/CA the sequence of events to avoid collisions is
- A:** NAV, RTS, CTS
 - B:** RTS, NAV, CTS
 - C:** NAV, CTS, RTS
 - D:** RTS, NAV, CTS
 - E:** RTS, CTS, NAV
- q) Bluetooth transmission can interfere with IEEE ... transmission.
- A:** 802.11a
 - B:** 802.11b
 - C:** 802.11g
 - D:** More than one of the above
 - E:** All of the above
- r) Which of the following am I allowed to put on my Web page without permission?
- A:** Beethoven's *Für Elize* as played by myself on the piano.
 - B:** A copy of Shakespeare's *Hamlet* from the book that I bought yesterday at the bookshop around the corner.
 - C:** A story from last week's *You*, that I find useful for my academic studies.
 - D:** More than one of the above
 - E:** None of the above

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MEMORANDUM: COS332

s) ATM differs from IP since ATM

A: Is connection-oriented.

B: Uses cells of a fixed size.

C: Ensures that all cells arrive in order.

D: More than one of the above

E: All of the above

t) Where are fragmented IP datagrams reassembled?

A: At the first router that notices that it is fragmented.

B: At the first router that can handle the total IP datagram.

C: At the final destination.

D: Whenever the TTL becomes 0.

E: Fragmented IP datagrams are discarded, rather than reassembled.

[20]

Question 2

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

a) 101000001010 [0]

b) 101001100010 [5]

c) 100110111111 [10]

d) 111101010010 [4]

e) 100101011000 [3]

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

[10]

7./...

Draw a waveform diagram that shows how the bit pattern 101100 is encoded using ...encoding.

- In each case say (i) how 0 and 1 are represented and, (ii) if applicable, show the level of the signal prior to the transmission of the given bit string. **[6]**

Suppose an Internet user wants to visit the web page associated with the URL `http://www.cs.up.ac.za`. Write down a minimal HTTP request consisting of three (3) lines that a browser can send to retrieve this page. **[4]**

Assume that the costs between nodes of a network are as follows:

Use Dijkstra's algorithm to determine the routes from A.

	Cost	Prior	#	Done	Candidates	X	B	C	D	E	F	G
1	A	BC	B	5 10 ? ? ? ?	A A ? ? ? ?							
2	AB	C	C	5 8 ? ? ? ?	A B ? ? ? ?							
3	ABC	DEG	D	5 8 10 12 ? 13	A B C C ? C							
4	ABCD	EG	E	5 8 10 12 ? 13	A B C C ? C							
5	ABCDE	GF	G	5 8 10 12 20 13	A B C C E C							
6	ABCDEG	F	F	5 8 10 12 14 13	A B C C G C							

TOTAL

[50]

END OF PAPER

Question 4

The IP address of one of the computers in your organisation is 192.192.192.192. Answer the questions below using dotted decimal notation.

- In which IP class does this address belong? [C] (1)
- What network mask will you use in this environment (not using subnetting)? [255.255.255.0] (1)
- Give the address of the network. [192.192.192.0] (1)
- What is the broadcast address for all hosts on this network? [192.192.192.255] (1)
- Your organisation, however, wishes to use (at least) 5 subnets with (at least) space for 20 computers on each subnet. In CIDR notation 10.10.0.0/16 means that the first 16 bits are to be used as the network portion of the address. Write down the address of the subnet on which 192.192.192.192 belongs, using CIDR notation. [192.192.192.27] (2)
- Give the netmask to be used for such subnetting. [255.255.255.224] (1)
- Give the broadcast address for the subnet containing 192.192.192.192. [192.192.192.223] (1)
- What are the (i) highest and (ii) lowest addresses that may be assigned to hosts on the subnet that contains 192.192.192.192? [192.192.192.193, 192.192.192.222] (2)

[10]

Question 5

Assume that the costs between nodes of a network are as follows:

A-C: 9 C-F: 8
A-D: 1 C-G: 1
A-G: 10 D-G: 4
B-F: 10 E-F: 9

Use Dijkstra's algorithm to determine the routes from A.

[10]

		Cost										Prior									
	Done	Candidates	X	B	C	D	E	F	G			B	C	D	E	F	G				
1		A		CDG	D	?	9	1	?	?	10	?	A	A	?	?	A				

2	AD	CG	G	?	9	1	?	?	5	?	A	A	?	?	D					
3	ADG	C	C	?	6	1	?	?	5	?	G	A	?	?	D					
4	ADGC	F	F	?	6	1	?	14	5	?	G	A	?	C	D					
5	ADGCF	BE	E	24	6	1	23	14	5	?	F	G	A	F	C	D				
6	ADGCFE	B	B	24	6	1	23	14	5	?	F	G	A	F	C	D				

14/...

Question 6

The dialogue below occurred between the e-mail program of a user, Jan Jacobs, and a POP3 server. Five lines have been omitted from the dialogue. These five lines have been marked [1], [2], [3], [4] and [5]. Complete the dialogue by writing them in the space provided on your answer sheet. [(1) user jan, (2) pass qq, (3) list, (4) dele 2, (5) retr 1/top 1 1] [10]

```
+OK POP3 nafplion v2000.69mdk server ready
[1]
+OK User name accepted, password please
[2]
+OK Mailbox open, 2 messages
[3]
+OK Mailbox scan listing follows
1 459
2 392
.
[4]
+OK Message deleted
[5]
+OK 459 octets
Return-Path: <santa@northpole.com>
Delivered-To: jan@up.ac.za
Received: from client (nafplion [10.1.1.1])
        by nafplion.up.ac.za (Postfix) with SMTP id 2B54D11AA98
        for <jan@nafplion.up.ac.za>; Thu, 4 Oct 2005 17:18:32 -0400 (EDT)
From: Santa <santa@northpole.com>
To: Jan Jacobs <jan@nafplion.up.ac.za>
Subject: 2005
Message-Id: <20051004211832.2B54D11AA98@nafplion.up.ac.za>
Date: Thu, 4 Oct 2005 17:18:32 -0400 (EDT)
Status: O
```

```
Hoi! Hoi! Hoi!
.
quit
+OK Sayonara
```

15/...

Question 7

Assume that the following six letters occur with the given frequencies:

A	20%	B	19%	C	9%
D	26%	E	5%	F	21%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- Write your answer in the space provided on your answer sheet. [A : 00 B : 111 C : 1101 D : 10 E : 1100 F : 01] (6)
- What message is encoded by the following bit string? 1011001111100 [DEBE] (2)
- How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? [17.66%] (2)

[10]

TOTAL

[100]

END OF PAPER

- 30) RIP (*Routing Information Protocol*) uses ... as its metric.
- ☐ A: Hop count
☐ B: Link speed
☐ C: Link congestion
☐ D: Actual link cost
☐ E: All of the above
- 31) BGP (*Border Gateway Protocol*) determines routes between
- ☐ A: Networks
☐ B: Domains
☐ C: Institutions
☒ D: Autonomous systems
☐ E: Hosts
- 32) The following FTP command may be used to upload a number of files to an FTP server:
- ☐ A: send
☐ B: upload
☐ C: xfer
☒ D: mput
☐ E: glob
- 33) If one uses a program running on a server from a workstation and X Windows is used to display the results graphically, the X server will execute on the
- ☒ A: Workstation
☐ B: Server
☐ C: Other Unix computer
☐ D: MIT computers
☐ E: More than one of the above

10./...

- 34) OSPF is a(n)
- ☐ A: EGP
☒ B: IGP
☐ C: RIP
☐ D: BGP
☐ E: More than one of the above
- 35) By whom are technical aspects of the Internet standardised?
- ☒ A: IETF
☐ B: ISO
☐ C: EIA
☐ D: ITU
☐ E: CCITT
- 36) A *de facto* standard is
- ☐ A: A standard proposed by the IETF.
☐ B: A standard proposed by ISO.
☐ C: A standard enforced by some authority.
☒ D: Something that has become a 'standard' because everybody uses it.
☐ E: Some standard that is ignored by everyone.
- 37) X.509 is a standard of the
- ☒ A: ITU
☐ B: IETF
☐ C: ISO
☐ D: IEEE
☐ E: EIA

11./...

- 38) Consider the following bit strings:
- 000111001x
 101000101y
 110001101z
- x, y and z are parity bits. Suppose odd parity is used. The values of x, y and z are
- ☐ A: 1, 0 and 0
☐ B: 1, 1 and 1
☐ C: 0, 1 and 0
☐ D: 0, 1 and 1
☒ E: None of the above
- 39) Unicode
- ☐ A: Is a 16-bit code.
☐ B: Makes provision for Cyrillic characters.
☐ C: Makes provision for Arabic script.
☐ D: Is used by some versions of Windows.
☒ E: All of the above
- 40) Use polynomial division to divide 1010 by 1011. The remainder is
- ☒ A: 011
☐ B: 001
☐ C: 111
☐ D: 1010
☐ E: None of the above

[40]

12./...

Question 2

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

- a) 001100011101 [0]
 b) 000011010111 [10]
 c) 111000101100 [12]
 d) 100101001100 [5]
 e) 010100111011 [6]

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

[10]

Question 3

- a) With which well-known ports are the following services usually associated?
- (i) HTTP [80]
 (ii) DNS [53]
 (iii) Telnet [23]
 (iv) SSH [22]
 (v) SMTP [25] (5)
- b) What are the following ports typically used for? (It is not necessary to give the corresponding port numbers.)
- (i) Well-known ports ['Standard' services, such as the Web, FTP, etc]
 (ii) Registered ports [Less common services for which a port has been allocated]
 (iii) Dynamic and/or Private Ports [Client-side and ad hoc applications] (3)
- c) Why are ports useful in firewalls? [Block access to specific services] (1)
- d) How many bits are used to identify a port in TCP/IP? [16] (1)

[10]

13./...

- 15) Which of the following is (are) *not* a function of ISO/OSI's layer 2?
- A: Contention control
 - B: Data delineation
 - ☒ C: Dialogue control
 - D: Error control
 - E: More than one of the above
- 16) When the only link between two nodes goes down in a network that uses Bellman-Ford routing, the nodes will
- ☒ A: Start to slowly count to infinity.
 - B: Immediately realise that the link is broken and adapt the tables to use a new route.
 - C: Switch to Dijkstra's algorithm.
 - D: Elect a new monitor.
- 17) A network uses the Bellman-Ford algorithm. The entry in node B's routing table for destination D is currently (A, 20). The cost associated with the link between B and C is 5. C's entry for destination D is currently (E, 5). After C has sent routing information to B, the entry at B for destination D changes to
- A: (A, 20)
 - B: (E, 5)
 - C: (E, 10)
 - D: (C, 5)
 - E: (B, 10)
 - ☒ F: None of the above

6/...

- 18) How many class B IP network addresses can (potentially) exist?
- ☒ A: $\pm 16\,000$
 - B: $\pm 64\,000$
 - C: $\pm 1\,000\,000$
 - D: $\pm 4\,000$
 - E: $\pm 8\,000$
- 19) IP fragmentation occurs because of
- ☒ A: Different MTUs on the Internet.
 - B: Broken links.
 - C: A shortage of IP addresses.
 - D: Lost packets.
- 20) ICMP is used to
- A: Report errors that occur on the IP layer.
 - B: Test network functions.
 - C: Modify routes.
 - D: More than one of the above
 - ☒ E: All of the above
- 21) ping works by sending the following ICMP packet:
- A: Ping request
 - ☒ B: Echo request
 - C: Information request
 - D: Source quench
 - E: All of the above

7/...

- 22) The path MTU can be determined by using
- A: ICMP's *Echo request* and *echo reply*
 - ☒ B: IP's *Don't fragment* and ICMP's *Destination unreachable: Fragmentation required but DF bit is set*
 - C: ICMP's *Get path MTU*
 - D: TCP's three-way handshake
 - E: HTTP's GET /MTU
- 23) Which ISO OSI layer provides process-to-process delivery?
- A: 2
 - B: 3
 - ☒ C: 4
 - D: 5
 - E: 6
 - F: 7
- 24) The three messages sent by TCP when a connection is established, are
- ☒ A: SYN, SYN+ACK, ACK
 - B: HELO, HELO, HELO
 - C: HELO, HELO, ACK
 - D: HELO, HELO+OK, OK
- 25) Flow control in TCP uses
- A: Stop-and-wait
 - B: The unrestricted approach
 - ☒ C: Window advertisements
 - D: RTS and CTS

8/...

- 26) The first line of an HTML 4 file should be
- A: <html>
 - B: <HTML>
 - ☒ C: <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
 - D: More than one of the above
- 27) What follows the line containing Host in an HTTP request?
- A: Content-type
 - B: GET / HTTP/1.0
 - ☒ C: An empty line
 - D: More than one of the above
- 28) Consider the problem of *transparency* in SMTP. Which of the following is true?
- A: Transparency is not an issue — any type of data can be e-mailed.
 - B: In most e-mail programs users should avoid typing a full stop on a line of its own, because that will prematurely terminate the body of the message and cause only the part prior to the full stop to be sent.
 - C: POP3 solves the SMTP transparency problem.
 - D: IMAP solves the SMTP transparency problem.
 - ☒ E: None of the above
- 29) Which statement about a virtual terminal protocol is *false*?
- A: A virtual terminal protocol allows a user to use a computer via a network in a manner similar to using a terminal locally at that computer.
 - B: Telnet is an example of a virtual terminal protocol.
 - C: SSH is an example of a virtual terminal protocol.
 - ☒ D: Virtual terminal protocols operate at ISO layer 6.
 - E: One can, in principle, connect to the local host with a virtual terminal protocol.

9/...

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 2 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Dr E Marais (RAU)

THIS PAPER CONSISTS OF 16 PAGES.

ANSWER ALL QUESTIONS.

Question 1

- 1) Current dial-up modems achieve speeds up to 56 kbps.
 - A: According to Moore's law we should see an increase in modem speeds every 18 months.
 - B: According to Nyquist's theorem speeds are unlikely to increase beyond this point.
 - ☒ C: According to Shannon's work speeds are unlikely to increase beyond this point.
 - D: According to Turing's results the maximum communication speed only depends on the number of states.
 - E: None of the above
- 2) *Signal constellation* refers to
 - ☒ A: A representation of points where valid data can occur when modulated by a modem.
 - B: The regeneration of a distorted signal.
 - C: The manner data is represented in digital encoding techniques, such as Manchester encoding.
 - D: Transmission of data by LEO satellites.
 - E: Transmission of data by geosynchronous satellites.

2./...

2

COURSE: Computer Science
MEMORANDUM: COS332

- 3) Consider ADSL. The 'A' signifies that it is asymmetric; in other words
 - ☒ A: The upload and download speeds are different
 - B: The electrical potential of one line is varied while the other is grounded.
 - C: It establishes a connection between a user and an ISP.
 - D: It differs from one country to another.
 - E: It is not fully digital.
- 4) Addressing in USB
 - A: Is based on IP addresses.
 - B: Is similar to Ethernet addressing.
 - ☒ C: Uses a 7-bit address
 - D: Is not necessary because USB does not perform routing.
 - E: Depends on the manufacturer of the USB device.
- 5) Free Space Optics (FSO) is a technology that
 - A: Uses a very flexible optical fibre.
 - B: Multiplexes many signals on a single optical fibre.
 - C: Uses unallocated bandwidth on an optical fibre for out of band signalling.
 - ☒ D: Does not use optical fibre.
 - E: More than one of the above
- 6) A UTP cable used for Ethernet, uses the following connector:
 - A: DB25
 - B: Telkom
 - ☒ C: RJ-45
 - D: X-45

3./...

3

COURSE: Computer Science
MEMORANDUM: COS332

- 7) Which of the following claims about USB is **false**?
 - A: It can communicate at 12 Mbps
 - B: It is shielded
 - ☒ C: It is primarily used to connect two computers
 - D: Data is transmitted serially
- 8) An Ethernet card that is in promiscuous mode will
 - A: Occasionally send obscene messages.
 - B: Continue to transmit even after a collision has occurred.
 - ☒ C: Accept messages not addressed to it.
 - D: Will not forward the token to another node once it has received it.
 - E: Will flood the network with traffic.
- 9) Which layer operates in terms of *frames*?
 - A: Physical
 - ☒ B: Data link
 - C: Transport
 - D: Network
 - E: Application
- 10) In the binary exponential backoff algorithm a station waits ... time slots to retransmit after the n th collision.
 - A: 2^n
 - B: $2^n - 1$
 - ☒ C: Anywhere from 0 to $2^n - 1$ (for sufficiently small values of n)
 - D: $1/2^n$
 - E: 16

4./...

4

COURSE: Computer Science
MEMORANDUM: COS332

- 11) The theoretical upper time bound in a CSMA/CD protocol to send frame n is ... slots.
 - A: 2^n
 - B: $\log_2 n$
 - C: $\log_n 2$
 - D: π^n
 - ☒ E: None of the above
- 12) When a station using HDLC receives an RNR frame it
 - A: Retransmits the last frame.
 - B: Retransmits the indicated frame.
 - C: Retransmits all the frames from the indicated frame onwards.
 - ☒ D: Waits.
 - E: Sends the next information frame.
- 13) In switched full-duplex Ethernet networks
 - A: Broadcast messages cannot be sent.
 - ☒ B: Collisions are virtually eliminated.
 - C: A single hub may be used.
 - D: Switches cannot be connected to one another.
 - E: Frames are, in fact, sent in half-duplex mode.
- 14) BSC accomplishes transparency by using
 - A: Bit stuffing
 - ☒ B: Byte stuffing
 - C: Known field lengths
 - D: All of the above
 - E: None of the above

5./...

University of Pretoria

SEMESTER TEST: 29 APRIL 2005

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 1 hour
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) A UTP cable used for Ethernet, uses the following connector:

A: DB25

B: Telkom

☒ **C:** RJ-45

D: X-45

b) Which of the following claims about USB is **false**?

A: It can communicate at 12 Mbps

B: It is shielded

☒ **C:** It is primarily used to connect two computers

D: Data is transmitted serially

2./...

COURSE: Computer Science
MEMORANDUM: COS332

c) Manchester encoding is better than NRZ because

- ☒ **A:** Manchester is self-synchronising.
- B:** The baud rate of Manchester is higher than that of NRZ.
- C:** The baud rate of Manchester is lower than that of NRZ.
- D:** The wires may be switched in the case of Manchester.
- E:** None of the above

d) If a communications channel uses QAM at a baud rate of 9600, data is transmitted at

- A:** 1200 bps.
- B:** 2400 bps.
- C:** 9600 bps.
- D:** 19200 bps.
- ☒ **E:** 28800 bps.

e) The Hayes command ATDT0,1234

- A:** Contains invalid characters.
- B:** Uses pulse dialling.
- ☒ **C:** Will wait a moment after dialling 0.
- D:** Is used to establish an Ethernet connection.

f) XML fits into layer ... of the ISO/OSI model.

- A:** 4
- B:** 5
- ☒ **C:** 6
- D:** 7

COURSE: Computer Science
MEMORANDUM: COS332

- g) At which ISO/OSI layer are communicating applications synchronised?
- A: 3
 - B: 4
 - ☒ C: 5
 - D: 6
- h) Which of the following is (are) *not* a function of ISO/OSI's layer 2?
- A: Contention control
 - B: Data delineation
 - ☒ C: Dialogue control
 - D: Error control
 - E: More than one of the above
- i) By whom are technical aspects of the Internet standardised?
- ☒ A: IETF
 - B: ISO
 - C: EIA
 - D: ITU
 - E: CCITT
- j) If the ASCII for 'A' is 41_{16} and for 'J' it is $4A_{16}$, which character is represented by 47_{16} ?
- A: 'F'
 - ☒ B: 'G'
 - C: 'H'
 - D: 'I'
 - E: 'J'

COURSE: Computer Science
MEMORANDUM: COS332

- k) The ASCII for '0' is 30_{16} . To convert a character k to a digit, the following formula(e) may therefore be used:

A: $k \cdot 00001111_2$

B: $k \cdot F0_{16}$

C: $k \oplus 00001111_2$

D: $k - 48_{10}$

E: More than one of the above

Note that $\oplus$ is used to denote the logical OR and $\cdot$ the logical AND.

- l) The following is a run-length encoded (RLE) bit string: 0,3,5,1. Which of the following may be the original string?

A: 1110000001

B: 01110000001

C: 100011110

D: 11110000

E: None of the above

- m) Which IEEE standard specifies the operation of wireless local area networks?

A: IEEE 802.3

B: IEEE 802.5

C: IEEE 802.10

D: IEEE 802.11

- n) Layer 6 of ISO/OSI is the ... layer.

A: Application

B: Datalink

C: Session

D: Presentation

E: Physical

COURSE: Computer Science
MEMORANDUM: COS332

o) Which of the following URLs is/are valid?

A: `http://www.cs.up.ac.za/`

B: `ftp://ftp.sun.ac.za`

C: `mailto://webmaster@up.ac.za`

D: More than one of the above

E: All of the above

p) A Hamming code capable of correcting single-bit errors and using n parity bits can, at most, contain ... data bits.

A: $2^n - n - 1$

B: $\frac{n!}{n}$

C: 10

D: $\frac{n!}{(n-2)!}$

E: $2 \times n$

q) PGP may be used for

A: Public key cryptography

B: Digital signatures

C: Symmetric encryption

D: All of the above

E: None of the above

r) If parity is used for error control, only errors that affect ... will be detected.

A: A single bit

B: An odd number of bits

C: An even number of bits

D: All of the above

E: None of the above

COURSE: Computer Science
MEMORANDUM: COS332

s) IEEE 802.3 defines

A: Ethernet

B: LLC

C: HDLC

D: Token ring

E: More than one of the above

t) When one node polls another, it wants to know whether

A: The other node has something to send.

B: The other node is communicating.

C: It may send something to the other node.

D: None of the above

[20]

Question 2

Assume that the following six letters occur with the given frequencies:

A 27%

B 3%

C 25%

D 32%

E 2%

F 11%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

a) Write your answer in the space provided on your answer sheet. **[A : 10, B : 0001, C : 01, D : 11, E : 0000, F : 001]** (6)

b) What message is encoded by the following bit string? **00000000011011 [EECAD]** (2)

[8]

7./...

COURSE: Computer Science
MEMORANDUM: COS332

Question 3

What is the *purpose* of each of the following?

- a) ISO/OSI's layer 3 [Routing]
- b) ISDN [Digital infrastructure for various services]
- c) DSL [High-speed digital subscriber connection]
- d) HTTP's HEAD [To retrieve message without body]
- e) Modem [Modulation of digital signals for transmission over analogue medium]
- f) Multiplexer [Sharing of communications line amongst devices]
- g) CRC [Integrity checking]

[7]

Question 4

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

- a) 101111110100 [0]
- b) 011111110111 [5]
- c) 010010001000 [7]
- d) 000011110111 [12]
- e) 010010010100 [5]

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0. **[10]**

8./...

COURSE: Computer Science
MEMORANDUM: COS332

Question 5

The dialogue below occurred between the e-mail program of a user, Jan Jacobs, and a POP3 server. Five lines have been omitted from the dialogue. These five lines have been marked [1], [2], [3], [4] and [5]. Complete the dialogue by writing them in the space provided on your answer sheet. [(1) user jan, (2) pass qq, (3) list, (4) dele 2, (5) retr 1/top 1 1] [5]

```
+OK POP3 nafplion v2000.69mdk server ready
[1]
+OK User name accepted, password please
[2]
+OK Mailbox open, 2 messages
[3]
+OK Mailbox scan listing follows
1 459
2 392
.
[4]
+OK Message deleted
[5]
+OK 459 octets
Return-Path: <santa@northpole.com>
Delivered-To: jan@up.ac.za
Received: from client (nafplion [10.1.1.1])
        by nafplion.up.ac.za (Postfix) with SMTP id 2B54D11AA98
        for <jan@nafplion.up.ac.za>; Thu,  4 Oct 2005 17:18:32 -0400 (EDT)
From: Santa <santa@northpole.com>
To: Jan Jacobs <jan@nafplion.up.ac.za>
Subject: 2005
Message-Id: <20051004211832.2B54D11AA98@nafplion.up.ac.za>
Date: Thu,  4 Oct 2005 17:18:32 -0400 (EDT)
Status:  0

Hoi! Hoi! Hoi!
.
quit
+OK Sayonara
```

TOTAL

[50]

END OF PAPER

University of Pretoria

JUNE EXAMINATION 2010

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 2 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Dr M Coetzee (UJ)

THIS PAPER CONSISTS OF 15 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED.
AN ANSWER SHEET IS PROVIDED SEPARATELY.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) SAT3 is a

- A: GEO satellite
- B: LEO satellite
- C: Performance measure
- ☒ D: Undersea cable
- E: Time zone in South Africa

2) INMARTSAT is a set of

- A: GEO satellites
- B: LEO satellites
- C: Performance measures
- ☒ D: Undersea cables
- E: Time zones in South Africa

2./...

COURSE: Computer Science
MEMORANDUM: COS332

- 3) UMTS has been designed to be a logical upgrade path for
- A: UTP
 - ☒ B: GSM
 - C: Neotel
 - D: Optical fibre
 - E: POTS
- 4) When data arrives in bursts with relatively long periods of silence between the bursts, the line exhibits a high degree of
- A: Latency
 - ☒ B: Jitter
 - C: Bandwidth
 - D: Propagation delay
 - E: Contention
- 5) Landing points of SEACOM include:
- A: Windhoek, Johannesburg and Pretoria
 - B: Cape Town, Walvis Bay, Luanda
 - C: Cape Town, Mtunzini, Luanda
 - ☒ D: Mtunzini, Luanda, Maputo
 - E: Miami, Cape Town, Durban
- 6) Three stations are waiting to transmit data on a medium where .2-persistent CSMA/CD is used. What is the probability of a successful transmission the moment when the medium becomes available?
- A: $3 \times 8 \times 0.2$
 - ☒ B: $3 \times (0.2 \times 0.8 \times 0.8)$
 - C: $8 \times (0.3 \times 0.8 \times 0.8)$
 - D: $0.2 + 0.2 + 0.2$
 - E: $0.2 + 0.8 + 0.8$

COURSE: Computer Science
MEMORANDUM: COS332

- 7) The end of the payload of an Ethernet (802.3) frame is indicated by a
- ☐ **A:** Size field in the header
 - B:** CRC
 - C:** Flag
 - D:** EOT character
 - E:** Preamble
- 8) When a token is lost in an IEEE 802.5 network, the network uses the following algorithm to regenerate the token:
- A:** Bully
 - B:** Largest
 - C:** Friend
 - D:** Sweet
 - ☐ **E:** None of the above
- 9) Suppose a node using SDLC has to transmit the data 01111110. The data will be observed on the line as
- ☐ **A:** 011111010
 - B:** 01111110
 - C:** 011111100
 - D:** 011111110
 - E:** 011111X10
- 10) CSMA/CA is similar to
- A:** 0-persistent CSMA.
 - ☐ **B:** Non-persistent CSMA.
 - C:** Aloha.
 - D:** MACA.
 - E:** MACAW.

COURSE: Computer Science
MEMORANDUM: COS332

11) IPv6

- A:** Is already in use on a number of networks.
- B:** Will be deployed globally on the Internet in 2010.
- C:** Was never intended to be actually deployed, because it was proposed in a 1 April RFC.
- D:** Will only be deployed after IPv5 has been widely deployed.
- E:** Will make the DNS obsolete.

12) Let S be the source address of an IP datagram. Let D be the destination address of an IP datagram. Let N be the netmask used on a router. Let $\&$ represent the logical and operation. If $S\&N = D\&N$ then the router will

- A:** Use source routing.
- B:** Forward the datagram to the next hop as indicated in the router's routing table.
- C:** Directly deliver the packet.
- D:** Move the packet from layer 3 to layer 4.
- E:** Discard the packet.

13) On which layer of ISO OSI does ICMP fit?

- A:** 3
- B:** 4
- C:** 5
- D:** 6
- E:** 7

COURSE: Computer Science
MEMORANDUM: COS332

- 14) Suppose an entry in node A's routing table indicates that node B is the next node on the path to Z. Suppose an entry in node B's routing table indicates that the next hop on the path to Z is A. The resulting problem is addressed by using the following IP header field:
- ☒ **A:** TTL
 - B:** Source address
 - C:** Destination address
 - D:** Network unreachable
 - E:** Been-there flag
- 15) IP packets may be fragmented because of
- A:** Limited size of TCP packets
 - ☒ **B:** Maximum PDU size
 - C:** Congestion
 - D:** Number of hops between source and destination
 - E:** Limited bandwidth
- 16) TCP logically transports
- A:** Fixed size packets
 - B:** Structured data
 - C:** Single characters
 - ☒ **D:** A stream of bytes
 - E:** A stream of bits
- 17) Which transport layer protocol(s) is/are used by DNS?
- A:** TCP
 - B:** UDP
 - ☒ **C:** TCP and UDP
 - D:** TCP/IP
 - E:** IP

COURSE: Computer Science
MEMORANDUM: COS332

- 18) Which field does *not* occur in the UDP header?
- A: Source port
 - B: Destination port
 - ☒ C: Sequence number
 - D: Length
 - E: Checksum
- 19) TCP packets are known as
- A: Cells
 - B: Frames
 - ☒ C: Segments
 - D: Assemblies
 - E: Pipelines
- 20) During a slow start the amount of data sent will initially typically
- A: Remain constant.
 - B: Grow logarithmically.
 - C: Grow linearly.
 - ☒ D: Grow exponentially.
 - E: Decrease linearly.
- 21) If the ASCII for 'A' is 41_{16} and for 'J' it is $4A_{16}$, which character is represented by 47_{16} ?
- A: 'F'
 - ☒ B: 'G'
 - C: 'H'
 - D: 'I'
 - E: 'J'

COURSE: Computer Science
MEMORANDUM: COS332

- 22) A half-open attack occurs when
- A:** Only the first step of a three-way handshake is performed.
 - ☒ **B:** Only the first two steps of a three-way handshake are performed.
 - C:** All three steps of a three-way handshake are performed.
 - D:** A four-way handshake is performed.
 - E:** No handshake is performed.
- 23) Which field does *not* occur in the TCP header?
- A:** Source port
 - B:** Window (size)
 - C:** FIN
 - ☒ **D:** TTL
 - E:** SYN
- 24) Which field does *not* occur in a UDP header?
- A:** Source port
 - B:** Length
 - ☒ **C:** Sequence number
 - D:** Checksum
- 25) If a TCP segment is lost, TCP reacts by
- A:** Retransmitting the segment.
 - B:** Decreasing the rate at which segments are sent.
 - C:** Performing another three-way handshake.
 - ☒ **D:** More than one of the above
 - E:** All of the above

COURSE: Computer Science
MEMORANDUM: COS332

- 26) Flow control in TCP uses
- A:** Stop-and-wait
 - B:** The unrestricted approach
 - ☒ **C:** Window advertisements
 - D:** RTS and CTS
- 27) The three messages sent by TCP when a connection is established, are
- ☒ **A:** SYN, SYN+ACK, ACK
 - B:** HELO, HELO, HELO
 - C:** HELO, HELO, ACK
 - D:** HELO, HELO+OK, OK
- 28) The routes used by EGPs (external gateway protocols) are typically determined by
- A:** Costs
 - ☒ **B:** Policies
 - C:** Hop-count
 - D:** IANA
 - E:** Network Information Centres (NICs)
- 29) Which protocol is often used for single-signon?
- A:** X.500
 - B:** CMIP
 - ☒ **C:** LDAP
 - D:** CMOT
 - E:** X.509

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MEMORANDUM: COS332

- 30) Which of the following is *not* a valid DNS record type?
- A: PTR
 - B: MX
 - C: AAAA
 - ☒ D: AA
 - E: A
- 31) A zone file with serial number 2001102002, according to convention, was probably created on
- A: 1 October 2002
 - ☒ B: 20 October 2001
 - C: 20 January 2010
 - D: 20 October 2002
 - E: 20 January 1910
- 32) `bind` is a program that provides ... services.
- A: directory
 - B: address
 - C: routing
 - ☒ D: name
 - E: None of the above
- 33) The `nslookup` command that changes the server used for name resolution is
- ☒ A: `server`
 - B: `host`
 - C: `root`
 - D: `resolve`
 - E: `use`

COURSE: Computer Science
MEMORANDUM: COS332

- 34) Which of the following is *not* a valid second-level domain in the `za` ccTLD?
- A: `gov`
 - B: `edu`
 - ☒ C: `com`
 - D: `law`
 - E: `nom`
- 35) The `za` ccTLD is controlled by
- A: AfriNIC
 - ☒ B: ZADNA
 - C: `.za`
 - D: Uniforum
 - E: `.co.za`
- 36) What is the name of the file that contains IP addresses and corresponding names that may be used for name resolution on almost every computer connected to the Internet?
- A: `names.txt`
 - B: `resolv.conf`
 - C: `names`
 - D: `resolv`
 - ☒ E: `hosts`
- 37) A master-slave protocol may solve the following problem on layer 2:
- ☒ A: Media access control
 - B: Error control
 - C: Routing
 - D: Polling
 - E: None of the above

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MEMORANDUM: COS332

- 38) The layer 2 data delineation problem is about
- A:** Limiting the size of layer 2 frames.
 - B:** Marking the boundaries of a frame.
 - C:** Picking a data code for data representation.
 - D:** Controlling access to a shared medium.
 - E:** Encoding data in a form that can be transmitted via layer 1.
- 39) IP datagrams support source routing. Source routing is
- A:** Where the packet is routed back to the source.
 - B:** Where all the packets from the same source are all sent together.
 - C:** Where the source determines the route of the packet.
 - D:** Frequently used on the Internet.
 - E:** More than one of the above
- 40) If SMTP cannot deliver a message, it may
- A:** Wait and try again later.
 - B:** Send the message to a secondary server.
 - C:** Return an error message to the sender.
 - D:** More than one of the above
 - E:** All of the above

[40]

12./...

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Question 2

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

- a) 001101010101 [3]
- b) 011110000110 [1]
- c) 000000010110 [4]
- d) 100010111010 [1]
- e) 001101101011 [5]

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

[10]

Question 3

Many protocols use a form of flow control where packets are acknowledged. If they are not acknowledged within a given time, they are resent. Such protocols also sometimes ensure that the recipient is not overwhelmed by a flood of messages that cannot be processed immediately. These protocols include header fields that are used to effect such flow control. Name (a) three header fields in TCP and (b) two header fields in SDLC that are used for this purpose. Also say what the contents of each of these header fields represents.

[10]

[a) Sequence number - number of first byte in segment
Acknowledgement number - number of next expected byte
Window size - available buffer size for receiving information
b) N(S) - number of this frame
N(R) - number of next expected frame]

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Question 4

You work at a NIC. The NIC has control over IP addresses 200.0.0.1 and 201.255.255.254, as well as all addresses between these two. The first address has been granted to a company known as Acme, along with the other addresses in its (class-based) block. Answer the questions below using dotted decimal notation.

- a) In which IP class does the first address belong? [C] (1)
- b) What network mask will Acme use (not using subnetting)? [255.255.255.0] (1)
- c) Give the address of Acme's network. [200.0.0.0] (1)
- d) What is the broadcast address for all hosts on Acme's network? [200.0.0.255] (1)
- e) Acme has grown to the extent that the NIC decides to grant them all addresses in the range given at the top of this question using supernetting (including the obvious network and broadcast addresses implied by this). What is Acme's network address now? Use CIDR notation to answer the question. [200.0.0.0/7] (2)
- f) What netmask will Acme now use? [254.0.0.0] (1)
- g) What broadcast address will Acme now use? [201.255.255.255] (1)
- h) Suppose Acme wants to split this new address into at least 25 subnets, where each subnet can handle at least a million computers. What will the network address be of the subnet on which 200.200.200.200 belongs? Use CIDR notation for your answer. [200.192.0.0/12] (2)

[10]

Question 5

CSMA, CSMA/CD and TDMA are three techniques to regulate media access control. Name any five (5) other techniques, and, for each one listed, name one real-world protocol that uses that MAC approach. [10]

[Token passing/token ring: IEEE 802.5 Token passing/token bus: IEEE 802.4
 MACAW: IEEE 802.11 (Pure) MA: Aloha TDMA: GSM Slotted MA: Slotted Aloha
 Master-slave: SDLC/HDLC CDMA: CDMA2000 FDMA: ? Non-persistent CSMA/?
 p-persistent CSMA/?]

14./...

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Question 6

Pick the correct protocol from the list below and write the corresponding letter next to each description.

- 1) [L] To forward email from the sender towards the recipient.
- 2) [A] A network management protocol developed for TCP/IP.
- 3) [H] A secure virtual terminal protocol.
- 4) [E] A protocol where the roles of client and server are sometimes interchanged.
- 5) [S] A windowing protocol popular in Unix and its derived systems.
- 6) [D] A protocol that performs zone transfers.
- 7) [P] A secure protocol that manages a user's mailboxes.
- 8) [J] A protocol that includes the POST command for form completion.
- 9) [O] An implementation of Dijkstra's algorithm.
- 10) [Q] A protocol often used to load operating system images via the network.

Use the following protocols:

A	SNMP	K	SCP
B	XML	L	SMTP
C	POP2	M	POP3
D	DNS	N	Telnet
E	FTP	O	OSPF
F	RIP	P	IMAP4
G	SSH	Q	TFTP
H	HDLC	R	NNP
I	BSC	S	X.11
J	HTTP	T	DHCP

[10]

15./...

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Question 7

Assume that the following six letters occur with the given frequencies:

A	7%	B	13%	C	32%
D	22%	E	11%	F	15%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- Write your answer in the space provided on your answer sheet. **[A : 000 B : 100 C : 11 D : 01 E : 001 F : 101]** (6)
- What message is encoded by the following bit string?
101011110011 **[FDCBC]** (2)
- How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? **[18%]** (2)

[10]

TOTAL

[100]

END OF PAPER

University of Pretoria

SEMESTER TEST: 16 APRIL 2010

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 50 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

- a) If SMTP cannot deliver a message, it may
- A:** Wait and try again later.
 - B:** Send the message to a secondary server.
 - C:** Return an error message to the sender.
 - D:** More than one of the above
 - ☐ **E:** All of the above
- b) IP datagrams support source routing. Source routing is
- A:** Where the packet is routed back to the source.
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 - D:** Frequently used on the Internet.
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2./...

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- c) The layer 2 data delineation problem is about
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- d) A master-slave protocol may solve the following problem on layer 2:
- A:** Media access control
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 - C:** Routing
 - D:** Polling
 - E:** None of the above
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- A:** `names.txt`
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- A:** AfriNIC
 - B:** ZADNA
 - C:** `.za`
 - D:** Uniforum
 - E:** `.co.za`

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- g) Which of the following is not a valid second-level domain in the `za` ccTLD?
- A:** gov
 - B:** edu
 - ☒ **C:** com
 - D:** law
 - E:** nom
- h) The `nslookup` command that changes the server used for name resolution is
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 - B:** host
 - C:** root
 - D:** resolve
 - E:** use
- i) `bind` is a program that provides ... services.
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 - ☒ **D:** name
 - E:** None of the above
- j) A zone file with serial number 2001102002, according to convention, was probably created on
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 - ☒ **B:** 20 October 2001
 - C:** 20 January 2010
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 - E:** 20 January 1910

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MEMORANDUM: COS332

- k) Which of the following is not a valid DNS record type?
- A:** PTR
 - B:** MX
 - C:** AAAA
 - D:** AA
 - E:** A
- l) Which protocol is often used for single-signon?
- A:** X.500
 - B:** CMIP
 - C:** LDAP
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- A:** Costs
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 - B:** HELO, HELO, HELO
 - C:** HELO, HELO, ACK
 - D:** HELO, HELO+OK, OK

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- o) Flow control in TCP uses
- A:** Stop-and-wait
 - B:** The unrestricted approach
 - ☒ **C:** Window advertisements
 - D:** RTS and CTS
- p) If a TCP segment is lost, TCP reacts by
- A:** Retransmitting the segment.
 - B:** Decreasing the rate at which segments are sent.
 - C:** Performing another three-way handshake.
 - ☒ **D:** More than one of the above
 - E:** All of the above
- q) Which field does *not* occur in a UDP header?
- A:** Source port
 - B:** Length
 - ☒ **C:** Sequence number
 - D:** Checksum
- r) Which field does not occur in the TCP header?
- A:** Source port
 - B:** Window (size)
 - C:** FIN
 - ☒ **D:** TTL
 - E:** SYN

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s) A half-open attack occurs when

A: Only the first step of a three-way handshake is performed.

B: Only the first two steps of a three-way handshake are performed.

C: All three steps of a three-way handshake are performed.

D: A four-way handshake is performed.

E: No handshake is performed.

t) If the ASCII for 'A' is 41_{16} and for 'J' it is $4A_{16}$, which character is represented by 47_{16} ?

A: 'F'

B: 'G'

C: 'H'

D: 'I'

E: 'J'

[20]

Question 2

The IP addresses used by a department in an organisation range from 192.168.132.153 to 192.168.132.158.

- What netmask does the department use? [255.255.255.248]
- What is the broadcast address of the department? [192.168.132.159]
- What is the network address of the department? [192.168.132.152]
- What is the network address of the department in CIDR notation? [192.168.132.152/29]

The next two questions assume that routing to the organisation occurs in a classful manner.

- In which class does the organisation's IP address belong? [C]
- What is the organisation's network address? [192.168.132.0]

7./...

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Question 2 (Continued)

Now assume that the organisation's ISP has assigned the address referred to in the previous question, as well as the subsequent three network addresses of that class to the organisation. The ISP treats it as a supernet.

- g) What is the organisation's network address according to the ISP? Use CIDR notation. (2)
[192.168.132.0/22]
- h) What is the organisation's broadcast address on this supernet? [192.168.135.255]
- i) What is the netmask that the ISP associates with this supernet (assuming it uses a net-mask)? [255.255.252.0] [10]

Question 3

Give the names of the following concepts as they are used in TCP/IP. Where applicable, give the name of the concept as it is used in the TCP or IP header. If the concept is used in both IPv4 and IPv6, use the IPv4 name (if the two protocols use different names). The use of abbreviations to answer the questions is acceptable.

- a) The number that identifies a running process (for message delivery). [Port]
- b) The number of hops before a datagram will be discarded. [TTL]
- c) Dividing large IP packets into smaller packets. [Fragmentation]
- d) The number used to identify a TCP segment, so that it can be acknowledged. [Sequence number]
- e) The TCP flag that informs the other party that the connection is to be terminated. [FIN]
- f) The routing metric used by RIP. [Hop count]
- g) The routing protocol that exchanges a so-called link-state database. [OSPF]
- h) The protocol that IP uses to inform the sender of a packet when that packet is discarded. [ICMP]
- i) The protocol that is used to translate IP addresses to MAC addresses. [ARP]
- j) SNMP software that executes on a node that is managed remotely. [Agent]

[10]

8./...

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Question 4

Assume that the costs between nodes of a network are as follows:

A–B: 1 B–G: 8
A–D: 8 C–D: 10
B–D: 3 C–E: 4
B–E: 5 D–E: 6
B–F: 2

Use Dijkstra's algorithm to determine the routes from A.

[10]

[																				
Cost											Prior									
#		Done		Candidates		X		B	C	D	E	F	G		B	C	D	E	F	G
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----																				
1		A		BD		B		1	?	8	?	?	?		A	?	A	?	?	?
2		AB		DEFG		F		1	?	4	6	3	9		A	?	B	B	B	B
3		ABF		DEG		D		1	?	4	6	3	9		A	?	B	B	B	B
4		ABFD		EGC		E		1	14	4	6	3	9		A	D	B	B	B	B
5		ABFDE		GC		G		1	10	4	6	3	9		A	E	B	B	B	B
6		ABFDEG		C		C		1	10	4	6	3	9		A	E	B	B	B	B
]																				

TOTAL

[50]

END OF PAPER

University of Pretoria

JUNE EXAMINATION 2012

COURSE: Computer Science
PAPER: COS332

TIME: 2 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Prof M Coetzee (UJ)

THIS PAPER CONSISTS OF 16 PAGES.

ANSWER ALL QUESTIONS.

NO CALCULATORS PERMITTED. AN ANSWER SHEET IS PROVIDED SEPARATELY. MARKS INTENTIONALLY ADD UP TO 102; MAXIMUM IS 100. WHEN A QUESTION CAN BE ANSWERED BY MEANS OF AN ACRONYM IT IS NOT NECESSARY TO PROVIDE THE PHRASE THAT THE ACRONYM REPRESENTS.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) Which IEEE standard specifies the operation of wireless local area networks?

- A: IEEE 802.3
- B: IEEE 802.5
- C: IEEE 802.10
- D: IEEE 802.11

2./...

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2) Ethernet addresses have to be unique. This is accomplished as follows:

- A: Manufacturers apply for a list of addresses from the IEEE that are then used in network interface cards.
- B: The IEEE assigns a unique manufacturer's code to each manufacturer. The manufacturer then uses this code as the first half of the address and assigns a unique number to each card manufactured by it that is then used as the second half of the address. Thus a unique address is formed.
- C: The addresses are so long that it is statistically improbable that two addresses on a network will be the same.
- D: The network administrator assigns a unique address to each card.

3) FDDI

- A: Uses optical fibre.
- B: Uses a ring topology.
- C: Communicates at 100 Mbps.
- D: All of the above
- E: More than one of the above

4) Which of the following is *not* a function of an active monitor in a token ring?

- A: Transmitting Active Monitor Present frames.
- B: Regenerating lost tokens.
- C: Eliminating jitter.
- D: Initiating periodic elections.
- E: Two or more of the functions above are not functions of an active monitor.

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5) A master-slave protocol may solve the following problem on layer 2:

- A: Media access control
- B: Error control
- C: Routing
- D: Polling
- E: None of the above

6) Which term does *not* fit with the rest?

- A: Datagram
- B: Cell
- C: Segment
- D: Header
- E: Packet

7) The first line of an HTML 4 file should be

- A: <html>
- B: <HTML>
- C: <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"
"http://www.w3.org/TR/html4/strict.dtd">
- D: More than one of the above

8) Which of the following is an accepted ASN.1 encoding scheme?

- A: ASCII
- B: Unicode
- C: BER
- D: XML
- E: Binary

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- 9) The ASCII value of A is 41_{16} . Which of the following is *not* a quoted-printable representation of "ACME"?
- A: ACME
 - B: =41CME
 - C: A=43=4EE
 - D: =41=43=4D=45
 - E: All of the above are valid representations.
- 10) An XML encoding of ASN.1 data is known as
- A: XASN.1
 - B: ASN.X
 - C: ASN.XML
 - D: <ASN.1></ASN.1>
 - E: XER
- 11) Huffman code is used to
- A: Compress data
 - B: Correct errors in data
 - C: Encrypt data
 - D: More than one of the above
 - E: All of the above
- 12) Suppose the first octet of a UTF-8 character is 193. How long (in octets) will the entire character be?
- A: 1
 - B: 2
 - C: 3
 - D: 4
 - E: 5
 - F: 6

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- 13) Suppose the first octet of a UTF-8 character is 254. How long (in octets) will the entire character be?
- A: 1
 - B: 2
 - C: 3
 - D: 4
 - E: 5
 - F: 6
- 14) RIP (*Routing Information Protocol*)
- A: Uses broadcast messages.
 - B: Uses UDP.
 - C: Is hard to configure.
 - D: More than one of the above
 - E: All of the above
- 15) DNS normally uses the following port:
- A: 22
 - B: 25
 - C: 53
 - D: 80
 - E: 110
- 16) Which of the following messages can be used in SMTP?
- A: HELO
 - B: SENDER
 - C: MESS
 - D: EHLO
 - E: More than one of the above

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- 17) To have text display larger in WML, one can use the following:
- A: <big>
 - B: <bigger>
 - C: <font++>
 - D: <sizeup>
 - E: More than one of the above
- 18) Which statement about a virtual terminal protocol is *false*?
- A: A virtual terminal protocol allows a user to use a computer via a network in a manner similar to using a terminal locally at that computer.
 - B: Telnet is an example of a virtual terminal protocol.
 - C: SSH is an example of a virtual terminal protocol.
 - D: Virtual terminal protocols operate at ISO layer 6.
 - E: One can, in principle, connect to the local host with a virtual terminal protocol.
- 19) RIP (*Routing Information Protocol*) uses ... as its metric.
- A: Hop count
 - B: Link speed
 - C: Link congestion
 - D: Actual link cost
 - E: All of the above
- 20) BGP (*Border Gateway Protocol*) determines routes between
- A: Networks
 - B: Domains
 - C: Institutions
 - D: Autonomous systems
 - E: Hosts

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21) Which of the following is (are) bootstrapping protocols?

- A: BOOTP
- B: DHCP
- C: CMIP
- D: More than one of the above
- E: All of the above

22) To express the TLD com as an FQDN one would write

- A: .com
- B: com.
- C: com.arpa
- D: [com]
- E: *.com

23) Which of the following is *not* a sponsored TLD?

- A: .mil
- B: .pro
- C: .aero
- D: .museum
- E: They are all sponsored TLDs.

24) If one uses a program running on a server from a workstation and X Windows is used to display the results graphically, the X server will execute on the

- A: Workstation
- B: Server
- C: Other Unix computer
- D: MIT computers
- E: More than one of the above

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- 25) OSPF is a(n)
- A: EGP
 - B: IGP
 - C: RIP
 - D: BGP
 - E: More than one of the above
- 26) Consider the IP address 10.10.10.10/16. In which class does it belong?
- A: A
 - B: B
 - C: C
 - D: D
 - E: E
- 27) ICMP is used to
- A: Report errors that occur on the IP layer.
 - B: Test network functions.
 - C: Modify routes.
 - D: More than one of the above
 - E: All of the above
- 28) The MTU (*Maximum Transmission Unit*) is
- A: The size of the largest packet that can be transmitted via the network.
 - B: The highest speed at which packets can be transmitted through the network.
 - C: The size of the largest TCP segment that can be transmitted via the network.
 - D: The cost of the most expensive communications circuit that can be used.
 - E: The highest speed at which bits can be transmitted through the network.

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29) A switch (used in the place of a wiring hub)

- A: Connects different networks and can convert protocols.
- B: Makes a connection only between computers that are busy communicating.
- C: Routes packets between nodes.
- D: Operates on ISO OSI layer 3.
- E: More than one of the above

30) UMTS has been designed to be a logical upgrade path for

- A: UTP
- B: GSM
- C: Neotel
- D: Optical fibre
- E: POTS

31) As a message moves down the protocol stack

- A: Headers are always added
- B: Footers are always added
- C: Payload is always transferred without modification
- D: More than one of the above
- E: None of the above

32) Which ISO OSI layer's functionality can be informally described as being a 'pipe'?

- A: 1
- B: 2
- C: 3
- D: 4
- E: 5
- F: 6
- G: 7

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- 33) A header added to a message at ISO OSI layer n will be removed at the destination at layer ...
- A: $n - 1$
 - B: n
 - C: $n + 1$
 - D: 1
 - E: 7
- 34) TCP packets are known as
- A: Cells
 - B: Frames
 - C: Segments
 - D: Assemblies
 - E: Pipelines
- 35) By whom are technical aspects of the Internet standardised?
- A: IETF
 - B: ISO
 - C: EIA
 - D: ITU
 - E: CCITT
- 36) The most important use(s) of RFCs is (are)
- A: Headers for e-mail
 - B: Remote File Conventions
 - C: Internet standards
 - D: All of the above
 - E: None of the above

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- 37) The ASCII for '0' is 30_{16} . To convert a character k to a digit, the following formula(e) may therefore be used:

A: $k \cdot 00001111_2$

B: $k \cdot F0_{16}$

C: $k \oplus 00001111_2$

D: $k - 48_{10}$

E: More than one of the above

Note that $\oplus$ is used to denote the logical OR and $\cdot$ the logical AND.

- 38) What is the name of the file that contains IP addresses and corresponding names that may be used for name resolution on almost every computer connected to the Internet?

A: names.txt

B: resolv.conf

C: names

D: resolv

E: hosts

- 39) bind is a program that provides ... services.

A: directory

B: address

C: routing

D: name

E: None of the above

- 40) Which of the following is *not* a valid DNS record type?

A: PTR

B: MX

C: AAAA

D: AA

E: A

COURSE: Computer Science
PAPER: COS332

Question 2

Suppose you manage the network 137.215.0.0. You want to divide it into subnets such that you can accommodate at least n subnetworks, and where each subnetwork can accommodate at least c computers. Say, for the various combinations of n and c listed below

- (i) What netmask you will use; (1 each)
 - (ii) What the address (using CIDR notation) of the subnetwork that contains 137.215.215.215 is; and (2 each)
 - (iii) What the broadcast address of the subnetwork containing 137.215.215.215 is. (1 each)
- a) $n = 100$ and $c = 300$
 - b) $n = 10$ and $c = 4000$
 - c) $n = 4000$ and $c = 10$
 - d) $n = 300$ and $c = 100$

[16]

Question 3

- a) Explain what is meant by *transparency* on the data link layer. (1)
- b) List four approaches that may be used to achieve transparency on the data link layer. For each provide an example of a protocol that uses the specific approach. (8)

[9]

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PAPER: COS332

Question 4

Complete the paragraphs that follow by writing the words (or phrases) that have been omitted in the space provided on your answer sheet.

For LANs a token ring protocol has been standardised as IEEE [1]. This standard corresponds to a large extent to the token ring protocol used by [2] in its earlier products.

One potential source of problems in such a network is the loss of the token. If a token is indeed lost it will be regenerated by the [3]. This node continually confirms its presense by sending [4] messages around the ring. If any other station has not seen such a message in a while that node may call for [5], for which the [6] algorithm will be used.

This algorithm consists of three phases. Phase 1 is the initiation of the process. During phase 2 each node either sends its [7] to its neighbour or the [8] of its own and the one received from its other neighbour.

The problem of nodes going down — and potentially breaking the ring — is solved by using a device known as a wiring [9]. Older versions of this device (and these token rings in general) are recognisable by their genderless connectors that are known as [10] connectors.

[10]

14./...

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PAPER: COS332

Question 5

- a) IPv4 packets contain a header field that causes the packet to be discarded when this field becomes 0. What is the name of the field? (1)
- b) IPv6 packets contain a header field that causes the packet to be discarded when this field becomes 0. What is the name of the field? (1)
- c) What is the process of breaking IP packets into smaller packets called? (1)
- d) What is the private class A network address in IPv4? Use CIDR notation for your answer. (1)
- e) What does the IPv4 address 0.0.0.0 indicate? (1)
- f) What does the IPv4 address 127.0.0.1 indicate? (1)
- g) Which command causes an ICMP Echo request to be sent to a target machine? (1)

15./...

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Question 6

- a) Which 'message' forms the second part of the three-way handshake to establish a TCP session? (1)
- b) Which 'message' forms the final part of the three-way handshake to terminate a TCP session? (1)
- c) What is the field in the TCP header called that indicates how much buffer space is available at the destination? (1)
- d) In order to thwart half-open attacks that are intended to consume a computers's available memory, the respondent in a TCP connection notes the fact that phase 1 of the three-way handshake is complete by placing a special value in its response to the request to connect; this value is known as a (1)
- e) When a TCP segment is lost what does TCP assume to be the most likely cause? (1)
- f) When a TCP segment is lost, TCP responds by using the ... algorithm to determine its future rate of transmission. (1)
- g) In contrast to TCP, which is connection-oriented, what word describes UDP? (1)
- h) Which layer 4 protocol is used by TFTP? (1)
- i) Which layer 4 protocol is used by FTP? (1)
- j) Which TCP/IP transport layer protocol is considered unreliable? (1)

[10]

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PAPER: COS332

Question 7

Assume that the following six letters occur with the given frequencies:

A	14%	B	25%	C	12%
D	6%	E	30%	F	13%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- Write your answer in the space provided on your answer sheet. (6)
- What message is encoded by the following bit string?
10011000001 (2)
- How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? (2)

[10]

TOTAL

[100]

END OF PAPER

University of Pretoria

SEMESTER TEST: 11 APRIL 2012

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 50 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) A domain hack occurs when

A: One uses parts of the domain name to spell a word (such as `del.icio.us`).

B: One overrides the DNS by entering information in the `hosts` file.

C: One answers a DNS query before the query has been issued in the hope that an unsuspecting host will cache the incorrect 'answer'.

D: One modifies a zone file without authorisation.

E: More than one of the above

b) Which of the following is (are) bootstrapping protocols?

A: BOOTP

B: DHCP

C: CMIP

D: More than one of the above

E: All of the above

2./...

COURSE: Computer Science
MEMORANDUM: COS332

- c) RIP (*Routing Information Protocol*) is based on the ... algorithm.
- A: Dijkstra
 - ☒ B: Bellman-Ford
 - C: Greedy
 - D: Bully
 - E: Homocide
- d) An application protocol will typically respond with a message in the ...-range when it successfully completed a request.
- A: 100s
 - ☒ B: 200s
 - C: 300s
 - D: 400s
 - E: 500s
- e) An application protocol will typically respond with a message in the ...-range when security or configuration errors prevented it from completing a request.
- A: 100s
 - B: 200s
 - C: 300s
 - D: 400s
 - ☒ E: 500s
- f) Which of the following second-level domains does not exist?
- A: school.za
 - B: edu.za
 - C: ac.za
 - ☒ D: tech.za
 - E: law.za

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g) The software providing services on the internet is often known as a

A: Dæmon

B: Spirit

C: Ghost

D: Waiter

E: Client

h) To express the TLD `com` as an FQDN one would write

A: `.com`

B: `com.`

C: `com.arpa`

D: `[com]`

E: `*.com`

i) In the DNS, SOA refers to

A: Service-oriented Architecture

B: IPv6

C: Start of Authority

D: Start of Address

E: Start of Aliases

j) Which of the following is *not* a sponsored TLD?

A: `.mil`

B: `.pro`

C: `.aero`

D: `.museum`

E: They are all sponsored TLDs.

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MEMORANDUM: COS332

k) The most important routing protocol at JINX is

A: RIP

☒ **B:** BGP

C: OSPF

D: Dijkstra

E: Bellman-Ford

l) Suppose a distributed database is used by a number of application protocols. On which OSI ISO layer should the two-phase commit protocol ideally be implemented?

A: 1

B: 2

C: 3

D: 4

☒ **E:** 5

F: 6

G: 7

m) Which of the following is *not* a valid MIME type?

A: multipart/mixed

B: image/jpeg

☒ **C:** text/text

D: example/cos332

E: multipart/alternative

COURSE: Computer Science
MEMORANDUM: COS332

n) An XML encoding of ASN.1 data is known as

A: XASN.1

B: ASN.X

C: ASN.XML

D: <ASN.1></ASN.1>

☒ **E:** XER

o) If the leading octet in a UTF-8 character is $E0_{16}$, how many octets will follow this octet to complete the character?

A: 0

B: 1

☒ **C:** 2

D: 3

E: 4

p) If the leading octet in a UTF-8 character is $E1_{16}$, how many octets will follow this octet to complete the character?

A: 0

B: 1

☒ **C:** 2

D: 3

E: 4

q) How many octets may be occupied by a UTF-16 character?

A: 2

B: 3

C: 4

D: More than one of the above

E: All of the above

COURSE: Computer Science
MEMORANDUM: COS332

r) EBNF is

- A:** An abbreviation for Extended Backus-Naur Form
- B:** Often used by CFPs to specify the content of data messages
- C:** Standardised by a CFP
- D:** More than one of the above
- E:** All of the above

s) Huffman code is used to

- A:** Compress data
- B:** Correct errors in data
- C:** Encrypt data
- D:** More than one of the above
- E:** All of the above

t) Which of the following compression algorithms may be best for data that contains sequences of characters, where the sequences are often repeated?

- A:** Lempel-Ziv
- B:** RLE
- C:** Bit stuffing
- D:** Vice grip
- E:** Bit bucket

[20]

7./...

For each of the following terms say (i) what phrase it abbreviates and (ii) what it is used for.

- [20]

Assume that the costs between nodes of a network are as follows:

Use Dijkstra's algorithm to determine the routes from A.

[10]

#	Done	Candidates	X	B	C	D	E	F	G	B	C	D	E	F	G
1	A	DE	D	?	?	3	4	?	?	?	?	A	A	?	?
2	AD	EBF	E	7	?	3	4	9	?	D	?	A	A	D	?
3	ADE	BFG	B	7	?	3	4	9 14		D	?	A	A	D	E
4	ADEB	FGC	F	7 17		3	4	8 14		D	B	A	A	B	E
5	ADEBF	GC	G	7 17		3	4	8 10		D	B	A	A	B	F
6	ADEBFG	C	C	7 17		3	4	8 10		D	B	A	A	B	F

TOTAL

[50]

END OF PAPER

University of Pretoria

SEMESTER TEST: 29 APRIL 2013

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 50 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) If the International Standards Organisation (ISO) adopts a standard

- A:** All member countries *have* to comply with the standard, but the rest of the world do not have to.
- B:** All countries in the world *have* to comply with the standard.
- C:** Almost all countries in the world enforce compliance with almost ISO standards, even though they don't have to.
- D:** ISO standards are irrelevant because nobody implements them.
- E:** None of the above

b) In data communications a *protocol* is always ...

- A:** A detailed description of the rules to be followed when communicating.
- B:** Just another name for a standard.
- C:** Defined in an RFC.
- D:** Associated with a specific TCP port.
- E:** More than one of the above

2./...

COURSE: Computer Science
MEMORANDUM: COS332

- c) Ideally, an ISO layer provides services to ... and uses services from
- A:** the layer below it; the layer above it
 - ☒ **B:** the layer above it; the layer below it
 - C:** its peer layer elsewhere in the network; the layer below it
 - D:** the layer below it; its peer layer elsewhere in the network
 - E:** a peer layer somewhere in the network; a peer layer somewhere in the network
- d) The primary reason for including the `host :` header in the HTTP protocol is to
- A:** Ensure that the `hosts` file cannot be misused.
 - B:** Make a clear distinction between client and host requests.
 - ☒ **C:** Enable virtual hosting.
 - D:** Enhance security by ensuring that the destination server is indeed the one the user wanted to visit because the name of the server can now be compared with the name provided in the `host :` header.
 - E:** Specify alternative hosts that may contain the content the requester wants.
- e) The most important routing protocol at JINX is
- A:** RIP
 - ☒ **B:** BGP
 - C:** OSPF
 - D:** Dijkstra
 - E:** Bellman-Ford
- f) Indicating where layer 2 frames start and end is known as
- ☒ **A:** Data delineation
 - B:** Data fragmentation
 - C:** Data separation
 - D:** Packet framing
 - E:** Flow control

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MEMORANDUM: COS332

- g) The file that contains the names, addresses and related information about a domain or subdomain in the DNS is known as a ... file.
- A:** DNS
 - B:** Domain
 - C:** Resolution
 - ☒ **D:** Zone
 - E:** Bind
- h) Root name servers in the DNS perform ... name resolution.
- A:** Iterative
 - B:** Recursive
 - C:** Non-authoritative
 - D:** All of the above
 - ☒ **E:** None of the above
- i) A root name server will resolve the following query:
- A:** Type NS for `com`
 - B:** Type A for `com`
 - ☒ **C:** Type NS for `org.`
 - D:** Type A for `org.`
 - E:** All of the above
- j) A *zone transfer* refers to the process of
- A:** Moving hosting from one service provider to another.
 - B:** Selling a domain name to someone else.
 - ☒ **C:** Copying DNS entries from a primary server to a secondary server.
 - D:** Accessing data outside the corporate network, or data in the demilitarised zone of the corporate network.
 - E:** Obtaining new name servers.

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MEMORANDUM: COS332

k) Which of the following is *not* an SNMP request?

- A:** Get
- B:** GetNext
- C:** GetBulk
- D:** Reboot
- E:** Set

l) Non-repudiation is implemented through

- A:** Public key cryptography
- B:** LDAP
- C:** SNMP
- D:** Routing table exchange
- E:** Single sign-on

m) Suppose an HTTP request contains the following in its header:

`Accept-Charset: Shift_JIS`

Suppose a Web site is visited that contains text in 'plain' ASCII and the server does not support Shift_JIS. What will happen?

- A:** The browser will display an error message to indicate that the content cannot be displayed.
- B:** Google Translate will be called to translate the page to a language that can be represented in Shift_JIS.
- C:** The server will serve ASCII and the browser will interpret it as Shift_JIS, resulting in a garbled display.
- D:** The Accept-Charset header given implies ISO-8859-1, so the pages should be served and displayed without problems.

5./...

COURSE: Computer Science
MEMORANDUM: COS332

n) EXIF data embedded in some images

- A:** Is metadata about the image.
- B:** Enforces digital rights management.
- C:** Is not necessarily authentic.
- D:** More than one of the above
- E:** All of the above

o) A MIME `multipart/alternative` message should consist of components that

- A:** Represent different messages from which the recipient may choose, such as alternative dates to schedule a meeting.
- B:** The same message, encoded in different (alternative) manners.
- C:** Include the “real” message and an alternative “cover” message that eavesdroppers will see — effectively hiding the real message.
- D:** Are used just like `multipart/digest` components, but `multipart/alternative` uses a different separation mechanism between the components.
- E:** Represent different messages intended for multiple parties; each recipient will only receive the part addressed to him/her.

p) The following utility may be used to decode messages that have been captured from a network to their original formats (images, sounds, executables, and so on) so that they may be viewed, played or otherwise used by their usual applications.

- A:** `winpcap`
- B:** `nslookup`
- C:** `mimencode`
- D:** `dd`
- E:** `bash`

COURSE: Computer Science
MEMORANDUM: COS332

- q) Which of the following is *not* a valid ASN.1 constructor?
- A: SEQUENCE
 - ☒ B: MESSAGE
 - C: CHOICE
 - D: SET
- r) An email message that contains a PDF document as an attachment may use the following MIME subtype to encapsulate the mail message and its attachment.
- ☒ A: multipart/mixed
 - B: multipart/digest
 - C: multipart/alternative
 - D: multipart/plain
 - E: text/plain
- s) The ideal layer to implement encryption for most situations is
- A: 7
 - B: 6
 - C: 4
 - D: 3
 - ☒ E: Dependent on the details of the specific case.
- t) `rsync` is a utility that
- A: Was written to simplify use of ‘the cloud’.
 - B: Marks synchronisation points that enable a network to resynchronise if synchronisation is lost.
 - C: Synchronises the clock on the local computer with an external clock.
 - D: Synchronises directories on the local computer only.
 - ☒ E: Synchronises directories — possibly via the network.

[20]

7./...

COURSE: Computer Science
MEMORANDUM: COS332

Question 2

Assume that the following six letters occur with the given frequencies:

A	28%	B	17%	C	15%
D	21%	E	6%	F	13%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- Write your answer in the space provided on your answer sheet. **[A : 10 B : 111 C : 110 D : 01 E : 000 F : 001]** (6)
- What message is encoded by the following bit string?
11011101110111000 **[CBDCBE]** (2)
- How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? **[16.333%]** (2)

[10]

Question 3

(For both questions below plan your answers so that they fit in the space available on the answer sheet.)

- Name any five SMTP commands and very briefly say what the purpose of each is. **[See RFC 5321 Section 4.1.1]** (5)
- Name any five HTTP response header entries and very briefly say what the purpose of each is. **[See RFC 2616 Sections 6 & 7]** (5)

[10]

8./...

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Question 4

Assume that the costs between nodes of a network are as follows:

A–E: 1 D–E: 1
B–C: 8 D–F: 9
C–D: 4 D–G: 6
C–F: 3 E–F: 9
C–G: 1 F–G: 7

Use Dijkstra's algorithm to determine the routes from A.

[10]

[																				
Cost											Prior									
#		Done		Candidates		X		B	C	D	E	F	G		B	C	D	E	F	G
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----																				
1		A		E		E		?	?	?	1	?	?		?	?	?	A	?	?
2		AE		DF		D		?	?	2	1	10	?		?	?	E	A	E	?
3		AED		FCG		C		?	6	2	1	10	8		?	D	E	A	E	D
4		AEDC		FGB		G		14	6	2	1	9	7		C	D	E	A	C	C
5		AEDCG		FB		F		14	6	2	1	9	7		C	D	E	A	C	C
6		AEDCGF		B		B		14	6	2	1	9	7		C	D	E	A	C	C
]																				

TOTAL

[50]

END OF PAPER

University of Pretoria

JUNE EXAMINATION 2014

COURSE: Computer Science
PAPER: COS332

TIME: 2 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Prof M Coetzee (UJ)

THIS PAPER CONSISTS OF 15 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED.

AN ANSWER SHEET IS PROVIDED SEPARATELY.

WHEN A QUESTION CAN BE ANSWERED BY MEANS OF AN ACRONYM IT IS NOT NECESSARY TO PROVIDE THE PHRASE THAT THE ACRONYM REPRESENTS.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) The layer 2 data delineation problem is about

- A: Limiting the size of layer 2 frames.
- ☒ B: Marking the boundaries of a frame.
- C: Picking a data code for data representation.
- D: Controlling access to a shared medium.
- E: Encoding data in a form that can be transmitted via layer 1.

2) A master-slave protocol may solve the following problem on layer 2:

- ☒ A: Media access control
- B: Error control
- C: Routing
- D: Polling
- E: None of the above

2/...

2

COURSE: Computer Science
PAPER: COS332

3) A protocol using a moving window for flow control and with a 6-bit field to number packets, may have at most ... packets outstanding.

- ☒ A: 32
- B: 64
- C: 128
- D: 31
- E: 63
- F: 127

4) When the only link between two nodes goes down in a network that uses Bellman-Ford routing, the nodes will

- A: Start to slowly count to infinity.
- B: Immediately realise that the link is broken and adapt the tables to use a new route.
- C: Switch to Dijkstra's algorithm.
- ☒ D: Elect a new monitor.

5) A network uses the Bellman-Ford algorithm. The entry in node B's routing table for destination D is currently (A, 20). The cost associated with the link between B and C is 5. C's entry for destination D is currently (E, 5). After C has sent routing information to B, the entry at B for destination D changes to

- A: (A, 20)
- B: (E, 5)
- C: (E, 10)
- ☒ D: (C, 5)
- E: (B, 10)
- F: None of the above

3/...

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6) Which of the following is an accepted ASN.1 encoding scheme?

- A: ASCII
- B: Unicode
- ☒ C: BER
- D: XML
- E: Binary

7) XBRL is the Extensible

- A: Binary Record Level
- ☒ B: Business Reporting Language
- C: Bot Request Logic
- D: Beautiful Rendering Language
- E: Border Row List

8) Assume a HTTP request includes the following:

Accept-Charset: iso-8859-1, utf-8;q=0.5, *;q=0.7
Suppose further that the server supports (only) UTF-8 and UTF-16. Which code will be used?

- A: ISO-8559-1
- B: UTF-8
- ☒ C: UTF-16
- D: ASCII
- E: None of the above

9) What follows the line containing Host in an HTTP request?

- A: Content-type
- B: GET / HTTP/1.0
- ☒ C: An empty line
- D: More than one of the above

4/...

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253

10) Suppose the first octet of a UTF-8 character is 254. How long (in octets) will the entire character be?

- A: 1
- B: 2
- C: 3
- D: 4
- E: 5
- ☒ F: 6

11) The HTTP header contains a Host field in order to support

- A: Security.
- ☒ B: Virtual hosting.
- C: HTML.
- D: Authentication.
- E: Access speed.

12) Protocols designed to determine network routes usually operate on ISO OSI layer ...

- ☒ A: 7
- B: 6
- C: 5
- D: 4
- E: 3

13) Which of the following pairs of protocols have similar applications?

- A: SMTP and SNMP
- B: HTTP and TCP
- ☒ C: Telnet and SSH
- D: POP3 and P3P
- E: DNS and XML

5/...

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PAPER: COS332

14) How may information be encoded that is supplied with an HTTP POST request?

- ☒ A: Key-value pairs, with spaces replaced by a + and URL-encoding applied afterwards.
- ☐ B: Plain text
- ☐ C: The native encoding of whatever data is communicated is used unmodified.
- ☐ D: HTML
- ☐ E: No data accompanies the HTTP POST message.

15) In practice, the parameter following an HTTP GET message may be ...

- ☐ A: A filename.
- ☐ B: The name of an executable script.
- ☐ C: Any parameter the server has been built to interpret.
- ☐ D: More than one of the above
- ☒ E: All of the above

16) RIP (*Routing Information Protocol*)

- ☐ A: Uses broadcast messages.
- ☐ B: Uses UDP.
- ☒ C: Is hard to configure.
- ☒ D: More than one of the above
- ☐ E: All of the above

17) If SMTP cannot deliver a message, it may

- ☐ A: Wait and try again later.
- ☐ B: Send the message to a secondary server.
- ☐ C: Return an error message to the sender.
- ☐ D: More than one of the above
- ☒ E: All of the above

6/...

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PAPER: COS332

18) A domain hack occurs when

- ☒ A: One uses parts of the domain name to spell a word (such as del.icio.us).
- ☐ B: One overrides the DNS by entering information in the hosts file.
- ☐ C: One answers a DNS query before the query has been issued in the hope that an unsuspecting host will cache the incorrect 'answer'.
- ☐ D: One modifies a zone file without authorisation.
- ☐ E: More than one of the above

19) A *zone transfer* refers to the process of

- ☐ A: Moving hosting from one service provider to another.
- ☐ B: Selling a domain name to someone else.
- ☒ C: Copying DNS entries from a primary server to a secondary server.
- ☐ D: Accessing data outside the corporate network, or data in the demilitarised zone of the corporate network.
- ☐ E: Obtaining new name servers.

20) The following network protocol(s) is (are) used for network management:

- ☐ A: SNMP
- ☐ B: CMIP
- ☐ C: CMOT
- ☒ D: More than one of the above
- ☐ E: All of the above

21) The IP address 222.222.222.222 belongs to class

- ☐ A: A
- ☒ B: B
- ☒ C: C
- ☐ D: D
- ☐ E: E

7/...

22) Flow control in TCP uses

- A: Stop-and-wait
- B: The unrestricted approach
- ☒ C: Window advertisements
- D: RTS and CTS

23) A typical ARP message looks as follows:

- A: Who has 10.1.1.1? Tell 10.1.1.2.
- B: Who has 00-04-38-68-da-00? Tell 00-11-43-e1-ba-4d.
- C: GET / HTTP / 1.1
- ☒ D: What is the IP of xx.co.za?
- E: What is the name of 196.7.147.37?

24) Suppose a distributed database is used by a number of application protocols. On which OSI ISO layer should the two-phase commit protocol ideally be implemented?

- A: 1
- B: 2
- C: 3
- D: 4
- ☒ E: 5
- F: 6
- G: 7

25) Which transport layer protocol(s) is/are used by DNS?

- A: TCP
- B: UDP
- ☒ C: TCP and UDP
- D: TCP/IP
- E: IP

8/...

26) Which field does *not* occur in the UDP header?

- ☒ A: Source port
- B: Destination port
- ☒ C: Sequence number
- D: Length
- E: Checksum

27) Which of the following URLs is/are valid?

- A: http://www.cs.up.ac.za/
- B: ftp://ftp.sun.ac.za
- C: mailto://webmaster@up.ac.za
- ☒ D: More than one of the above
- E: All of the above

28) Who ultimately decides which TLDs are available on the Internet?

- A: NSF
- B: ISO
- C: Uniform
- D: UP
- ☒ E: ICANN

29) Who of the following people is famous for (amongst others) his work on the early Internet (then the ARPANET)?

- A: Bill Gates
- ☒ B: Vinton Cerf
- C: Steve Jobs
- D: Tim Berners-Lee
- E: Mark Zuckerberg

9/...

30) Which of the following sites was *not* one of the original four connected to the ARPANET?

- A: UCLA
- ☒ B: MIT
- C: SRI
- D: UCSB
- E: University of Utah

31) What role(s) do RFCs fulfil for the Internet?

- A: Define standards
- B: Provide information
- C: Opinion polls
- ☒ D: More than one of the above
- E: All of the above

32) When a new DNS TLD is in *landrush* it means

- A: Only trademark holders are allowed to register domains in that TLD.
- B: Exactly the same as when the TLD is in a general available state.
- C: Normal users are allowed to register trademarked names in that TLD that would not have been possible under normal conditions.
- ☒ D: The domain recently started accepting registrations from the general public — possibly at a higher cost than the eventual annual fees will be.
- E: Its use has not been approved yet.

33) The *whois* command should (eventually) query the ... records.

- A: ICANN
- B: IANA
- ☒ C: Registrar's
- D: TLD registry
- E: *who.is*

10/...

34) When does a DNS zone transfer occur?

- A: When the zone owner moves to a different ISP.
- ☒ B: When a secondary nameserver is synchronised with a primary nameserver.
- C: When an owner sells a TLD to someone else.
- D: When a server is moved outside the DMZ.
- E: During name resolution.

35) A DNS SOA TTL entry indicates the time ... (in seconds).

- ☒ A: A machine is allowed to use a resolved name before the name has to be resolved again.
- B: A secondary nameserver is allowed to use names before synchronising with the primary server.
- C: The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- ☒ D: The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.
- E: The time a client who wants to resolve a name should wait before sending another resolution request.

36) A DNS SOA RETRY entry indicates the time ... (in seconds).

- A: A machine is allowed to use a resolved name before the name has to be resolved again.
- B: A secondary nameserver is allowed to use names before synchronising with the primary server.
- C: The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- ☒ D: The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.
- E: The time a client who wants to resolve a name should wait before sending another resolution request.

11/...

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PAPER: COS332

37) Which of the following is *not* a valid nslookup command?

- A: ls example.com
B: set type=any
C: server 8.8.8.8
D: exit
E: resolve up.ac.za

38) The structure used to store data used by SNMP is known as ...

- A: DSNMP
B: X.509
C: MIB
D: NMP
E: NIC

39) A toy "telephone" consisting of two tin cans connected by a piece of string is able to communicate in ... mode.

- A: Simplex
B: Half-duplex
C: Full duplex
D: Half-simplex
E: Full simplex

40) Which of the following is a useful Android app for determining information about networks?

- A: LanDroid
B: Nexus
C: iCloud
D: DNS
E: IMDB

[40]

12/...

COURSE: Computer Science
PAPER: COS332

Question 2

Consider each of the following host IP addresses.

- | | | |
|----------------------------------|-------------|-----------|
| 1) 161.156.114.180 ($n = 104$) | 161.156.0.0 | 113.0.0.0 |
| 2) 113.176.201.141 ($n = 27$) | | |
| 3) 41.58.129.89 ($n = 221$) | | |

For each address

a) Provide the *network address* of the network on which the host resides;

b) Identify the class of the network;

c) Divide (or combine) the network into (at least) n subnets (or supernets), with each subnet (or supernet) as large as possible for the values of n given above.

(i) Provide the resulting netmask;

(ii) Provide the *network address* of the subnet (or supernet) on which the host now resides; and

(iii) Provide the *broadcast address* of the subnet (or supernet) on which the host now resides.

$111\ 0000$
 $000111\ (5 \times 3 = 15)$

[15]

Question 3

List (name) five (5) features of TCP that make TCP reliable.

[5]

13/...

Question 4

- a) Suppose a user enters a URL — say `www.cs.up.ac.za` — in her favourite browser. This causes a message to be transmitted down the protocol stack of her computer with (almost) every layer in the stack adding a header. For layers 7, 6, 4, and 3 name one field in the header added by that layer *and* provide the correct value that will be used in that field. (Note that headers on the higher layers often provide a name followed by a value (such as *Subject: Examination*); consider the part to the left of the colon as the parameter name and the part to the right as the value. In contrast, lower level protocols often use a positional approach, where we know that a certain sequence of bits will contain, say, the checksum. This name will not be transmitted but the value — such as `1dba3410` — will be.

Note that this example will normally not involve an explicit layer 6, but layer 7 will include information that logically belongs on layer 6. Use such values to answer the question.

Hint: Answer this question using parameter names for which you know what their values will be.

- b) Name a similar header field and value for layer 4 of the response to the request described above.

Question 5

Assume that the costs between nodes of a network are as follows:

A-C: 10 B-G: 6
A-G: 8 C-E: 9
B-C: 1 C-G: 7
B-D: 9 D-E: 6
B-E: 1 D-F: 9

Use Dijkstra's algorithm to determine the routes from A.

11 10 18 12 27 8 CAEBDA

10
[10]
14/...

Question 6

The dialogue below occurred between the email program of a user and a server. Ten fragments have been omitted from the dialogue. These fragments have been marked [1], [2], [3], etc. Complete the dialogue by writing these fragments in the space provided on your answer sheet. [10]

220 nafplion ESMTP Exim 4.80 Fri, 30 May 2014 00:30:40 +0200

[1] sparta HELLO

250 [2]-nafplion Hello sparta.mso [10.1.1.14]

250-SIZE 52428800

250-8BITIME

250-PIPELINING

250 HELP

MAIL FROM [3]: <molivier@nafplion.mso>

250 OK

MAIL TO [4]: <root@nafplion.mso>

250 Accepted

DATA

354 Enter message, ending with [5] by itself

From: "Martin Olivier" <molivier@nafplion.mso>

To: <root@nafplion.mso>

Subject [6]: Examination

Date: Fri, 30 May 2014 00:30:45 +0200

Message-ID: <000001cf7b8d\$a4e28eb0\$eea7ac10\$[7]>

MIME-Version: 1.0

Content-Type: [8]/[9];

. [10] = "-----NextPart_000_0001_01CF7B9E.686BACD0"

X-Mailer: Microsoft Outlook 15.0

Thread-Index: Ac97jYvlgKpUKNsmRRGQwXrN+5o/Jq==

Content-Language: en-za

15/...

Question 7

Assume that the following six letters occur with the given frequencies:

A	11%	B	9%	C	19%
D	20%	E	7%	F	34%

2.3 bits

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- a) Write your answer in the space provided on your answer sheet. (6)
- b) What message is encoded by the following bit string? (2)
1 1 1 0 0 1 0 1 0 1 0 1 1 1 1
- c) How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed-length code? (2)

$$0.57 \div 3 \times 100 = 19\%$$

[10] 10

TOTAL

[100]

END OF PAPER

A = 100
B = 1011
C = 00
D = 01
E = 1010
F = 11

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Question 3

For each of the following terms say (i) what phrase it abbreviates and (ii) what it is used for.

- a) RJ45
- b) SSH
- c) POP3
- d) LDAP
- e) ISP

[10]

Question 4

Assume that the costs between nodes of a network are as follows:

A-C: 10 B-G: 6
A-G: 8 C-E: 9
B-C: 1 C-G: 7
B-D: 9 D-E: 6
B-E: 1 D-F: 9

Use Dijkstra's algorithm to determine the routes from A.

[10]

TOTAL

[50]

END OF PAPER

University of Pretoria

SEMESTER TEST: 9 APRIL 2014

COURSE: Computer Science
PAPER: COS332

TIME: 50 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 8 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

- a) Who of the following people is famous for (amongst others) his work on the early Internet (then the ARPANET)?

A: Bill Gates
B: Vinton Cerf ✓
C: Steve Jobs
D: Tim Berners-Lee
E: Mark Zuckerberg

- b) Which of the following sites was *not* one of the original four connected to the ARPANET?

A: UCLA
B: MIT ✓
C: SRI
D: UCSB
E: University of Utah

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PAPER: COS332

- c) Suppose a distributed database is used by a number of application protocols. On which OSI ISO layer should the two-phase commit protocol ideally be implemented?

A: 1

B: 2

C: 3

D: 4

E: 5 ✓

F: 6

G: 7

- d) Which ISO OSI layer's functionality can be informally described as being a 'pipe'?

A: 1

B: 2

C: 3

D: 4 ✓

E: 5

F: 6

G: 7

- e) What role(s) do RFCs fulfil for the Internet?

A: Define standards

B: Provide information

C: Opinion polls

D: More than one of the above ✓

E: All of the above

3./...

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- f) How may information be encoded that is supplied with an HTTP POST request?

A: Key-value pairs, with spaces replaced by a + and URL-encoding applied afterwards. ✓

B: Plain text

C: The native encoding of whatever data is communicated is used unmodified.

D: HTML

E: No data accompanies the HTTP POST message.

- g) In practice, the parameter following an HTTP GET message may be ...

A: A filename.

B: The name of an executable script.

C: Any parameter the server has been built to interpret.

D: More than one of the above

E: All of the above ✓

- h) The HTTP header contains a Host field in order to support

A: Security.

B: Virtual hosting. ✓

C: HTML.

D: Authentication.

E: Access speed.

- i) Protocols designed to determine network routes usually operate on ISO OSI layer ...

A: 7 ✓

B: 6

C: 5

D: 4

E: 3

4./...

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PAPER: COS332

j) Who ultimately decides which TLDs are available on the Internet?

- A: NSF
- B: ISO
- C: Uniforum
- D: UP
- E: ICANN ✓

k) When a new DNS TLD is in *landrush* it means

- A: Only trademark holders are allowed to register domains in that TLD.
- B: Exactly the same as when the TLD is in a general available state.
- C: Normal users are allowed to register trademarked names in that TLD that would not have been possible under normal conditions.
- D: The domain recently started accepting registrations from the general public — possibly at a higher cost than the eventual annual fees will be. ✓
- E: Its use has not been approved yet.

l) The `whois` command should (eventually) query the ... records.

- A: ICANN
- B: IANA
- C: Registrar's ✓
- D: TLD registry
- E: `who.is`

m) When does a DNS zone transfer occur?

- A: When the zone owner moves to a different ISP.
- B: When a secondary nameserver is synchronised with a primary nameserver. ✓
- C: When an owner sells a TLD to someone else.
- D: When a server is moved outside the DMZ.
- E: During name resolution.

5./...

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n) A DNS SOA TTL entry indicates the time ... (in seconds).

- A: A machine is allowed to use a resolved name before the name has to be resolved again. ✓
- B: A secondary nameserver is allowed to use names before synchronising with the primary server.
- C: The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- D: The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.
- E: The time a client who wants to resolve a name should wait before sending another resolution request.

o) A DNS SOA RETRY entry indicates the time ... (in seconds).

- A: A machine is allowed to use a resolved name before the name has to be resolved again.
- B: A secondary nameserver is allowed to use names before synchronising with the primary server.
- C: The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- D: The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed. ✓
- E: The time a client who wants to resolve a name should wait before sending another resolution request.

p) Which of the following is *not* a valid `nslookup` command?

- A: `ls example.com`
- B: `set type=any`
- C: `server 8.8.8.8`
- D: `exit`
- E: `resolve up.ac.za` ✓

6./...

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- q) Assume you are writing a server that is intended to be accessed by a Telnet client.
- A: Normally the server should echo every character received, because the client may not be able to echo typed characters locally. ✓
 - B: The server should not echo received input because echoing will cause characters to be displayed twice on modern Telnet clients.
 - C: The server should query the Telnet client to determine whether it echoes locally; the server should then only echo if the client will not echo.
 - D: The server should issue an ANSI escape sequence to disable local echoing at the client; the server should then echo everything received.
 - E: A good compromise is to echo every second character received; the user should be still be able to determine the intended output even if some characters are missing and others are repeated.
- r) The structure used to store data used by SNMP is known as ...
- A: DSNMP
 - B: X.509
 - C: MIB ✓
 - D: NMP
 - E: NIC
- s) A toy "telephone" consisting of two tin cans connected by a piece of string is able to communicate in ... mode.
- A: Simplex
 - B: Half-duplex ✓
 - C: Full duplex
 - D: Half-simplex
 - E: Full simplex

7/...

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- t) Which of the following is a useful Android app for determining information about networks?
- A: LanDroid ✓
 - B: Nexus
 - C: iCloud
 - D: DNS
 - E: IMDB

[20]

Question 2

Your PC is connected to a DNS server d provided by your ISP. Whenever your PC wants to resolve a FQDN it simply asks d to it; d does it and responds with the answer. Assume d has just been activated and therefore has no names cached at this point. Suppose your PC wants to send an email to $y@x.com$, and requests d to resolve the FQDN.

Assume that r is the root server chosen, c is the chosen $.com$ nameserver and x is the chosen $x.com$ nameserver.

Draw a diagram to indicate the messages that are sent between between these five machines. Represent the five machines (PC, d , r , c , x) using circles. Represent the messages using arcs between the circles. Label the "request" messages with the type of request (such as A, AAAA, SOA), as well as the (partial) domain name that forms part of the message. For the "response" messages indicate what the response is where possible. (Hint: sometimes the response may be another server, such as x .) Number the messages (requests and responses) in the order in which they are sent.

[10]

8./...

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TIME: 2 hours
MARKS: 100

EXAMINER: 1. Prof MS Olivier
2. Dr DT van der Haar (UJ)

THIS PAPER CONSISTS OF 16 PAGES.

ANSWER ALL QUESTIONS.

NO CALCULATORS PERMITTED.

AN ANSWER SHEET IS PROVIDED SEPARATELY.

WHEN A QUESTION CAN BE ANSWERED BY MEANS OF AN ACRONYM IT IS NOT NECESSARY TO PROVIDE THE PHRASE THAT THE ACRONYM REPRESENTS.

STUDENTS ARE ALLOWED TO WRITE ON THE EXAMINATION PAPER.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1) A switch (used in the place of a hub)

- A: Connects different networks and can convert protocols.
- B: Makes a connection only between computers that are busy communicating.
- C: Routes packets between nodes.
- D: Operates on ISO OSI layer 3.
- E: More than one of the above

2) Which of the following is (are) *not* a function of ISO/OSI's layer 2?

- A: Contention control
- B: Data delineation
- C: Dialogue control
- D: Error control
- E: More than one of the above

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Question 1 (Continued)

3) Layer 6 of ISO/OSI is the ... layer.

- A: Application
- B: Datalink
- C: Session
- D: Presentation
- E: Physical

4) The ISO OSI session layer is positioned (exactly) between the ... and ... layers.

- A: Network, Application
- B: Transport, Application
- C: Network, Presentation
- D: Transport, Presentation
- E: Presentation, Application

5) What is the name of ISO layer 2?

- A: Data link layer
- B: Presentation layer
- C: Network layer
- D: Data layer
- E: More than one of the above

6) Layers 1 to 4 of the ISO OSI model are known as the

- A: Network-oriented layers
- B: Application-oriented layers
- C: TCP/IP-oriented layers
- D: More than one of the above
- E: None of the above

Question 1 (Continued)

- 7) By whom are technical aspects of the Internet standardised?
- A: IETF
 - B: ISO
 - C: EIA
 - D: ITU
 - E: CCITT
- 8) Which of the following URLs is/are valid?
- A: `http://www.cs.up.ac.za/`
 - B: `ftp://ftp.sun.ac.za`
 - C: `mailto://webmaster@up.ac.za`
 - D: More than one of the above
 - E: All of the above
- 9) X.509 is a standard of the
- A: ITU
 - B: IETF
 - C: ISO
 - D: IEEE
 - E: EIA
- 10) If a group of people want to establish a new DNS top-level domain, who will approve or deny such a proposal?
- A: Jon Postel
 - B: ICANN
 - C: IETF
 - D: ISO
 - E: The president of the USA

Question 1 (Continued)

- 11) What is the name of the file that contains IP addresses and corresponding names that be used for name resolution on almost every computer connected to the Internet?
- A: `names.txt`
 - B: `resolv.conf`
 - C: `names`
 - D: `resolv`
 - E: `hosts`
- 12) bind is a program that provides ... services.
- A: directory
 - B: address
 - C: routing
 - D: name
 - E: None of the above
- 13) If the International Standards Organisation (ISO) adopts a standard
- A: All member countries *have* to comply with the standard, but the rest of the world not have to.
 - B: All countries in the world *have* to comply with the standard.
 - C: Almost all countries in the world enforce compliance with almost ISO standards, though they don't have to.
 - D: ISO standards are irrelevant because nobody implements them.
 - E: None of the above

Question 1 (Continued)

- 14) rsync is a utility that
- A: Was written to simplify use of 'the cloud'.
 - B: Marks synchronisation points that enable a network to resynchronise if synchronisation is lost.
 - C: Synchronises the clock on the local computer with an external clock.
 - D: Synchronises directories on the local computer only.
 - E: Synchronises directories — possibly via the network.
- 15) When does a DNS zone transfer occur?
- A: When the zone owner moves to a different ISP.
 - B: When a secondary nameserver is synchronised with a primary nameserver.
 - C: When an owner sells a TLD to someone else.
 - D: When a server is moved outside the DMZ.
 - E: During name resolution.
- 16) Which of the following is *not* a valid nslookup command?
- A: ls example.com
 - B: set type=any
 - C: server 8.8.8.8
 - D: exit
 - E: resolve up.ac.za
- 17) HDLC accomplishes transparency by using
- A: Bit stuffing
 - B: Byte stuffing
 - C: Known field lengths
 - D: All of the above
 - E: None of the above

Question 1 (Continued)

- 18) Ethernet addresses have to be unique. This is accomplished as follows:
- A: Manufacturers apply for a list of addresses from the IEEE that are then used in network interface cards.
 - B: The IEEE assigns a unique manufacturer's code to each manufacturer. The manufacturer then uses this code as the first half of the address and assigns a unique number to each card manufactured by it that is then used as the second half of the address. Thus a unique address is formed.
 - C: The addresses are so long that it is statistically improbable that two addresses on the network will be the same.
 - D: The network administrator assigns a unique address to each card.
- 19) A token is reserved as follows on a token ring: The node wishing to reserve the token
- A: Sends a message to the monitor.
 - B: Sends a special reservation message around the ring.
 - C: Simply increases its priority.
 - D: Just 'grabs' the token even though it may not be its turn.
 - E: None of the above
- 20) When no monitor's presence is detected on an 802.5 network
- A: The hub appoints a new monitor.
 - B: The previous monitor resumes its duties.
 - C: The station with the highest priority starts acting as monitor.
 - D: A new monitor is elected by the remaining nodes.
 - E: The network simply continues operating because it does not really need a monitor.

Question 1 (Continued)

- 21) Suppose a node using SDLC has to transmit the data 011111110. The data will be observed on the line as
- A: 011111010
 - B: 011111110
 - C: 0111111100
 - D: 0111111110
 - E: 0111111X10
- 22) A multicast packet
- A: Should be delivered to all subscribed parties.
 - B: In TCP/IP has an IP address with a first byte between 224 and 239, both included.
 - C: Is routed like a class C packet.
 - D: More than one of the above
 - E: All of the above
- 23) Which of the following is not a valid MIME type?
- A: application
 - B: audio
 - C: data
 - D: text
 - E: vnd

8/...

Question 1 (Continued)

- 24) Assume a HTTP request includes the following:
Accept-Charset: iso-8859-1, utf-8;q=0.5, *;q=0.7
Suppose further that the server supports (only) UTF-8 and UTF-16. Which code was used?
- A: ISO-8559-1
 - B: UTF-8
 - C: UTF-16
 - D: ASCII
 - E: None of the above
- 25) Which of the following is *not* a valid ASN.1 constructor?
- A: SEQUENCE
 - B: MESSAGE
 - C: CHOICE
 - D: SET
- 26) EBNF is
- A: An abbreviation for Extended Backus-Naur Form
 - B: Often used by CFPs to specify the content of data messages
 - C: Standardised by a CFP
 - D: More than one of the above
 - E: All of the above

Question 1 (Continued)

- 27) How may information be encoded that is supplied with an HTTP POST request?
- A: Key-value pairs, with spaces replaced by a + and URL-encoding applied afterwards.
 - B: Plain text
 - C: The native encoding of whatever data is communicated is used unmodified.
 - D: HTML
 - E: No data accompanies the HTTP POST message.
- 28) IP address 172.16.5.79 is a ... address.
- A: Class A
 - B: Class B
 - C: Class C
 - D: Class D
 - E: Class E
- 29) IP fragmentation occurs because of
- A: Different MTUs on the Internet.
 - B: Broken links.
 - C: A shortage of IP addresses.
 - D: Lost packets.
- 30) The MTU (*Maximum Transmission Unit*) is
- A: The size of the largest packet that can be transmitted via the network.
 - B: The highest speed at which packets can be transmitted through the network.
 - C: The size of the largest TCP segment that can be transmitted via the network.
 - D: The cost of the most expensive communications circuit that can be used.
 - E: The highest speed at which bits can be transmitted through the network.

Question 1 (Continued)

- 31) `tracert` (or `tracert`) can work using
- A: IP's TTL and ICMP's *Time exceeded*
 - B: IP's TTL and ICMP's *Destination unreachable*
 - C: ICMP's Router advertisement
 - D: IP's *Source route*
 - E: All of the above
- 32) An ARP cache is maintained on
- A: Each host connected to an Ethernet bus.
 - B: A special ARP cache server.
 - C: Layer 4 of the protocol stack.
 - D: The RARP server.
 - E: The hub.
- 33) What is the purpose of the netmask?
- A: Allows a host to send frames to the router
 - B: Used by routers to identify the network portion from the host portion of an IP address, to enable proper routing of packets
 - C: Used by administrators to join IP address blocks
 - D: Used exclusively by routers to determine whether a frame must be forwarded or blocked
- 34) On which layer of ISO OSI does ICMP fit?
- A: 3
 - B: 4
 - C: 5
 - D: 6
 - E: 7

Question 1 (Continued)

- 35) The three messages sent by TCP when a connection is established, are
- A: SYN, SYN+ACK, ACK
 - B: HELO, HELO, HELO
 - C: HELO, HELO, ACK
 - D: HELO, HELO+OK, OK
- 36) Which field does *not* occur in a UDP header?
- A: Source port
 - B: Length
 - C: Sequence number
 - D: Checksum
- 37) The problem of session hijacking is solved by
- A: Using a random initial TCP sequence number.
 - B: Setting the TCP SYN flag.
 - C: Load shedding.
 - D: Not disbanding the Scorpions.
 - E: Reintroducing the death penalty (TTL).
- 38) Which field does *not* occur in the TCP header?
- A: Source port
 - B: Window (size)
 - C: FIN
 - D: TTL
 - E: SYN

Question 1 (Continued)

- 39) The first line in a WAP page is
- A: <wap>
 - B: <card>
 - C: <vnd.wap.wml>
 - D: <wml>
 - E: <phone>
- 40) A domain hack occurs when
- A: One uses parts of the domain name to spell a word (such as del.icio.us).
 - B: One overrides the DNS by entering information in the hosts file.
 - C: One answers a DNS query before the query has been issued in the hope that an unsuspecting host will cache the incorrect 'answer'.
 - D: One modifies a zone file without authorisation.
 - E: More than one of the above

University of Pretoria

8 JUNE 2016

COURSE: Computer Science
PAPER: COS332

EXAMINER: 1. Prof MS Olivier
2. Dr DT van der Haar (UI)

THIS PAPER CONSISTS OF 17 PAGES.

TIME: 3 hours
MARKS: 100

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Question 1 (Continued)

- 2) A switch (used in the place of a hub)
- A: Connects different networks and can convert protocols.
B: Makes a connection only between computers that are busy communicating.
C: Routes packets between nodes.
D: Operates on ISO OSI layer 3.
E: More than one of the above
- 3) An E1 line communicates at
- A: the same speed as a T1 line
B: 1.5 Mbps
C: 2.048 Mbps
D: 10.000 Mbps
E: All of the above

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

- 1) Manchester encoding is better than NRZ because
- A: Manchester is self-synchronising.
B: The baud rate of Manchester is higher than that of NRZ.
C: The baud rate of Manchester is lower than that of NRZ.
D: The wires may be switched in the case of Manchester.
E: None of the above
- 4) Consider ADSL. The 'A' signifies that it is asymmetric; in other words
- A: The upload and download speeds are different
B: The electrical potential of one line is varied while the other is grounded.
C: It establishes a connection between a user and an ISP.
D: It differs from one country to another.
E: It is not fully digital.
- 5) Which of the following countries is *not* directly linked to SEACOM?
- A: South Africa
B: Mozambique
C: DRC
D: Egypt
E: India

2/...

3/...

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Question 1 (Continued)

- 6) Which of the following is (are) *not* a function of ISO/OSI's layer 2?

☒ A: Media access control
☐ B: Data delineation
☐ C: Dialogue control
☐ D: Error control
☐ E: More than one of the above

- 7) A header added to a message at ISO OSI layer n will be removed at the destination at layer

...
☐ A: $n - 1$
☒ B: n
☐ C: $n + 1$
☐ D: 1
☐ E: 7

- 8) Layers 1 to 4 of the ISO OSI model are known as the

☒ A: Network-oriented layers
☐ B: Application-oriented layers
☐ C: TCP/IP-oriented layers
☐ D: More than one of the above
☐ E: None of the above

- 9) If parity is used for error control, only errors that affect ... will be detected.

☐ A: A single bit
☐ B: An odd number of bits
☐ C: An even number of bits
☒ D: All of the above
☐ E: None of the above

4./...

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Question 1 (Continued)

- 10) A protocol is

☒ A: A set of agreed-upon rules for message interchange.
☐ B: Always described in an RFC.
☐ C: Always open for interpretation.
☐ D: Something that precedes a metacol.
☐ E: A set of diagrams that indicate the sequence of messages that are interchanged.

- 11) The `za` ccTLD is controlled by

☐ A: AfriNIC
☒ B: ZADNA
☐ C: `.za`
☐ D: Uniforum
☐ E: `.co.za`

- 12) `bind` is a program that provides ... services.

☐ A: directory
☐ B: address
☒ C: routing
☐ D: name
☐ E: None of the above

- 13) The `whois` command should (eventually) query the ... records.

☐ A: ICANN
☐ B: IANA
☒ C: Registrar's
☐ D: TLD registry
☐ E: `who.is`

5./...

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Question 1 (Continued)

- 14) Which IEEE standard specifies the operation of wireless local area networks?

A: IEEE 802.3
B: IEEE 802.5
C: IEEE 802.10
D: IEEE 802.11

- 15) The Ethernet address FF:FF:FF:FF:FF:FF is

A: A MAC address
B: A Broadcast address
C: A Group address
D: Invalid
E: None of the above

- 16) An Ethernet frame may contain ... octets of data.

A: 0 – 1500
B: 46 – ∞
C: 46 – 1500
D: 2 – 1500
E: 150 – 4800

- 17) When no monitor's presence is detected on an 802.5 network

A: The hub appoints a new monitor.
B: The previous monitor resumes its duties.
C: The station with the highest priority starts acting as monitor.
D: A new monitor is elected by the remaining nodes.
E: The network simply continues operating because it does not really need a monitor.

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Question 1 (Continued)

- 18) Suppose a node using SDLC has to transmit the data 01111110. The data will be observed on the line as

A: 0111111010
B: 011111110
C: 0111111100
D: 0111111110
E: 0111111X10

- 19) ARP translates a(n) ... address to a(n) ... address.

A: IP, MAC
B: MAC, IP
C: TCP, IP
D: IPv4, IPv6
E: MAC, CAM

- 20) Suppose the first octet of a UTF-8 character is 193. How long (in octets) will the entire character be?

A: 1
B: 2
C: 3
D: 4
E: 5
F: 6

7/...

Question 1 (Continued)

21) On which layer is SSL/TLS usually positioned?

- A: 6
- B: 5
- C: 'Just above 5'
- D: 'Just above 4'
- ☒ E: 4

22) Huffman code is used to

- A: Compress data
- B: Correct errors in data
- C: Encrypt data
- ☒ D: More than one of the above
- E: All of the above

23) To use POP3 anonymously, one uses the command

- A: USER ANON
- ☒ B: USER ANONYMOUS
- C: NOUSER
- D: USER *
- E: None of the above

24) IP address 172.16.5.79 is a ... address.

- A: Class A
- ☒ B: Class B
- C: Class C
- D: Class D
- E: Class E

8/...

Question 1 (Continued)

25) IP fragmentation occurs because of

- A: Different MTUs on the Internet.
- B: Broken links.
- C: A shortage of IP addresses.
- ☒ D: Lost packets.

26) ICMP is used to

- A: Report errors that occur on the IP layer.
- B: Test network functions.
- ☒ C: Modify routes.
- D: More than one of the above
- E: All of the above

27) ping works by sending the following ICMP packet:

- A: Ping request
- ☒ B: Echo request
- C: Information request
- D: Source quench
- E: All of the above

28) Which of the following IP addresses does *not* belong to a private IP address block?

- A: 10.11.12.13
- ☒ B: 172.16.17.18
- C: 192.168.169.170
- D: 238.239.240.241

9/...

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Question 1 (Continued)

- 29) What is the purpose of the netmask?
- A: Allows a host to send frames to the router
 - B: Used by routers to identify the network portion from the host portion of an IP address to enable proper routing of packets
 - C: Used by administrators to join IP address blocks
 - ☒ D: Used exclusively by routers to determine whether a frame must be forwarded or blocked
- 30) A typical ARP message looks as follows:
- A: Who has 10.1.1.1? Tell 10.1.1.2.
 - B: Who has 00-04-38-68-da-00? Tell 00-11-43-e1-ba-4d.
 - ☒ C: GET / HTTP/1.1
 - D: What is the IP of xx.co.za?
 - E: What is the name of 196.7.147.37?
- 31) An AS (Autonomous System) is usually a
- A: Personal computer.
 - B: Network controlled by a single authority.
 - ☒ C: Mainframe computer.
 - D: ccTLD.
 - E: DNS registrar.
- 32) Which protocol has reliability as one of its main goals?
- A: IP
 - ☒ B: TCP
 - C: UDP
 - D: More than one of the above
 - E: All of the above

10./...

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Question 1 (Continued)

- 33) During a slow start the amount of data sent will initially typically
- A: Remain constant.
 - B: Grow logarithmically.
 - C: Grow linearly.
 - ☒ D: Grow exponentially.
 - E: Decrease linearly.
- 34) Which of the following considerations is the most important for BGP (*Border Gateway Protocol*)?
- A: Number of hops
 - B: Nature of routes inside organisations
 - C: Routing policies
 - ☒ D: Dijkstra's algorithm
 - E: ICMP information about network conditions
- 35) Consider the problem of *transparency* in SMTP. Which of the following is true?
- A: Transparency is not an issue — any type of data can be e-mailed.
 - B: In most e-mail programs users should avoid typing a full stop on a line of its own, because that will prematurely terminate the body of the message and cause only the part prior to the full stop to be sent.
 - C: POP3 solves the SMTP transparency problem.
 - D: IMAP solves the SMTP transparency problem.
 - ☒ E: None of the above

11./...

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Question 1 (Continued)

36) Which of the following is a valid DNS root server?

- ☒ A: f.root-servers.net
- ☐ B: .com
- ☐ C: p.root-servers.net
- ☐ D: f.root-server.net
- ☐ E: More than one of the above

37) If SMTP cannot deliver a message, it may

- ☐ A: Wait and try again later.
- ☐ B: Send the message to a secondary server.
- ☐ C: Return an error message to the sender.
- ☐ D: More than one of the above
- ☒ E: All of the above

38) An application protocol will typically respond with a message in the ...-range when security or configuration errors prevented it from completing a request.

- ☒ A: 100s
- ☐ B: 200s
- ☐ C: 300s
- ☐ D: 400s
- ☐ E: 500s

39) The most important routing protocol at JINX is

- ☐ A: RIP
- ☐ B: BGP
- ☐ C: OSPF
- ☒ D: Dijkstra
- ☐ E: Bellman-Ford

12/...

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Question 1 (Continued)

40) A multicast packet

- ☐ A: Should be delivered to all subscribed parties.
- ☐ B: In TCP/IP has an IP address with a first byte between 224 and 239, both included.
- ☐ C: Is routed like a class C packet.
- ☒ D: More than one of the above
- ☐ E: All of the above

[40]

Question 2

a) With which well-known ports are the following services usually associated?

- (i) HTTP 80
- (ii) DNS 53
- (iii) Telnet 23
- (iv) SSH 22
- (v) SMTP 25

(5)

b) What are the following ports typically used for? (It is not necessary to give the corresponding port numbers.)

- (i) Well-known ports Application
- (ii) Registered ports Private Application
- (iii) Dynamic and/or Private Ports Testing

(3)

c) Why are ports useful in firewalls? for running data

(1)

d) How many bits are used to identify a port in TCP/IP? 16

(1)

[10]

13/...

Question 3

Assume some computer X has one network interface with MAC address 10-20-30-40-50-60 and IPv4 address 10.20.30.40. Assume, for the purposes of this question that 10.0.0.0/8 is a routable, public network address. Whenever a question below refers to an IP address (or even just an address) IPv6 is implied.

- What is the loopback address of X ? $fe::80$ (1)
- What unspecified address will X use whenever necessary? $10.20.30.40$ (1)
- What is the link-local address of X ? No privacy-protection mechanisms are to be used. $fe::1$ (2)
- What is the IPv4 mapped address of X ? $10.20.30.40$ (2)
- What would the IPv4 compatible address of X have been? (2)
- If X wants to use 6to4 tunnelling, what would its network address be? (Use CIDR-notation.) B (3)
- If X were to use Teredo tunnelling, what prefix would indicate that it was using such tunnelling? $2001::1$ (2)
- What (destination) prefix would X use when it transmits a multicast packet? 0001 (2)

[15]

14./...

Question 4

The addendum contains copies of 13 packets captured with Wireshark on a spanning port of a managed switch. These 13 packets are in the sequence in which they have been captured. The capture lasted for only a few seconds; these 13 packets are just a few of the thousands of packets that were captured during this period. Packets that are not relevant to this question have been omitted. Some details of these 13 packets that are not relevant to this question have been hidden or omitted. Line numbers are missing it indicates that one or more lines have been removed from the packet. Where line numbers are missing it indicates that one or more lines have been removed because they were not deemed relevant. Answer the questions below about regarding the packets contained in the addendum.

- Consider packet 1.
 - Which application layer protocol is used? $IPV4$ (1)
 - Which transport layer protocol is used? UDP (1)
 - Which network layer protocol is used? (4)
 - Which data link layer protocol is used? (4)
- Details about a number of network interfaces have been captured. What are the vendor IDs of interfaces in these packets of the interfaces supplied by each of the following manufacturers?
 - ASRock (3)
 - Raspberry Pi (3)
 - TP-Link (3)
- Consider packets 1 and 2.
 - Explain the action of node 10.1.1.31. Why does it transmit packet 2? (1)
 - The TTL fields of packets 1 and 2 are quite different (128 vs 64). What is the (most likely) explanation for this difference? (1)
 - The destination ports of packets 1 and 2 are the same (53). Why? (1)
 - The source ports of packets 1 and 2 are quite different (63933 and 8312, respectively). Briefly explain this phenomenon. (4) (1)

15./...

Question 4 (Continued)

- d) Consider packet 3. Assume that it is a response to packet 2.
- (i) What information on layer 4 supports that packet 3 is a response to packet 2?
 - (ii) What information on layer 3 may support that packet 3 is a response to packet 2?
 - (iii) What information on layer 2 may support that packet 3 is a response to packet 2?
 - (iv) Is the application layer response in packet 3 iterative or recursive? Why? (4)
- e) Consider packet 4.
- (i) All the addresses, as well as the destination ports in packets 2 and 4 are identical. Briefly explain why the source ports are not.
 - (ii) The TTL fields of packets 2 and 4 are also identical (64). Would one expect this, or is it coincidental? Explain briefly.
 - (iii) It seems that it was not necessary for 10.1.1.31 to send this packet. Why?
 - (iv) Why do you think 10.1.1.31 sent this packet even though it was not really necessary? (4)
- f) Consider packet 6. To which of the packets provided in the addendum is it (most likely) a response? (1)
- g) Consider packet 7. In what way does it differ from packet 1 on the application layer? (1)
- h) Consider packet 9.

- (i) The trace contained no response to the query issued in packet 8. Given this fact, what feature of the DNS does packet 9 exploit? Briefly explain.
- (ii) While packets 8 and 9 are very similar, they are sent to different destinations (8.8.8.8 and 192.5.4.1, respectively). Explain why both packets use the same destination address on layer 2. (2)

16./...

Question 4 (Continued)

- i) Assume packet 10 is a response to packet 9.
- (i) Packet 10 does not contain the answer to the query contained in packet 9. Justify this claim.
 - (ii) What type of information is provided as 'answer' in packet 10?
 - (iii) What should one conclude from the fact that a proper answer is not provided?
 - (iv) Is the IPv6 address provided in line 53 of packet 10 somehow derived from an IPv4 address? Why (not)? (4)
- j) Consider the merit of sending packet 11 at this time.
- (i) Why may packet 11 provide useful information? (2)
 - (ii) Why may packet 11 be problematic? [26]

Question 5

Consider the packets captured on a link between two hosts that are provided in the addendum and marked packet A through packet J. (See the previous question about more details about how they were captured.) Answer the questions below based on these traces. For ease of reference you may assume that 10.1.1.15 is computer X and 54.164.213.10 is computer Y.

- a) What is the purpose of packets A, B and C? Justify your answer. (1)
- b) Consider packet D.
 - (i) What is its application layer purpose?
 - (ii) What is `squid` (line 41)?
 - (iii) The TCP ACK flag is set. What is acknowledged? (3)

17./...

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Question 5 (Continued)

- c) Packet E does not carry any payload — its length is 0.
- (i) For what purpose was E sent?
 - (ii) The TCP Seq field has the value 1. Is this correct? Why (not)?
 - (iii) The TCP Ack field has the value 618. Is this correct? Why (not)?
 - (iv) The calculated window size advertisement is 15872. Is this (approximately) what you expected? Why (not)? (4)
- d) What is the MIME type of the payload in packet F? (1)
- e) Which node (X or Y) initiates termination of the connection? Justify your answer. (1)

[10]

TOTAL

[100]

END OF PAPER

Packet 3

```

1 Frame 299052: 124 bytes on wire (992 bits), 124 bytes captured (992 bits) on ir
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberf_ca:1f:1e
3 Destination: Raspberf_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
10 Total Length: 110
11 Identification: 0x15ef (5615)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 55
15 Protocol: UDP (17)
16 Header checksum: 0x5255 [validation disabled]
17 Source: 8.8.8.8
18 Destination: 10.1.1.31
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 8312 (8312)
22 Source Port: 53
23 Destination Port: 8312
24 Length: 90
25 Checksum: 0xaaaa [validation disabled]
26 [Stream index: 338]
27 Domain Name System (response)
28 [Request In: 297958]
29 [Time: 0.276987000 seconds]
30 Transaction ID: 0x80b0
31 Flags: 0x8190 Standard query response, No error
32 Questions: 1
33 Answer RRs: 2
34 Authority RRs: 0
35 Additional RRs: 1
36 Queries
37 mail.cs.up.ac.za: type A, class IN
38 Name: mail.cs.up.ac.za
39 [Name Length: 16]
40 [Label Count: 5]
41 Type: A (Host Address) (1)
42 Class: IN (0x0001)
43 Answers
44 mail.cs.up.ac.za: type CNAME, class IN, cname hermes.cs.up.ac.za
45 hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239

```

Packet 4

```

1 Frame 299074: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46
3 Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 75
11 Identification: 0xbec7 (48871)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0xa08b [validation disabled]
17 Source: 10.1.1.31
18 Destination: 8.8.8.8
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 8535 (8535), Dst Port: 53 (53)
22 Source Port: 8535
23 Destination Port: 53
24 Length: 55
25 Checksum: 0x902e [validation disabled]
26 [Stream index: 339]
27 Domain Name System (query)
28 Transaction ID: 0x20e7
29 Flags: 0x0110 Standard query
30 Questions: 1
31 Answer RRs: 0
32 Authority RRs: 0
33 Additional RRs: 1
34 Queries
35   hermes.cs.up.ac.za: type A, class IN
36   Name: hermes.cs.up.ac.za
37   [Name Length: 18]
38   [Label Count: 5]
39   Type: A (Host Address) (1)
40   Class: IN (0x0001)

```


Packet 5

```

1 Frame 300334: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on 1r
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
3 Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.1.31
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
10 Total Length: 91
11 Identification: 0xcb07 (51975)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 55
15 Protocol: UDP (17)
16 Header checksum: 0x9d4f [validation disabled]
17 Source: 8.8.8.8
18 Destination: 10.1.1.31
19 [Source GeoIP: Unknown]
20 [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 8535 (8535)
22 Source Port: 53
23 Destination Port: 8535
24 Length: 71
25 Checksum: 0x9812 [validation disabled]
26 [Stream index: 339]
27 Domain Name System (response)
28 [Request In: 299075]
29 [Time: 0.487202000 seconds]
30 Transaction ID: 0x20e7
31 Flags: 0x8190 Standard query response, No error
32 Questions: 1
33 Answer RRs: 1
34 Authority RRs: 0
35 Additional RRs: 1
36 Queries
37 hermes.cs.up.ac.za: type A, class IN
38 Name: hermes.cs.up.ac.za
39 [Name Length: 18]
40 [Label Count: 5]
41 Type: A (Host Address) (1)
42 Class: IN (0x0001)
43 Answers
44 hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239

```


Packet 6

```

1 Frame 300334: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on 1r
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
3 Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
10 Total Length: 91
11 Identification: 0xcb07 (51975)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 55
15 Protocol: UDP (17)
16 Header checksum: 0x9d4f [validation disabled]
17 Source: 8.8.8.8
18 Destination: 10.1.1.31
19 [Source GeoIP: Unknown]
20 [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 8535 (8535)
22 Source Port: 53
23 Destination Port: 8535
24 Length: 71
25 Checksum: 0x9812 [validation disabled]
26 [Stream index: 339]
27 Domain Name System (response)
28 [Request In: 299075]
29 [Time: 0.487202000 seconds]
30 Transaction ID: 0x20e7
31 Flags: 0x8190 Standard query response, No error
32 Questions: 1
33 Answer RRs: 1
34 Authority RRs: 0
35 Additional RRs: 1
36 Queries
37   hermes.cs.up.ac.za: type A, class IN
38     Name: hermes.cs.up.ac.za
39     [Name Length: 18]
40     [Label Count: 5]
41     Type: A (Host Address) (1)
42     Class: IN (0x0001)
43 Answers
44   hermes.cs.up.ac.za: type A, class IN, addr 137.215.40.239

```

Packet 7

```

1 Frame 300357: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on inte
2 Ethernet II, Src: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e), Dst: Raspberr_ca:1f:1e
3 Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.2, Dst: 10.1.1.31
7 0100 ... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 62
11 Identification: 0x349e (13470)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 128
15 Protocol: UDP (17)
16 Header checksum: 0xefe [validation disabled]
17 Source: 10.1.1.2
18 Destination: 10.1.1.31
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 63934 (63934), Dst Port: 53 (53)
22 Source Port: 63934
23 Destination Port: 53
24 Length: 42
25 Checksum: 0x81b1 [validation disabled]
26 [Stream index: 340]
27 Domain Name System (query)
28 Transaction ID: 0x0013
29 Flags: 0x0100 Standard query
30 Questions: 1
31 Answer RRs: 0
32 Authority RRs: 0
33 Additional RRs: 0
34 Queries
35 mail.cs.up.ac.za: type AAAA, class IN
36 Name: mail.cs.up.ac.za
37 [Name Length: 16]
38 [Label Count: 5]
39 Type: AAAA (IPv6 Address) (28)
40 Class: IN (0x0001)

```

Packet 8

```
1 Frame 300371: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46
3 Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.1.31, Dst: 8.8.8.8
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total length: 75
11 Identification: 0xbf02 (48898)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0xa070 [validation disabled]
17 Source: 10.1.1.31
18 Destination: 8.8.8.8
19 [Source GeoIP: Unknown]
20 [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 9533 (9533), Dst Port: 53 (53)
22 Source Port: 9533
23 Destination Port: 53
24 Length: 55
25 Checksum: 0x784f [validation disabled]
26 [Stream index: 341]
27 Domain Name System (query)
28 Transaction ID: 0x34c5
29 Flags: 0x0110 Standard query
30 Questions: 1
31 Answer RRs: 0
32 Authority RRs: 0
33 Additional RRs: 1
34 Queries
35   hermes.cs.up.ac.za: type AAAA, class IN
36   Name: hermes.cs.up.ac.za
37   [Name Length: 18]
38   [Label Count: 5]
39   Type: AAAA (IPv6 Address) (28)
40   Class: IN (0x0001)
```


Packet 9

```

1 Frame 303165: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on inte
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46
3 Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.1.31, Dst: 192.5.4.1
7 0100 ... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 75
11 Identification: 0x449d (17565)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0x66df [validation disabled]
17 Source: 10.1.1.31
18 Destination: 192.5.4.1
19 [Source GeoIP: Unknown]
20 [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 54556 (54556), Dst Port: 53 (53)
22 Source Port: 54556
23 Destination Port: 53
24 Length: 55
25 Checksum: 0x1cd3 [validation disabled]
26 [Stream index: 344]
27 Domain Name System (query)
28 [Response In: 303964]
29 Transaction ID: 0x2d6b
30 Flags: 0x0010 Standard query
31 Questions: 1
32 Answer RRs: 0
33 Authority RRs: 0
34 Additional RRs: 1
35 Queries
36   hermes.cs.up.ac.za: type AAAA, class IN
37     Name: hermes.cs.up.ac.za
38     [Name Length: 18]
39     [Label Count: 5]
40     Type: AAAA (IPv6 Address) (28)
41     Class: IN (0x0001)

```

Packet 10

```

1 Frame 303963: 302 bytes on wire (2416 bits), 302 bytes captured (2416 bits) on
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
3 Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 192.5.4.1, Dst: 10.1.1.31
7 0100 .... = Version: 4
8 ... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x10 (DSCP: Unknown, ECN: Not-ECT)
10 Total Length: 288
11 Identification: 0x4375 (17269)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 55
15 Protocol: UDP (17)
16 Header checksum: 0x7022 [validation disabled]
17 Source: 192.5.4.1
18 Destination: 10.1.1.31
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 54556 (54556)
22 Source Port: 53
23 Destination Port: 54556
24 Length: 268
27 Domain Name System (response)
28 [Request In: 303165]
29 [Time: 0.335626000 seconds]
30 Transaction ID: 0x2d6b
31 Flags: 0x8010 Standard query response, No error
32 Questions: 1
33 Answer RRs: 0
34 Authority RRs: 4
35 Additional RRs: 6
36 Queries
37 hermes.cs.up.ac.za: type AAAA, class IN
38 Name: hermes.cs.up.ac.za
39 [Name Length: 18]
40 [Label Count: 5]
41 Type: AAAA (IPv6 Address) (28)
42 Class: IN (0x0001)
43 Authoritative nameservers
44 ac.za: type NS, class IN, ns za-ns.anycast.pch.net
45 ac.za: type NS, class IN, ns uchpx.uct.ac.za
46 ac.za: type NS, class IN, ns disa.tenet.ac.za
47 ac.za: type NS, class IN, ns ns0.is.co.za
48 Additional records
49 ns0.is.co.za: type A, class IN, addr 196.4.160.17
50 disa.tenet.ac.za: type A, class IN, addr 196.21.79.50
51 disa.tenet.ac.za: type AAAA, class IN, addr 2001:4200:ffff:a::1
52 uchpx.uct.ac.za: type A, class IN, addr 137.158.128.1
53 uchpx.uct.ac.za: type AAAA, class IN, addr 2001:43f8:75::3

```

Packet 11

```

1 Frame 303972: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46 (
3 Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
7 0100 ... = Version: 4
8 ... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 75
11 Identification: 0xbf0b (48907)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0xa067 [validation disabled]
17 Source: 10.1.1.31
18 Destination: 8.8.8.8
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 37222 (37222), Dst Port: 53 (53)
22 Source Port: 37222
23 Destination Port: 53
24 Length: 55
25 Checksum: 0x0183 [validation disabled]
26 [Stream index: 346]
27 Domain Name System (query)
28 Transaction ID: 0x3f68
29 Flags: 0x0110 Standard query
30 Questions: 1
31 Answer RRs: 0
32 Authority RRs: 0
33 Additional RRs: 1
34 Queries
35 hermes.cs.up.ac.za: type AAAA, class IN
36 Name: hermes.cs.up.ac.za
37 [Name Length: 18]
38 [Label Count: 5]
39 Type: AAAA (IPv6 Address) (28)
40 Class: IN (0x0001)

```


Packet 12

```

1 Frame 304879: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Raspberr_ca:1f:1e
3 Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4 Source: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.1.1.31
7 0100 ... = Version: 4
8 ... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x0c (DSCP: Unknown, ECN: Not-ECT)
10 Total Length: 124
11 Identification: 0x2418 (9240)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 55
15 Protocol: UDP (17)
16 Header checksum: 0x441e [validation disabled]
17 Source: 8.8.8.8
18 Destination: 10.1.1.31
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 37222 (37222)
22 Source Port: 53
23 Destination Port: 37222
24 Length: 104
25 Checksum: 0x5def [validation disabled]
26 [Stream index: 346]
27 Domain Name System (response)
28 [Request In: 303973]
29 [Time: 0.403078000 seconds]
30 Transaction ID: 0x3f68
31 Flags: 0x8190 Standard query response, No error
32 Questions: 1
33 Answer RRs: 0
34 Authority RRs: 1
35 Additional RRs: 1
36 Queries
37   hermes.cs.up.ac.za: type AAAA, class IN
38     Name: hermes.cs.up.ac.za
39     [Name Length: 18]
40     [Label Count: 5]
41     Type: AAAA (IPv6 Address) (28)
42     Class: IN (0x0001)
43     Authoritative nameservers
44     cs.up.ac.za: type SOA, class IN, mname ns2.up.ac.za

```


Packet 13

```

1 Frame 304887: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: AsrockIn_29:d3:3e
3 Destination: AsrockIn_29:d3:3e (d0:50:99:29:d3:3e)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.31, Dst: 10.1.1.2
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 132
11 Identification: 0x19d0 (6608)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0x4a77 [validation disabled]
17 Source: 10.1.1.31
18 Destination: 10.1.1.2
19 [Source GeoIP: Unknown]
20 [Destination GeoIP: Unknown]
21 User Datagram Protocol, Src Port: 53 (53), Dst Port: 63934 (63934)
22 Source Port: 53
23 Destination Port: 63934
24 Length: 112
25 Checksum: 0xceb2 [validation disabled]
26 [Stream index: 340]
27 Domain Name System (response)
28 [Request In: 300358]
29 [Time: 1.948991000 seconds]
30 Transaction ID: 0x0013
31 Flags: 0x8180 Standard query response, No error
32 Questions: 1
33 Answer RRs: 1
34 Authority RRs: 1
35 Additional RRs: 0
36 Queries
37 mail.cs.up.ac.za: type AAAA, class IN
38 Name: mail.cs.up.ac.za
39 [Name Length: 16]
40 [Label Count: 5]
41 Type: AAAA (IPv6 Address) (28)
42 Class: IN (0x0001)
43 Answers
44 mail.cs.up.ac.za: type CNAME, class IN, cname hermes.cs.up.ac.za
45 Authoritative nameservers
46 cs.up.ac.za: type SOA, class IN, mname ns2.up.ac.za

```

Packets for Question 5

Packet A

```

1 Frame 549655: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on inte
2 Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:40
3 Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
4 Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
5   Source Port: 52400
6   Destination Port: 80
7   [Stream index: 252]
8   [TCP Segment Len: 0]
9   Sequence number: 0 (relative sequence number)
10  Acknowledgment number: 0
11  Header Length: 40 bytes
12  Flags: 0x002 (SYN)
13    000. .... .... = Reserved: Not set
14    ...0 .... .... = Nonce: Not set
15    .... 0... .... = Congestion Window Reduced (CWR): Not set
16    .... .0.. .... = ECN-Echo: Not set
17    .... ..0. .... = Urgent: Not set
18    .... ...0 .... = Acknowledgment: Not set
19    .... .... 0... = Push: Not set
20    .... .... .0.. = Reset: Not set
21    .... .... .1. = Syn: Set
22    .... .... ...0 = Fin: Not set
23    [TCP Flags: *****S*]
24    Window size value: 29200
25    [Calculated window size: 29200]
26    Checksum: 0x827b [validation disabled]
27    Urgent pointer: 0
28    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-

```

Packet B

```

1 Frame 550473: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on inte
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4c
3 Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
4 Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
5   Source Port: 80
6   Destination Port: 52400
7   [Stream index: 252]
8   [TCP Segment Len: 0]
9   Sequence number: 0      (relative sequence number)
10  Acknowledgment number: 1 (relative ack number)
11  Header Length: 40 bytes
12  Flags: 0x012 (SYN, ACK)
13    000. .... .... = Reserved: Not set
14    ...0 .... .... = Nonce: Not set
15    .... 0... .... = Congestion Window Reduced (CWR): Not set
16    .... .0.. .... = ECN-Echo: Not set
17    .... ..0. .... = Urgent: Not set
18    .... ...1 .... = Acknowledgment: Set
19    .... .... 0... = Push: Not set
20    .... .... .0.. = Reset: Not set
21    .... .... .1. = Syn: Set
22    .... .... .0 = Fin: Not set
23    [TCP Flags: *****A**S*]
24    Window size value: 17898
25    [Calculated window size: 17898]
26    Checksum: 0x01c2 [validation disabled]
27    Urgent pointer: 0
28    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-

```


Packet C

```

1 Frame 550475: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface
2 Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:41
3 Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
4 Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
5     Source Port: 52400
6     Destination Port: 80
7     [Stream index: 252]
8     [TCP Segment Len: 0]
9     Sequence number: 1 (relative sequence number)
10    Acknowledgment number: 1 (relative ack number)
11    Header Length: 32 bytes
12    Flags: 0x010 (ACK)
13        000. .... = Reserved: Not set
14        ...0 .... = Nonce: Not set
15        .... 0... = Congestion Window Reduced (CWR): Not set
16        .... .0.. = ECN-Echo: Not set
17        .... ..0. = Urgent: Not set
18        .... ...1 = Acknowledgment: Set
19        .... .... = Push: Not set
20        .... ...0 = Reset: Not set
21        .... ..0. = Syn: Not set
22        .... ...0 = Fin: Not set
23        [TCP Flags: *****A****]
24    Window size value: 229
25    [Calculated window size: 29312]
26    [Window size scaling factor: 128]
27    Checksum: 0x7542 [validation disabled]
28    Urgent pointer: 0
29    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

```

Packet D

```

1 Frame 550478: 683 bytes on wire (5464 bits), 683 bytes captured (5464 bits) on
2 Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Tp-LinkT_e7:76:4f
3 Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
4 Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
5   Source Port: 52400
6   Destination Port: 80
7   [Stream index: 252]
8   [TCP Segment Len: 617]
9   Sequence number: 1 (relative sequence number)
10  [Next sequence number: 618 (relative sequence number)]
11  Acknowledgment number: 1 (relative ack number)
12  Header Length: 32 bytes
13  Flags: 0x018 (PSH, ACK)
14    000. .... = Reserved: Not set
15    ...0 .... = Nonce: Not set
16    .... 0... = Congestion Window Reduced (CWR): Not set
17    .... .0.. = ECN-Echo: Not set
18    .... .0.  = Urgent: Not set
19    .... .1.  = Acknowledgment: Set
20    .... 1... = Push: Set
21    .... .0.. = Reset: Not set
22    .... ..0. = Syn: Not set
23    .... ...0 = Fin: Not set
24    [TCP Flags: *****AP***]
25    Window size value: 229
26    [Calculated window size: 29312]
27    [Window size scaling factor: 128]
28    Checksum: 0x5c86 [validation disabled]
29    Urgent pointer: 0
30    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
32 Hypertext Transfer Protocol
33   GET /livecountping/142650410/630826641504?__=ghidbpv89h2c&routed=1&jid&site
34   Host: lc48.dsr.livifyre.com\r\n
35   User-Agent: Mozilla/5.0 (Windows NT 10.0) AppleWebKit/537.36 (KHTML, like C
36   Origin: http://www.thelancet.com\r\n
37   Accept: */*\r\n
38   Referer: http://www.thelancet.com/journals/laninf/article/PIIS1473-3099(14)
39   Accept-Encoding: gzip, deflate, sdch\r\n
40   Accept-Language: en-GB,en;q=0.8,en-US;q=0.6,af;q=0.4,en-2A;q=0.2\r\n
41   Via: 1.1 styx (squid/3.3.8)\r\n
42   X-Forwarded-For: 10.1.1.23\r\n
43   Cache-Control: max-age=0\r\n
44   Connection: keep-alive\r\n
45   \r\n

```

Packet E

1 Frame 551367: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface
 2 Ethernet II, Src: Tp-Link_T7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byte_e9:68:4c:
 3 Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
 4 Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
 5 Source Port: 80
 6 Destination Port: 52400
 7 [Stream index: 252]
 8 [TCP Segment Len: 0]
 9 Sequence number: 1 (relative sequence number)
 10 Acknowledgment number: 618 (relative ack number)
 11 Header Length: 32 bytes
 12 Flags: 0x010 (ACK)
 13 000. - Reserved: Not set
 14 ...0 - Nonce: Not set
 15 0... - Congestion Window Reduced (CWR): Not set
 16 0... - ECN-Echo: Not set
 17 0... - Urgent: Not set
 18 1... - Acknowledgment: Set
 19 0... - Push: Not set
 20 0... - Reset: Not set
 21 0... - Syn: Not set
 22 0... - Fin: Not set
 23 [TCP Flags:A....]
 24 Window size value: 31
 25 [Calculated window size: 15872]
 26 [Window size scaling factor: 512]
 27 Checksum: 0x7327 [validation disabled]
 28 Urgent pointer: 0
 29 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

Packet F

```

1 Frame 551378: 657 bytes on wire (5256 bits), 657 bytes captured (5256 bits) on
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4c
3 Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
4 Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
5   Source Port: 80
6   Destination Port: 52400
7   [Stream index: 252]
8   [TCP Segment Len: 591]
9   Sequence number: 1 (relative sequence number)
10  [Next sequence number: 592 (relative sequence number)]
11  Acknowledgment number: 618 (relative ack number)
12  Header Length: 32 bytes
13  Flags: 0x018 (PSH, ACK)
14    000. .... .... = Reserved: Not set
15    .... .... .... = Nonce: Not set
16    .... 0... .... = Congestion Window Reduced (CWR): Not set
17    .... 0... .... = ECN-Echo: Not set
18    .... ..0. .... = Urgent: Not set
19    .... ...1 .... = Acknowledgment: Set
20    .... .... 1... = Push: Set
21    .... .... 0.. = Reset: Not set
22    .... .... ..0. = Syn: Not set
23    .... .... ...0 = Fin: Not set
24    [TCP Flags: *****AP***]
25    Window size value: 31
26    [Calculated window size: 15872]
27    [Window size scaling factor: 512]
28    Checksum: 0x8e38 [validation disabled]
29    Urgent pointer: 0
30    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
32 Hypertext Transfer Protocol
33   HTTP/1.1 200 OK\r\n
34   Server: Livecount/1.1 (Simple 4.0) lc48.dsr.livefyre.com\r\n
37   Access-Control-Allow-Origin: http://www.thelancet.com\r\n
38   Access-Control-Max-Age: 1728000\r\n
39   Access-Control-Allow-Methods: GET, OPTIONS\r\n
40   Access-Control-Allow-Headers: Origin, Accept-Language, Accept-Encoding, X-
41   Access-Control-Allow-Credentials: true\r\n
42   Connection: close\r\n
43   Content-Type: application/json\r\n
44   \r\n

```


Packet G

```

1 Frame 551360: 66 bytes on wire (528 bits): 66 bytes captured (528 bits) on interface
2 Ethernet 17, Src: Giga-byt_e9:68:40 (74:dd:35:e9:68:40), Dst: Tp-LinkT_e7:76:40
3 Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
4 Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
5   Source Port: 52400
6   Destination Port: 80
7   [Stream Index: 252]
8   [TCP Segment Len: 0]
9   Sequence number: 618      (relative sequence number)
10  Acknowledgment number: 592  (relative ack number)
11  Header Length: 32 bytes
12  Flags: 0x010 (ACK)
13    000. .... .... = Reserved: Not set
14    ...0 .... .... = Nonce: Not set
15    .... 0... .... = Congestion Window Reduced (CWR): Not set
16    .... .0... .... = ECH-Echo: Not set
17    .... ..0. .... = Urgent: Not set
18    .... ...1 .... = Acknowledgment: Set
19    .... .... 0... = Push: Not set
20    .... .... .0.. = Reset: Not set
21    .... .... ..0. = Syn: Not set
22    .... .... ...0 = Fin: Not set
23    [TCP Flags: *****A****]
24    Window size value: 238
25    [Calculated window size: 30464]
26    [Window size scaling factor: 128]
27    Checksum: 0x6fab [validation disabled]
28    Urgent pointer: 0
29    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

```

Packet H

```
1 Frame 552818: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface
2 Ethernet II, Src: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46), Dst: Giga-Byt_e9:68:4c
3 Internet Protocol Version 4, Src: 54.164.213.10, Dst: 10.1.1.15
4 Transmission Control Protocol, Src Port: 80 (80), Dst Port: 52400 (52400), Seq:
5     Source Port: 80
6     Destination Port: 52400
7     [Stream index: 252]
8     [TCP Segment Len: 0]
9     Sequence number: 592      (relative sequence number)
10    Acknowledgment number: 618  (relative ack number)
11    Header Length: 32 bytes
12    Flags: 0x011 (FIN, ACK)
13        000, .... .... = Reserved: Not set
14        ...0 .... .... = Nonce: Not set
15        .... 0... .... = Congestion Window Reduced (CWR): Not set
16        .... .0.. .... = ECN-Echo: Not set
17        .... .0. .... = Urgent: Not set
18        .... ...1 .... = Acknowledgment: Set
19        .... .... 0... = Push: Not set
20        .... .... .0.. = Reset: Not set
21        .... .... ..0. = Syn: Not set
22        .... .... ...1 = Fin: Set
23        [TCP Flags: *****A***F]
24    Window size value: 31
25    [Calculated window size: 15872]
26    [Window size scaling factor: 512]
27    Checksum: 0x6aae [validation disabled]
28    Urgent pointer: 0
29    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Packet 1

```
1 Frame 552820: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on inte
2 Ethernet II, Src: Giga-Byt_e9:68:40 (74:d4:35:e9:68:40), Dst: Ip-LinkI_e7:76:46
3 Internet Protocol Version 4, Src: 10.1.1.15, Dst: 54.164.213.10
4 Transmission Control Protocol, Src Port: 52400 (52400), Dst Port: 80 (80), Seq:
5   Source Port: 52400
6   Destination Port: 80
7   [Stream index: 252]
8   [TCP Segment Len: 0]
9   Sequence number: 618      (relative sequence number)
10  Acknowledgment number: 593  (relative ack number)
11  Header Length: 32 bytes
12  Flags: 0x011 (FIN, ACK)
13    000. .... = Reserved: Not set
14    ...0 .... = Nonce: Not set
15    .... 0... = Congestion Window Reduced (CWR): Not set
16    .... 0... = ECN-Echo: Not set
17    .... 0... = Urgent: Not set
18    .... 1... = Acknowledgment: Set
19    .... 0... = Push: Not set
20    .... 0... = Reset: Not set
21    .... 0... = Syn: Not set
22    .... 1... = Fin: Set
23    [TCP Flags: *****A***F]
24    Window size value: 238
25    [Calculated window size: 30464]
26    [Window size scaling factor: 128]
27    Checksum: 0x6c2f [validation disabled]
28    Urgent pointer: 0
29    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```


Packets for Question 4

Packet 1

```
1 Frame 297947: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on inte
2 Ethernet II, Src: Asrockin_29:d3:3e (d0:50:99:29:d3:3e), Dst: Raspberr_ca:1f:1e
3   Destination: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
4   Source: Asrockin_29:d3:3e (d0:50:99:29:d3:3e)
5   Type: IPv4 (0x0800)
6   Internet Protocol Version 4, Src: 10.1.1.2, Dst: 10.1.1.31
7   0100 .... = Version: 4
8   .... 0101 = Header Length: 20 bytes
9   Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10  Total Length: 62
11  Identification: 0x33e2 (13282)
12  Flags: 0x00
13  Fragment offset: 0
14  Time to live: 128
15  Protocol: UDP (17)
16  Header checksum: 0xff64 [validation disabled]
17  Source: 10.1.1.2
18  Destination: 10.1.1.31
19  [Source GeoIP: Unknown]
20  [Destination GeoIP: Unknown]
21  User Datagram Protocol, Src Port: 63933 (63933), Dst Port: 53 (53)
22  Source Port: 63933
23  Destination Port: 53
24  Length: 42
25  Checksum: 0x81ce [validation disabled]
26  [Stream index: 337]
27  Domain Name System (query)
28  Transaction ID: 0x0012
29  Flags: 0x0100 Standard query
30  Questions: 1
31  Answer RRs: 0
32  Authority RRs: 0
33  Additional RRs: 0
34  Queries
35      mail.cs.up.ac.rs: type A, class IN
36      Name: mail.cs.up.ac.rs
37      [Name Length: 16]
38      [Label Count: 5]
39      Type: A (Host Address) (1)
40      Class: IN (0x0001)
```

Packet 2

```

1 Frame 297957: 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on Inte
2 Ethernet II, Src: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e), Dst: Tp-LinkT_e7:76:46
3 Destination: Tp-LinkT_e7:76:46 (30:b5:c2:e7:76:46)
4 Source: Raspberr_ca:1f:1e (b8:27:eb:ca:1f:1e)
5 Type: IPv4 (0x0800)
6 Internet Protocol Version 4, Src: 10.1.1.31, Dst: 8.8.8.8
7 0100 .... = Version: 4
8 .... 0101 = Header Length: 20 bytes
9 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
10 Total Length: 73
11 Identification: 0xbec2 (48866)
12 Flags: 0x00
13 Fragment offset: 0
14 Time to live: 64
15 Protocol: UDP (17)
16 Header checksum: 0xa092 [validation disabled]
17 Source: 10.1.1.31
18 Destination: 8.8.8.8
19 [Source GeolP: Unknown]
20 [Destination GeolP: Unknown]
21 User Datagram Protocol, Src Port: 8312 (8312), Dst Port: 53 (53)
22 Source Port: 8312
23 Destination Port: 53
24 Length: 53
25 Checksum: 0xabbl [validation disabled]
26 [Stream index: 338]
27 Domain Name System (query)
28 Transaction ID: 0x80b0
29 Flags: 0x0110 Standard query
30 Questions: 1
31 Answer RRs: 0
32 Authority RRs: 0
33 Additional RRs: 1
34 Queries
35 mail.cs.up.ac.za: type A, class IN
36 Name: mail.cs.up.ac.za
37 [Name Length: 16]
38 [Label Count: 5]
39 Type: A (Host Address) (1)
40 Class: IN (0x0001)

```


University of Pretoria

SEMESTER TEST: 12 APRIL 2016

COURSE: Computer Science
MEMORANDUM: COS332

TIME: 60 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 13 PAGES.

ANSWER ALL QUESTIONS.
NO CALCULATORS PERMITTED

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

a) Layer 6 of ISO/OSI is the ... layer.

A: Application

B: Datalink

C: Session

☒ **D:** Presentation

E: Physical

b) The ISO OSI session layer is positioned (exactly) between the ... and ... layers.

A: Network, Application

B: Transport, Application

C: Network, Presentation

☒ **D:** Transport, Presentation

E: Presentation, Application

2./...

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c) What is the name of ISO layer 2?

- ☒ **A:** Data link layer
- B:** Presentation layer
- C:** Network layer
- D:** Data layer
- E:** More than one of the above

d) Layers 1 to 4 of the ISO OSI model are known as the

- A:** Network-oriented layers
- B:** Application-oriented layers
- C:** TCP/IP-oriented layers
- D:** More than one of the above
- ☒ **E:** None of the above

e) Suppose a distributed database is used by a number of application protocols. On which OSI ISO layer should the two-phase commit protocol ideally be implemented?

- A:** 1
- B:** 2
- C:** 3
- D:** 4
- ☒ **E:** 5
- F:** 6
- G:** 7

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f) By whom are technical aspects of the Internet standardised?

A: IETF

B: ISO

C: EIA

D: ITU

E: CCITT

g) Ideally, an ISO layer provides services to ... and uses services from

A: the layer below it; the layer above it

B: the layer above it; the layer below it

C: its peer layer elsewhere in the network; the layer below it

D: the layer below it; its peer layer elsewhere in the network

E: a peer layer somewhere in the network; a peer layer somewhere in the network

h) When a new DNS TLD is in *landrush* it means

A: Only trademark holders are allowed to register domains in that TLD.

B: Exactly the same as when the TLD is in a general available state.

C: Normal users are allowed to register trademarked names in that TLD that would not have been possible under normal conditions.

D: The domain recently started accepting registrations from the general public — possibly at a higher cost than the eventual annual fees will be.

E: Its use has not been approved yet.

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i) A DNS SOA RETRY entry indicates the time ... (in seconds).

A: A machine is allowed to use a resolved name before the name has to be resolved again.

B: A secondary nameserver is allowed to use names before synchronising with the primary server.

C: The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.

☒ **D:** The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.

E: The time a client who wants to resolve a name should wait before sending another resolution request.

j) Which of the following is not a valid MIME type?

A: application

B: audio

☒ **C:** data

D: text

E: vnd

k) Suppose the first octet of a UTF-8 character is 193. How long (in octets) will the entire character be?

A: 1

☒ **B:** 2

C: 3

D: 4

E: 5

F: 6

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- l) Which of the following is a valid MIME type?
- A:** html
 - B:** xml
 - C:** jpg
 - ☒ **D:** model
 - E:** msword
- m) Which of the following is *not* a valid MIME type?
- A:** multipart/mixed
 - B:** image/jpeg
 - ☒ **C:** text/text
 - D:** example/cos332
 - E:** multipart/alternative
- n) EBNF is
- A:** An abbreviation for Extended Backus-Naur Form
 - B:** Often used by CFPs to specify the content of data messages
 - C:** Standardised by a CFP
 - ☒ **D:** More than one of the above
 - E:** All of the above
- o) The ideal layer to implement encryption for most situations is
- A:** 7
 - B:** 6
 - C:** 4
 - D:** 3
 - ☒ **E:** Dependent on the details of the specific case.

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- p) HTTP is used, amongst others, by Web browsers, WAP browsers, and applications that download maps and music, as well as many other applications. HTTP is on the

☒ **A:** Application layer

B: Bottom half of the application layer

C: Top half of the application layer

D: Presentation layer

E: None of the above

- q) How may information be encoded that is supplied with an HTTP `POST` request?

☒ **A:** Key-value pairs, with spaces replaced by a + and URL-encoding applied afterwards.

B: Plain text

C: The native encoding of whatever data is communicated is used unmodified.

D: HTML

E: No data accompanies the HTTP `POST` message.

- r) Flow control in TCP uses

A: Stop-and-wait

B: The unrestricted approach

☒ **C:** Window advertisements

D: RTS and CTS

- s) Which transport layer protocol(s) is/are used by DNS?

A: TCP

B: UDP

☒ **C:** TCP and UDP

D: TCP/IP

E: IP

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t) TFTP

A: Has strong access control.

☒ **B:** Uses UDP.

C: Cannot be used with DNS.

D: More than one of the above

E: All of the above

u) What is the name of the server that is commonly used on Unix systems to share files with Windows clients?

A: Tango

B: Foxtrot

☒ **C:** Samba

D: Waltz

E: Jive

v) The ccTLD of `who.is` is that of

A: Israel

B: Isle of Man

☒ **C:** Iceland

D: International Software

E: Ipso Facto

w) Which of the following second-level domains does not exist?

A: `school.za`

B: `edu.za`

C: `ac.za`

☒ **D:** `tech.za`

E: `law.za`

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- x) The primary reason for including the `host :` header in the HTTP protocol is to
- A:** Ensure that the `hosts` file cannot be misused.
 - B:** Make a clear distinction between client and host requests.
 - ☒ **C:** Enable virtual hosting.
 - D:** Enhance security by ensuring that the destination server is indeed the one the user wanted to visit because the name of the server can now be compared with the name provided in the `host :` header.
 - E:** Specify alternative hosts that may contain the content the requester wants.
- y) Which of the following is *not* an SNMP request?
- A:** Get
 - B:** GetNext
 - C:** GetBulk
 - ☒ **D:** Reboot
 - E:** Set

[25]

Question 2

Name five (5) fields that occur in a TCP header but not in a UDP header. Briefly explain why each field is necessary in a TCP header but not in a UDP header. (For the purposes of this question FIN is not an acceptable answer.)

[5]

9./...

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Question 3

Assume that computers on an internet are addressed using addresses (A records) consisting of numbers from 1 to 100. Assume node 1 is a root name server and you use node 100 to perform *recursive* name resolution. The following are (partial) zone files at some nodes running *iterative* DNS servers. The files are incomplete since some rows and columns have been omitted; these omissions do not affect the answers to the questions. No caching occurs.

Node 1			Node 13			Node 21			Node 22		
com.	NS	10	ac	NS	20	xx	NS	66	xx	NS	77
org.	NS	11	co	NS	21	yy	NS	55	yy	NS	88
mil.	NS	12	org	NS	22	zz	NS	44	zz	NS	99
za.	NS	13	net	NS	23						

Node 55			Node 66			Node 77		
@	NS	55	@	NS	66	@	NS	77
			@	MX,1	98	@	MX,1	99
www	A	50	www	A	70	www	A	80
ftp	A	51	ftp	A	71	ftp	A	81
mail	A	52	mail	A	72	mail	A	82
arthur	A	53	arthur	A	73	arthur	A	83
						john	NS	55

The following series of `nslookup` queries is executed in sequence from top to bottom. Provide the address (that is, number) returned by each of the queries. The queries are indicated by a Roman number in square brackets (such as “[iii]”). The commands that do not return responses are not numbered. Write N/A if the query will not return an address.

```

nslookup> www.xx.org.za      [i]    [80]    > ftp.xx.co.za          [vi]   [98]
> arthur.xx.co.za           [ii]   [73]    > john.xx.org.za       [vii]  [98]
> set type=NS
> com.                       [iii]  [10]    > john.xx.org.za       [viii] [55]
> co.za                     [iv]   [N/A]   > server 1
> set type=A
> ftp.xx.co.za              [v]    [N/A]   > com.                  [ix]   [N/A]
> server 100
> set type=MX
>                             > server 77
>                             > www.xx.org.za        [x]    [80]
>                             >

```

[10]

10./...

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Question 4

[555 bytes are consumed from A's buffer.
1500 bytes are consumed from B's buffer.

A

Source port	55
Destination port	66
Next byte to send	2223
Next byte expected	3334
My window size	555
Peer's window size	1500
Data to send	10000

B

Source port	66
Destination port	55
Next byte to send	3334
Next byte expected	2223
My window size	1500
Peer's window size	555
Data to send	10000

Payload size: 555 bytes

SRC:66,DEST:55,SEQ:3334, ACK:2223, WinAdv:1500

A

Source port	55
Destination port	66
Next byte to send	2223
Next byte expected	3889
My window size	0
Peer's window size	1500
Data to send	10000

A sends data to B.

Payload size: 1000 bytes

SRC:55, DEST:66, SEQ:2223, ACK:3889, WinAdv:0

B

Source port	66
Destination port	55
Next byte to send	3889
Next byte expected	3223
My window size	500
Peer's window size	0
Data to send	9445

Payload size: 0 bytes

SRC:66,DEST:55,SEQ:3889, ACK:3223, WinAdv:500

60 bytes are consumed from A's buffer.

A

Source port	55
Destination port	66
Next byte to send	3223
Next byte expected	3889
My window size	60
Peer's window size	500
Data to send	9000

A

Source port	55
Destination port	66
Next byte to send	3223
Next byte expected	3889
My window size	60
Peer's window size	500
Data to send	9000

A sends data to B.

Payload size: 500 bytes

SRC:55, DEST:66, SEQ:3223, ACK:3889, WinAdv:60

B

Source port	66
Destination port	55
Next byte to send	3889
Next byte expected	3723
My window size	0
Peer's window size	60
Data to send	9445

Payload size: 60 bytes

SRC:66,DEST:55,SEQ:3889, ACK:3723, WinAdv:0

A

Source port	55
Destination port	66
Next byte to send	3723
Next byte expected	3949
My window size	0
Peer's window size	0
Data to send	8500

- a) Source port of segment (i) = [66]
- b) Source port of segment (ii) = [55]
- c) Source port of segment (iii) = [66]
- d) Sequence number of segment (i) [3334]
- e) Sequence number of segment (iii) [3889]
- f) Acknowledgement number of segment (iv) [3889]
- g) Window advertisement of segment (i) [1500]
- h) Window advertisement of segment (iii) [500]
- i) Data length of segment (iv) [500]
- j) Data length of segment (v) [60]

]

Process A is given TCP socket 55 to connect to TCP socket 66 on server B (where these ports are used for some non-standard purposes). After a while both A and B has thousands of bytes of data ready to send to the other computer. A last successfully sent byte 2222, while B last successfully sent byte 3333. Both windows are currently full, so neither computer can transmit. Node A consumes 555 bytes from its buffer, while node B consumes 1500 bytes from its buffer. Both computers will send as much data as possible during any given transmission, but neither computer will transmit more than a 1000 bytes in a single segment.

Normally A and B will communicate in full duplex. However, for the purposes of this question assume that messages are sent one at a time. The events that happen are as follows:

(i) B sends data to A; (ii) A sends data to B; (iii) B sends an acknowledgement to A; A consumes 60 bytes from its buffer; (iv) A sends data to B; and (v) B sends a message to A.

Your task is to provide the values of the following header fields of the five segments listed above.

- a) Source port of segment (i)
- b) Source port of segment (ii)

- c) Source port of segment (iii)
- d) Sequence number of segment (i)
- e) Sequence number of segment (iii)
- f) Acknowledgement number of segment (iv)
- g) Window advertisement of segment (i)
- h) Window advertisement of segment (iii)
- i) Data length of segment (iv)
- j) Data length of segment (v)

[10]

TOTAL

[50]

END OF PAPER

Multiple Choice Questions

- 1) If the International Standards Organisation (ISO) adopts a standard
 - A. All member countries have to comply with the standard, but the rest of the world do not have to.
 - B. All countries in the world have to comply with the standard.
 - C. Almost all countries in the world enforce compliance with almost ISO standards, even though they don't have to.
 - D. D: ISO standards are irrelevant because nobody implements them.
 - E. None of the above
- 2) In data communications a protocol is always ...
 - A. A detailed description of the rules to be followed when communicating.
 - B. Just another name for a standard.
 - C. Defined in an RFC.
 - D. Associated with a specific TCP port.
 - E. More than one of the above
- 3) Ideally, an ISO layer provides services to ... and uses services from
 - A. the layer below it; the layer above it
 - B. the layer above it; the layer below it
 - C. its peer layer elsewhere in the network; the layer below it
 - D. the layer below it; its peer layer elsewhere in the network
 - E. a peer layer somewhere in the network; a peer layer somewhere in the network
- 4) The primary reason for including the host: header in the HTTP protocol is to
 - A. Ensure that the hosts file cannot be misused.
 - B. Make a clear distinction between client and host requests.
 - C. Enable virtual hosting.
 - D. Enhance security by ensuring that the destination server is indeed the one the user wanted to visit because the name of the server can now be compared with the name provided in the host: header.
 - E. Specify alternative hosts that may contain the content the requester wants.
- 5) The most important routing protocol at JINX is
 - A. RIP
 - B. BGP
 - C. OSPF
 - D. Dijkstra
 - E. Bellman-Ford
- 6) Indicating where layer 2 frames start and end is known as
 - A. Data delineation
 - B. Data fragmentation
 - C. Data separation
 - D. Packet framing
 - E. Flow control

- 7) The file that contains the names, addresses and related information about a domain or subdomain in the DNS is known as a ... file.
- A. DNS
 - B. Domain
 - C. Resolution
 - D. Zone
 - E. Bind
- 8) Root name servers in the DNS perform ... name resolution.
- A. Iterative
 - B. Recursive
 - C. Non-authoritative
 - D. All of the above
 - E. None of the above
- 9) A root name server will resolve the following query
- A. Type NS for com
 - B. Type A for com
 - C. Type NS for org.
 - D. Type A for org.
 - E. All of the above
- 10) A zone transfer refers to the process of
- A. Moving hosting from one service provider to another.
 - B. Selling a domain name to someone else.
 - C. Copying DNS entries from a primary server to a secondary server.
 - D. Accessing data outside the corporate network, or data in the demilitarised zone of the corporate network.
 - E. Obtaining new name servers.
- 11) Which of the following is not an SNMP request?
- A. Get
 - B. GetNext
 - C. GetBulk
 - D. Reboot
 - E. Set
- 12) Non-repudiation is implemented through
- A. Public key cryptography
 - B. LDAP
 - C. SNMP
 - D. Routing table exchange
 - E. Single sign-on
- 13) Suppose an HTTP request contains the following in its header:
- Accept-Charset:Shift JIS

Suppose a Web site is visited that contains text in 'plain' ASCII and the server does not support Shift JIS. What will happen?

- A. The browser will display an error message to indicate that the content cannot be displayed.
- B. Google Translate will be called to translate the page to a language that can be represented in Shift JIS.
- C. The server will serve ASCII and the browser will interpret it as Shift JIS, resulting in a garbled display.
- D. The Accept-Charset header given implies ISO-8859-1, so the pages should be served and displayed without problems.

14) EXIF data embedded in some images

- A. Is metadata about the image.
- B. Enforces digital rights management.
- C. Is not necessarily authentic.
- D. More than one of the above
- E. All of the above

15) A MIME multipart/alternative message should consist of components that

- A. Represent different messages from which the recipient may choose, such as alternative dates to schedule a meeting.
- B. The same message, encoded in different (alternative) manners.
- C. Include the "real " message and an alternative "cover" message that eavesdroppers will see — effectively hiding the real message.
- D. Are used just like multipart/digest components, but multipart/alternative uses a different separation mechanism between the components. E: Represent different messages intended for multiple parties; each recipient will only receive the part addressed to him/her.

16) The following utility may be used to decode messages that have been captured from a network to their original formats (images, sounds, executables, and so on) so that they may be viewed, played or otherwise used by their usual applications.

- A. winpcap
- B. nslookup
- C. mimencode
- D. dd
- E. bash

17) Which of the following is not a valid ASN.1 constructor?

- A. SEQUENCE
- B. MESSAGE
- C. CHOICE
- D. SET

18) An email message that contains a PDF document as an attachment may use the following MIME subtype to encapsulate the mail message and its attachment.

- A. multipart/mixed

- B. multipart/digest
- C. multipart/alternative
- D. multipart/plain
- E. text/plain

19) The ideal layer to implement encryption for most situations is

- A. 7
- B. 6
- C. 4
- D. 3
- E. Dependent on the details of the specific case.

20) rsync is a utility that

- A. Was written to simplify use of 'the cloud'.
- B. Marks synchronisation points that enable a network to resynchronise if synchronisation is lost.
- C. Synchronises the clock on the local computer with an external clock.
- D. Synchronises directories on the local computer only.
- E. Synchronises directories — possibly via the network.

21) Who of the following people is famous for (amongst others) his work on the early Internet (then the ARPANET)?

- A. Bill Gates
- B. Vinton Cerf
- C. Steve Jobs
- D. Tim Berners-Lee
- E. Mark Zuckerberg

22) Which of the following sites was not one of the original four connected to the ARPANET?

- A. UCLA
- B. MIT
- C. SRI
- D. UCSB
- E. University of Utah

23) Suppose a distributed database is used by a number of application protocols. On which 'OSI ISO layer should the two-phase commit protocol ideally be implemented?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6
- G. 7

24) Which ISO OSI layer's functionality can be informally described as being a 'pipe'?

- A. 1
- B. 2

- C. 3
- D. 4
- E. 5
- F. 6
- G. 7

25) What role(s) do RFCs fulfil for the Internet?

- A. Define standards
- B. Provide information
- C. Opinion polls
- D. More than one of the above
- E. All of the above

26) How may information be encoded that is supplied with an HTTP POST request?

- A. Key-value pairs, with spaces replaced by a + and URL-encoding applied afterwards.
- B. Plain text
- C. The native encoding of whatever data is communicated is used unmodified.
- D. HTML
- E. No data accompanies the HTTP POST message.

27) In practice, the parameter following an HTTP GET message may be ...

- A. filename.
- B. The name of an executable script.
- C. Any parameter the server has been built to interpret.
- D. More than one of the above
- E. All of the above

28) Protocols designed to determine network routes usually operate on ISO OSI layer ...

- A. 7
- B. 6
- C. 5
- D. 4
- E. 3

29) Who ultimately decides which TLDs are available on the internet?

- A. NSF
- B. ISO
- C. Uniforum
- D. UP
- E. ICANN

30) When a new DNS TLD is in landrush it means

- A. Only trademark holders are allowed to register domains in that TLD.
- B. Exactly the same as when the TLD is in a general available state.
- C. Normal users are allowed to register trademarked names in that TLD that would not have been possible under normal conditions.
- D. The domain recently started accepting registrations from the general public - possibly at a higher cost than the eventual annual fees will be.

E. Its use has not been approved yet.

31) The whois command should (eventually) query the ... records.

- A. ICANN
- B. IANA
- C. Registrar's
- D. TLD registry
- E. who. Is

32) When does a DNS zone transfer occur?

- A. When the zone owner moves to a different ISP.
- B. When a secondary nameserver is synchronised with a primary nameserver.
- C. When an owner sells a TLD to someone else.
- D. When a server is moved outside the DMZ.
- E. During name resolution.

33) A DNS SOA TTL entry indicates the time... (in seconds).

- A. A machine is allowed to use a resolved name before the name has to be resolved again.
- B. A secondary name server is allowed to use names before synchronising with the primary server,
- C. The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- D. The time a secondary name server should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.
- E. The time a client who wants to resolve a name should wait before sending another resolution request.

34) A DNS SOA RETRY entry indicates the time ... (in seconds).

- A. A machine is allowed to use a resolved name before the name has to be resolved again.
- B. A secondary nameserver is allowed to use names before synchronising with the primary server.
- C. The time a secondary nameserver is allowed to exceed the time limit imposed on it by the primary server if it cannot establish contact with the primary server.
- D. The time a secondary nameserver should wait to obtain a current copy of DNS entries after a previous synchronisation attempt failed.
- E. The time a client who wants to resolve a name should wait before sending another resolution request.

35) Which of the following is not a valid nslookup command?

- A. Is example.com
- B. set type=any
- C. server 8.8.8.8
- D. exit
- E. resolve up.ac.za

36) Assume you are writing a server that is intended to be accessed by a Telnet client.

- A. Normally the server should echo every character received, because the client may not be able to echo typed characters locally.
- B. The server should not echo received input because echoing will cause characters to be displayed twice on modern Telnet clients.
- C. The server should query the Telnet client to determine whether it echoes locally; the server should then only echo if the client will not echo.
- D. The server should issue an ANSI escape sequence to disable local echoing at the client; the server should then echo everything received.
- E. A good compromise is to echo every second character received; the user should be still be able to determine the intended output even if some characters are missing and others are repeated.

37) The structure used to store data used by SNMP is known as ...

- A. DSNMP
- B. X.509
- C. MIB
- D. NMP
- E. NIC

38) A toy "telephone" consisting of two tin cans connected by a piece of string is able to communicate in ... mode.

- A. Simplex
- B. Half-duplex
- C. Full duplex
- D. Half-simplex
- E. Full simplex

39) Which of the following is a useful Android app for determining information about networks?

- A. LanDroid
- B. Nexus
- C. iCloud
- D. DNS
- E. IMDB

40) A switch (used in the place of a hub)

- A. Connects different networks and can convert protocols.
- B. Makes a connection only between computers that are busy communicating
- C. Routes packets between nodes
- D. Operates on ISO OSI layer 3
- E. More than one of the above

41) Which of the following is (are) not a function of ISO/OSI's layer 2?

- A. Contention control
- B. Data delineation
- C. Dialogue control
- D. Error control
- E. More than one of the above

Long Questions

Question 1

- a) With which well-known port are the following services usually associated?
- | | | |
|------|--------|-----------|
| i. | HTTP | 80 |
| ii. | DNS | 53 |
| iii. | Telnet | 23 |
| iv. | SSH | 22 |
| v. | SMTP | 25 |
- b) What are the following ports typically used for? (it is not necessary to give the corresponding port numbers.)
- Well-known ports
Standard services, such as Web, FTP, etc.
 - Registered ports
Less common services for which a port has been allocated
 - Dynamic and/or Private Ports
Client-side and ad hoc applications
- c) Why are ports useful in firewalls?
The block access to specific services
- d) How many bits are used to identify a port in TCP/IP?
16 bits

Question 2

- a) How would one normally write the following IPv6 address?
0876:0000:0000:0000:0000:0000:0005:4321:h
876::5:4321
- b) What is the IPv6 loopback address?
::1
- c) Write 137.215.98.101 as an IPv6 address using a combined colon-hexadecimal/dotted-decimal notation.
::ffff:137.215.98.101
- d) Write 137.215.98.101 as an IPv6 address using only colon-hexadecimal notation.
::ffff:89d7:6265
- e) Suppose the vendor-id of Xerox is 00:00:01. Suppose you use their card 123456h produced with this vendor-id. What is the MAC address of the card?
00:00:01:12:34:56 Or 00-00-01-12-34-56

- f) Suppose you want to connect to port 8080 of the server at 12::abcd. How would you enter this address in a browser?
[12::abcd]:8080

Question 3

- a) Suppose UP decides to split the 16-bit host portion of their IP address into a 11-bit sub and 5-bit host portion.
- How many subnets will be supported?
1025 - 2048
 - How many nodes will be supported per subnet?
 - Provide the new netmask.
 - On which subnet will 137.215.98.25 be? (Provide just the number of the subset; not subset address.)
 - What will the host number of 137.215.98.25 be?
 - Write down the (internal) network address of 137.215.98.25 using CIDR notation.
 - What is the broadcast address of the network that question (iv) refers to?
 - Write down the first (smallest) IP address of a node on subnet 1.
 - Write down the last (largest) IP address of a node on subnet 1.
- b) Provide the following 'special' IPv4 addresses.
- Localhost
127.0.0.1
 - UP's network address in CIDR format
137.215.0.0/16
 - Requester issuing a DHCP request
0.0.0.0
 - The private 'class C' address block using CIDR notation?
192.168.0.0/16

- v. The broadcast address for network 10.0.0.0.
10.255.255.255

Question 4 (Other examples in 2013 Semester test, 2012 Exam)

Assume that the following six letters occur with the given frequencies:

A 11% | B 9% | C 19%

D 20% | E 7% | F 34%

Now design a Huffman-code for these letters. In each case, label the lighter subtree with a 0 and the heavier subtree with a 1.

- a) Write your answer on the space provided on your answer sheet.

A - 100

B - 1011

C - 00

D - 01

E - 1010

F - 11

- b) What message is encoded by the following bit string?

111001010101101

FAEBD

- c) How much bandwidth (expressed as a percentage) will, on average, be saved by using this code rather than the shortest possible fixed length code?

$0.57/3 \times 100 = 19\%$

Question 5 (Other examples in 2013 Semester Test, 2014 Exam, 2014 Semester Test)

Assume that the costs between nodes of a network are as follows:

A - D: 3

B - F: 1

A - E: 4

D - F: 6

B - C: 10

E - G: 10

B - D: 4

F - G: 2

B - E: 10

Use Dijkstra's algorithm to determine the routes from A.

#	Done	Candidates	X	Cost						Prior					
				B	C	D	E	F	G	B	C	D	E	F	G
1	A	DE	D	?	?	3	4	?	?	?	?	A	A	?	?
2	AD	EBF	E	7	?	3	4	9	?	D	?	A	A	D	?
3	ADE	BFG	B	7	?	3	4	9	14	D	?	A	A	D	E

4	ADEB	FGC	F	7	17	3	4	8	14	D	B	A	A	B	E
5	ADEBF	GC	G	7	17	3	4	8	10	D	B	A	A	B	F
6	ADEBFG	C	C	7	17	3	4	8	10	D	B	A	A	B	F

Question 6

(For both questions below plan your answers so that they fit in the space available on the answer sheet.)

- a) Name any five SMTP commands and very briefly say what the purpose of each is.
- HELO – Sent by a client to identify itself, usually with a domain name.**
 - MAIL FROM – Identifies the sender of the message.**
 - RCPT TO – Identifies the message recipients.**
 - RSET – Nullifies the entire message transaction and resets the buffer.**
 - QUIT – Terminates the session.**
- b) Name any five HTTP response header entries and very briefly say what the purpose of each is.
- Age - conveys the sender's estimate of the amount of time since the response was generated at the origin server.**
 - Server - contains information about the software used by the origin server to handle the request.**
 - Location - used to redirect the recipient to a location other than the Request-URI for completion.**
 - ETag - provides the current value of the entity tag for the requested variant.**
 - Accept-Ranges - allows the server to indicate its acceptance of range requests for a resource.**

Question 7

Assume some computer X has one network interface with MAC address 10-20-30-40-50-60 and IPv4 address 10.20.30.40. Assume, for the purpose of this question that 10.0.0.0/8 is a routable, public network address. Whenever a question below refers to an IP address (or even just an address) IPv6 is implied.

- a) What is the loopback address of X?
::1/128
- b) What unspecified address will X use whenever necessary?
::0/128
- c) What is the link-local address of X? No privacy-protection mechanisms are to be used.
Fe80::1220:30ff:fe40:5060
- d) What is the IPv4 mapped address of X?
::ffff.10.20.30.40
- e) What would the IPv4 compatible address of X have been?
::0.10.20.30.40
- f) If X wants to use 6to4 tunnelling, what would its network address be? (Use CDIR-notation)
2002:a14:1e28::/48
- g) If X were to use Teredo tunnelling, what prefix would indicate that it was using such tunnelling?
2001::/32
- h) What (destination) prefix would X use when it transmits a multicast packet?
ff00::/8

Question 8

Name five (5) fields that occur in a TCP header but not in a UDP header. Briefly explain why each field is necessary in a TCP header but not in a UDP header. (For the purposes of this question FIN is not an acceptable answer).

- 1) Acknowledgement Number – UDP does not require acknowledgement of data transfer between the sender and receiver.**
- 2) Window Size – UDP does not need to regulate the amount of data to send before requiring acknowledgement as it does not make use of acknowledgement.**
- 3) Sequence Number – UDP does not order data to be sent in a certain manner.**
- 4) Urgent Pointer – UDP does not prioritize data**

5) Control Flags – UDP does not manage data flow for specific situations

Question 9 (Other examples in 2016 Big Class Test 1)

Process A is given TCP socket 55 to connect to TCP socket 66 on server B (where these ports are used for some non-standard purposes). After a while both A and B has thousands of bytes of data ready to send to the other computer. A last successfully sent byte 2222, while B last successfully sent byte 3333. Both windows are currently full, so neither computer can transmit. Node A consumes 555 bytes from its buffer, while node B consumes 1500 bytes from its buffer. Both computers will send as much data as possible during any given transmission, but neither computer will transmit more than a 1000 bytes in a single segment. Normally A and B will communicate in full duplex. However, for the purposes of this question assume that messages are sent one at a time. The events that happen are as follows: (i) B sends data to A; (ii) A sends data to B; (iii) B sends an acknowledgement to A; A consumes 60 bytes from its buffer; (iv) A sends data to B; and (v) B sends a message to A. Your task is to provide the values of the following header fields of the five segments listed above.

- | | |
|---|------|
| a) Source port of segment (i) | 66 |
| b) Source port of segment (ii) | 55 |
| c) Source port of segment (iii) | 66 |
| d) Sequence number of segment (i) | 3334 |
| e) Sequence number of segment (iii) | 3889 |
| f) Acknowledgement number of segment (iv) | 3889 |
| g) Window advertisement of segment (i) | 1500 |
| h) Window advertisement of segment (iii) | 500 |
| i) Data length of segment (iv) | 500 |
| j) Data length of segment (v) | 60 |

Question 10

Assume that computers on an internet are addressed using addresses (A records) consisting of numbers from 1 to 100. Assume node 1 is a root name server and you use node 100 to perform recursive name resolution. The following are (partial) zone files at some nodes running iterative DNS servers. The files are incomplete since some rows and columns have been omitted; these omissions do not affect the answers to the questions. No caching occurs.

Node 1			Node 13			Node 21			Node 22		
com.	NS	10	ac	NS	20	xx	NS	66	xx	NS	77
org.	NS	11	co	NS	21	yy	NS	55	yy	NS	88
mil.	NS	12	org	NS	22	zz	NS	44	zz	NS	99
za.	NS	13	net	NS	23						

Node 55			Node 66			Node 77		
@	NS	55	@	NS	66	@	NS	77
www	A	50	@	MX,1	98	@	MX,1	99
ftp	A	51	www	A	70	www	A	80
mail	A	52	ftp	A	71	ftp	A	81
arthur	A	53	mail	A	72	mail	A	82
			arthur	A	73	arthur	A	83
						john	NS	55

The following series of nslookup queries is executed in sequence from top to bottom. Provide the address (that is, number) returned by each of the queries. The queries are indicated by a Roman number in square brackets (such as "[iii]"). The commands that do not return responses are not numbered. Write N/A if the query will not return an address.

nslookup > [www.xx.org.za](#) [i] **80**

> arthur.xx.co.za [ii] **73**

> [ftp.xx.co.za](#) [vi] **98**

> set type=NS

> john.xx.org.za [vii] **98**

> com. [iii] **10**

> set type=A

> co.za [iv] **N/A**

> john.xx.org.za [viii] **55**

> set type=A

> server 1

> [ftp.xx.co.za](#) [v] **N/A**

> com. [ix] **N/A**

> server 100

> server 77

> set type=MX

> [www.xx.org.za](#) [x] **80**

Question 11

For each of the following terms say (i) what phrase it abbreviates and (ii) what it is used for.

- RFC – **Request for comments; internet standards and related issues**
- UTF – **Unicode Transformation Format; encoding of (Unicode) characters**
- RJ45 – **Recommended Jack 45; plug for Ethernet, etc.**
- ASCII – **American Standard Code for Information Interchange; representation of characters**
- SSH – **Secure Shell; Virtual terminal with encrypted communication**
- POP3 – **Post Office Protocol 3; protocol to retrieve email from server**
- LDAP – **Lightweight Directory Access Protocol; Directory management/SSO**
- CINX – **Cape Town Internet Exchange; the ISPA peering point in Cape Town**

- i) ISP – **Internet Service Provider; a party that provides another party with access to the internet**
- j) ICANN – **Internet Committee for Assigned Names and Numbers; body that decides about names (such as TLDs and numbers (such as well-known ports))**

Question 12

Your PC is connected to a DNS server d provided by your ISP. Whenever your PC wants to resolve a FQDN it simply asks d to it; d does it and responds with the answer. Assume d has just been activated and therefore has no names cached at this point. Suppose your PC wants to send an email to $y@x.com$. and requests d to resolve the FQDN.

Assume that r is the root server chosen, c is the chosen. com nameserver and $x\cdot$ is the chosen $x.com$ nameserver.

Draw a diagram to indicate the messages that are sent between these five machines. Represent the five machines (PC, d , r , c , x) using circles. Represent the messages using arcs between the circles. Label the "request" messages with the type of request (such as A, AAAA, SOA), as well as the (partial) domain name that forms part of the message. For the "response" messages indicate what the response is where possible. (Hint: sometimes the response may be another server, such as x .) Number the messages (requests and responses) in the order in which they are sent.

Question 13

List (name) five (5) features of TCP that make TCP reliable.

- 1) **TCP tracks which packets are received by the other end**
- 2) **TCP uses 3-way hand-shake**
- 3) **TCP is connection orientated**
- 4) **TCP sends streams of data**
- 5) **It will not deliver any out of order packets**

Question 14

Suppose you manage the network 137.215.0.0. You want to divide it into subnets such that you can accommodate at least n subnetworks, and where each subnetwork can accommodate at least c computers. Say, for the various combinations of n and c listed below.

- i. What netmask you will use;
- ii. What address (using CIDR notation) of the subnetwork that contains 137.215.215.215 is; and
- iii. What the broadcast address of the subnetwork containing 137.215.215.215 is.
 - a) $n = 100$ and $c = 300$
 - i.
 - b) $n = 10$ and $c = 4000$

c) $n = 4000$ and $c = 10$

d) $n = 300$ and $c = 100$

Question 15

- a) Explain what is meant by *transparency* on the data link layer.
- b) List four approaches that may be used to achieve transparency on the data link layer. For each provide an example of a protocol that uses the specific approach.

Question 16

The IP addresses used by a department in an organisation range from 192.168.132.153 to 192.168.132.158.

- a) What netmask does the department use?
255.255.255.248
- b) What is the broadcast address of the department?
192.168.132.159
- c) What is the network address of the department?
192.168.132.152
- d) What is the network address of the department in CIDR notation?
192.168.132.152/29

The next two questions assume that routing to the organisation occurs in a classful manner.

- e) In which class does the organisation's IP address belong?
C
- f) What is the organisation's network address?
192.168.132.0

Now assume that the organisation's ISP has assigned the address referred to in the previous question, as well as the subsequent three network addresses of that class to the organisation. The ISP treats it as a supernet.

- g) What is the organisation's network address according to the ISP? Use CIDR notation.

192.168.132.0/22

- h) What is the organisation's broadcast address on this supernet?

192.168.135.255

- i) What is the netmask that the ISP associates with this supernet (assuming it uses a netmask)?

255.255.252.0

Question 17

Give the names of the following concepts as they are used in TCP/IP. Where applicable, give the name of the concept as it is used in the TCP or IP header. If the concept is used in both IPv4 and IPv6, use the IPv4 name (if the two protocols use different names). The use of abbreviations to answer the questions is acceptable.

- a) The number that identifies a running process (for message delivery).

Port

- b) The number of hops before a datagram will be discarded.

TTL

- c) Dividing large IP packets into smaller packets.

Fragmentation

- d) The number used to identify a TCP segment, so that it can be acknowledged.

Sequence number

- e) The TCP flag that informs the other party that the connection is to be terminated.

FIN

- f) The routing metric used by RIP.

Hop count

- g) The routing protocol that exchanges a so-called link-state database.

OSPF

- h) The protocol that IP uses to inform the sender of a packet when that packet is discarded.

ICMP

- i) The protocol that is used to translate IP addresses to MAC addresses.

ARP

- j) SNMP software that executes on a node that is managed remotely.

Agent

Question 18

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left and even parity is used. You receive the following characters:

- a) 001101010101 **3**
- b) 011110000110 **1**
- c) 000000010110 **4**
- d) 100010111010 **1**
- e) 001101101011 **5**

Indicate in the appropriate space on your answer sheet which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

Question 19

You work at a NIC. The NIC has control over IP addresses 200.0.0.1 and 201.255.255.254, as well as all addresses between these two. The first address has been granted to a company known as Acme, along with the other addresses in its (class-based) block. Answer the questions below using dotted decimal notation.

- a) In which IP class does the first address belong?
C
- b) What network mask will Acme use (not using subnetting)?
255.255.255.0
- c) Give the address of Acme's network.
200.0.0.0
- d) What is the broadcast address for all hosts on Acme's network?
200.0.0.255
- e) Acme has grown to the extent that the NIC decides to grant them all addresses in the range given at the top of this question using supernetting (including the obvious network and broadcast addresses implied by this). What is Acme's network address now? Use CIDR notation to answer the question.
200.0.0.0/7
- f) What netmask will Acme now use?
254.0.0.0

- g) What broadcast address will Acme now use?
201.255.255.255
- h) Suppose Acme wants to split this new address into at least 25 subnets, where each subnet can handle at least a million computers. What will the network address be of the subnet on which 200.200.200.200 belongs? Use CIDR notation for your answer.
200.192.0.0/12

Question 20

CSMA, CSMA/CD and TDMA are three techniques to regulate media access control. Name any five (5) other techniques, and, for each one listed, name one real-world protocol that uses that MAC approach.

CDMA: CDMA2000

Master-slave: SDLC/HDLC

TDMA: GSM

(Pure) MA: Aloha

Slotted MA: Slotted Aloha

Question 21

An organisation, Acme, has a computer C with IPv4 address 150.150.150.150 on its network. Acme decides to subnet its network into at least 20 subnets that can each handle up to 1500 nodes.

- a) Which IPv4 class of address class does Acme use?
B
- b) What is the address of the network Acme uses?
150.150.0.0
- c) What is the broadcast address of this network (before subnetting)?
150.150.255.255
- d) What is Acme's netmask (before subnetting)?
255.255.0.0
- e) What is Acme's netmask (after subnetting)?
255.255.248.0
- f) What is the address of the subnet on which C is now located?
150.150.144.0
- g) On which subnet is C now located?

10

- h) What is the broadcast address of the subnet on which C is now located?

150.150.151.255

Question 22

Consider each of the following host IP addresses.

- 1) 161.156.114.180 (n = 104)
- 2) 113.176.201.141 (n = 27)
- 3) 41.58.129.89 (n = 221)

For each address

- a) Provide the network address of the network on which the host resides;
1) 161.156.0.0
2) 113.0.0.0
3) 41.0.0.0
- b) Identify the class of the network;
1) Class B
2) Class A
3) Class A
- c) Divide (or combine) the network into (at least) n subsets (or supernets), with each subnet (or supernet) as large as possible for the values of n given above.
 - i. Provide the resulting netmask;
1) 255.255.154.0
2)
 - ii. Provide the network address of the subnet (or supernet) on which the host now resides; and
 - iii. Provide the broadcast address of the subnet (or supernet) on which the host now resides.

Question 23

- a) IPv4 packets contain a header field that causes the packet to be discarded when this field becomes 0. What is the name of the field?

Time To Live

- b) IPv6 packets contain a header field that causes the packet to be discarded when this field becomes 0. What is the name of the field?

Hop Limit

- c) What is the process of breaking IP packets into smaller packets called?
IP Fragmentation
- d) What is the private class A network address in IPv4? Use CIDR notation for your answer.
10.0.0.0/8
- e) What does the IPv4 address 0.0.0.0 indicate?
All IPv4 addresses on the local machine
- f) What does the IPv4 address 127.0.0.1 indicate?
Localhost
- g) Which command causes an ICMP Echo request to be sent to a target machine?
Ping

Question 24

Pick the correct protocol from the list below and write the corresponding letter next to each description.

- 1) To forward email from the sender towards the recipient.
L - SMTP
- 2) A network management protocol developed for TCP/IP.
A - SNMP
- 3) A secure virtual terminal protocol.
H - HDLC
- 4) A protocol where the roles of client and server are sometimes interchanged.
E - FTP
- 5) A windowing protocol popular in Unix and its derived systems.
S - X.11
- 6) A protocol that performs zone transfers.
D - DNS
- 7) A secure protocol that manages a user's mailboxes.
P - IMAP4
- 8) A protocol that includes the POST command for form completion.
J - HTTP
- 9) An implementation of Dijkstra's algorithm.
O - OSPF
- 10) A protocol often used to load operating system images via the network.

Q – TFTP

Question 25

a) Consider packet 1.

i) Which application layer protocol is used?

DNS

ii) Which transport layer protocol is used?

UDP

iii) Which network layer protocol is used?

IPv4

iv) Which data link layer protocol is used?

Ethernet

b) What are the vendor IDs of interfaces in these packets of the interfaces supplied by each of the following manufacturers?

i) ASRock

29:d3:3e

ii) Raspberry-Pi

ca:1f:1e

iii) TP-Link

e7:76:46

c) Consider packets 1 and 2.

i) Explain the action of node 10.1.1.31. Why does it transmit packet 2?

10.1.1.31 received a DNS query from 10.1.1.2, and is sending a DNS query to 8.8.8.8 to answer the query it received.

ii) The TTL fields of packets 1 and 2 are quite different (128 vs 64). What is the (most likely) explanation for this difference?

The systems most likely just have different default values for TTL. (Different OSs).

- iii) The destination ports of packets 1 and 2 are the same (53). Why?
They are both using the DNS protocol to send queries, which uses port 53.
 - iv) The source ports of packets 1 and 2 are quite different (63933 and 8312, respectively). Briefly explain this phenomenon.
Packet 1 was sent from a registered port (there was a free port in the registered ports at that node), while packet 2 was sent from a dynamic port (it's possible there were no free ports in the registered ports at that node, so it requested one from the dynamic ports).
- d) Consider packet 3. Assume that it is a response to packet 2.
 - i) What information on layer 4 supports that packet 3 is a response to packet 2?
The destination port in packet 3 is the source port from packet 2.
 - ii) What information on layer 3 may support that packet 3 is a response to packet 2?
The destination IP address is the source IP address from packet 3.
 - iii) What information on layer 2 may support that packet 3 is a response to packet 2?
The destination MAC address is the source MAC address from packet 3.
- e) Consider packet 4.
 - i) All the addresses, as well as the destination ports in packets 2 and 4 are identical. Briefly explain why the source ports are not.
The source ports are temporary, and are randomly assigned when a process needs a port. The destination ports are the same because that is required for the DNS protocol.
 - ii) The TTL fields of packets 2 and 4 are also identical (64). Would one expect this, or is it coincidental? Explain briefly.
One would expect this, as both packets are from the same protocol on the same machine - it likely has 64 as the default TTL value for all outgoing UDP packets.
 - iii) It seems that it was not necessary for 10.1.1.31 to send this packet. Why?
Because this packet was sending a DNS query to resolve hermes.cs.up.ac.za. However, this FQDN was resolved in a previous DNS query (Packet 3), and therefore should be cached.

- iv) Why do you think that 10.1.1.31 sent this packet even though it was not really necessary?

A user explicitly sent the query.

- f) Consider packet 6. To which of the packets provided in the addendum is it (most likely) a response?

Packet 4.

- g) Consider packet 7. In which way does it differ from packet 1 on the application layer?

Packet 7 issued a DNS query for an IPv6 address (Type AAAA), while Packet 1 issued a DNS query for an IPv4 address (Type A).

- h) Consider packet 9

- i) The trace contained no response to the query issued in packet 8. Given this fact, what feature of the DNS does packet 9 exploit? Briefly explain.

It is exploiting DNS's ability to 'fall back' to another DNS server if it does not receive a response to the first query.

- ii) While packets 8 and 9 are very similar, they are sent to different destinations (8.8.8.8 and 192.5.4.1, respectively). Explain why both packets use the same destination address on layer 2.

Because the layer-2 destination (30:b5:c2:e7:76:46) is the next hop on the route to their destinations (8.8.8.8 and 192.5.4.1). It is likely the router connecting the local network to the Internet.

- i) Assume packet 10 is a response to packet 9.

- i) Packet 10 does not contain an answer to the query contained in packet 9. Justify this claim.

It did not return the IP address of hermes.cs.up.ac.za, but just a list of name servers that might be able to resolve hermes.cs.up.ac.za.

- ii) What type of information is provided as 'answer' in packet 10?

A list of name servers that might be able to resolve hermes.cs.up.ac.za.

- iii) What should one conclude from the fact that a proper answer is not provided?

The nameserver at 192.5.4.1 did not contain an entry for hermes.cs.up.ac.za.

j) Consider the merit of sending packet 11 at this time.

i) Why may packet 11 provide useful information?

It is possible that the initial query to 8.8.8.8 (packet 8) may have been lost, rather than ignored at the destination. Hence the query might actually be answered correctly this time.

ii) Why may packet 11 be problematic?

The packet may again be lost, or 8.8.8.8 may not have an answer for the DNS query (and therefore not answer the request).

Question 26

a) What is the purpose of packets A, B and C? Justify your answer.

To begin the TCP connection.

Packet A sends SYN to node Y. Packet B sends SYN+ACK back to node X, which then responds with ACK - a typical TCP connection handshake.

b) Consider packet D.

i) What is it's application layer purpose?

To send a GET request.

ii) What is squid (Line 41)?

A proxy server.

iii) The TCP ACK flag is set. What is acknowledged?

Packet C.

c) Packet E does not carry any payload - it's length is 0.

i) For what purposes was E sent?

To acknowledge Packet D.

ii) The TCP SEQ field has the value 1. Is this correct? Why (not)?

Yes. The SYN+ACK packet from the initial handshake had a payload size of 1 byte, which is why the sequence number was incremented by 1.

iii) The TCP ACK field has the value 618. Is this correct? Why (not)?

Yes. The ACK packet sent as part of the initial handshake (1 byte), plus the GET request sent (617 bytes), makes up the value 618 that is acknowledged.

- iv) The calculated window size advertisement is 15872. Is this (approximately) what you expected? Why (not)?

Not really. The last window size received was 17898, and only 618 bytes have been sent to node Y since the last window advertisement was received. However, it is probable that node Y received packets from other nodes in the meantime.

- d) What is the MIME type of the payload in packet F?

application/json

- e) Which node (X or Y)? Initiates termination of the connection? Justify your answer.

Node Y. It is the first node to set the FIN flag.